

**To determine the prognostic accuracy of the Heart Score as a predictor for
Major Adverse Cardiac Events in patients presenting with Chest Pain to
the Emergency Department in a tertiary care Hospital**



**A DISSERTATION SUBMITTED TO THE NATIONAL BOARD OF
EXAMINATIONS NEW DELHI**

EMERGENCY MEDICINE

(2022-2025)

BY

DR.SHAIK KARISHMA BANU

(REG.NO:152-46133-221-243387)

UNDER THE SUPERVISION AND GUIDANCE OF

DR.PRAVEEN KUMAR KULKARNI

CONSULTANT OF GENERAL MEDICINE

KRISHNA INSTITUTE OF MEDICAL SCIENCES

MINISTER ROAD

SECUNDERABAD

500003

DECLARATION BY THE CANDIDATE

I, DR. SHAIK KARISHMA BANU, hereby declare that the dissertation titled '**To determine the prognostic accuracy of the Heart Score as a predictor for Major Adverse Cardiac Events in patients presenting with Chest Pain to the Emergency Department in a tertiary care Hospital**' has been a genuine and bonafide work prepared by me under the guidance of **DR. PRAVEEN KULKARNI, CONSULTANT, GENERAL MEDICINE, KIMS HOSPITAL, SECUNDERABAD** and is submitted to the national board of Examination, New Delhi, in partial fulfillment to the regulations governing the award of Diplomate of National Board in Emergency Medicine. I also declare that this dissertation has not been submitted previously by any University for the award of any degree or diploma.



DR.SHAIK KARISHMA BANU

Place:Hyderabad

Date: 26/3/2025

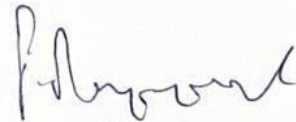
CERTIFICATE OF THE HEAD OF THE INSTITUTION

This is to certify that the dissertation entitled '**To determine the prognostic accuracy of the Heart Score as a predictor for Major Adverse Cardiac Events in patients presenting with Chest Pain to the Emergency Department in a tertiary care Hospital**' is a bonafide work done by **DR.SHAIK KARISHMA BANU** (REG.NO:152-46133-221-243387) during the period of May 2023- November 2024 at KIMS Hospital, SECUNDERABAD, in partial fulfillment of the regulations governing the Diplomate of National Board in Emergency Medicine degree

I have great pleasure in forwarding this dissertation to National Board of Examination New Delhi

Place: Hyderabad

Date: 26/3/2025



Signature of the head of institution

Dr. P. RAJA GOPAL

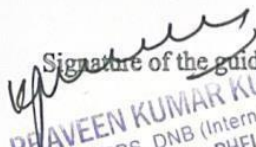
B.Sc, MBBS, MS, MCh, MNAMS (CTS), FIAS
Director, Medical Education

Krishna Institute of Medical Sciences Ltd.
Minister Road, Secunderabad-500003

CERTIFICATE BY THE GUIDE

This is to certify that the dissertation entitled '**To determine the prognostic accuracy of the Heart Score as a predictor for Major Adverse Cardiac Events in patients presenting with Chest Pain to the Emergency Department in a tertiary care Hospital**' is a bonafide work done by **DR. SHAIK KARISHMA BANU** (REG.NO:152-46133-221-243387) during the period of May 2023- November 2024 at KIMS Hospital, SECUNDERABAD, in partial fulfillment of the regulations governing the Diplomate of National Board in Emergency Medicine degree I have great pleasure in forwarding this dissertation to National Board of Examination New Delhi

Place : Hyderabad
Date : 26/03/2025

Signature of the guide

DR. PRAVEEN KUMAR KULKARNI
MBBS, DNB (Internal Medicine)
Diplomate Diabetology, PHFI (CCEBDM)
Reg. No: 83696
Consultant Physician
HOD - EMS/EMD
Krishna Institute of Medical Sciences Ltd
Minister Road, Secunderabad-500003.

CERTIFICATE BY THE CO GUIDE

This is to certify that the dissertation entitled '**To determine the prognostic accuracy of the Heart Score as a predictor for Major Adverse Cardiac Events in patients presenting with Chest Pain to the Emergency Department in a tertiary care Hospital**' is a bonafide work done by **DR. SHAIK KARISHMA BANU** (REG.NO:152-46133-221-243387) during the period of May 2023-Nov-2024 at KIMS Hospital, SECUNDERABAD, in partial fulfillment of the regulations governing the Diplomate of National Board in Emergency Medicine degree. I have great pleasure in forwarding this dissertation to National Board of Examination New Delhi

Place : Hyderabad
Date. 26/3/2025

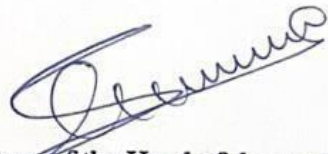
Signature of the co-guide

D. Reddy Chowdary


CERTIFICATE BY THE HEAD OF DEPARTMENT

This is to certify that the dissertation entitled “**To determine the prognostic accuracy of the Heart Score as a predictor for Major Adverse Cardiac Events in patients presenting with Chest Pain to the Emergency Department in a tertiary care Hospital**” is a bonafide work done by **DR. SHAIK KARISHMA BANU** (REG.NO:152-46133-221-243387) during the period of May 2023- Nov-2024 at KIMS Hospital, SECUNDERABAD, in partial fulfillment of the regulations governing the Diplomate of National Board in Emergency Medicine degree have great pleasure in forwarding this dissertation to National Board of Examination New Delhi

Place : Hyderabad
Date 26/3/2025


Signature of the Head of department
Dr. S.SRINIVAS
MBBS MD
Regd No: 01049
Senior Consultant & HOD
Department Of Emergency Medicine
KIMS Hospital, Secunderabad - 003

DECLARATION CUM UNDERTAKING FOR FRESH THESIS

I DR. SHAIK KARISHMA BANU, hereby declare that this dissertation entitled “To determine the prognostic accuracy of the Heart Score as a predictor for Major Adverse Cardiac Events in patients presenting with Chest Pain to the Emergency Department in a tertiary care Hospital” is a “bonafide” in nature and was carried out by me under the guidance and supervision of my guide DR.PRAVEEN KULKARNI and co-guide DR.RADHIKA CHOWDARY. The interpretations put forth are based on my reading and understanding of the original texts and they are not published in the form of books, articles or monographs or articles. The other books, articles and websites, which I have made use of are acknowledged at the respective place in the text.

Place: Hyderabad
Date: 26/3/25

Dr. Shaik Karishma Banu
(Candidate Signature)

Candidate Name: Dr. Shaik Karishma Banu

Specialty: DNB EMERGENCY MEDICINE

Registration Number: 152-46133-221-243387

COUNTERSIGNED & VERIFIED

[Signature]
(Guide Signature)

Guide Name:

Guide Designation:

Institution:

Dr. PRAVEEN KUMAR KULKARNI
MBBS, DNB (Internal Medicine)
Diplomate Diabetology, PKFI (CCEBDM)
Reg. No: 60888
Consultant Physician
HOD - EMC/EMD
Krishna Institute of Medical Sciences Ltd.
Minister Road, Secunderabad-500003.

Dr. Radhika Chowdary
(Co-Guide Signature)
Co-Guide Name:
Co-Guide Designation:
Institution:
Stamp bearing name:



[Signature]
(HOI Signature)
HOI Name:
HOI Designation:
Institution:
Stamp bearing name:

Dr. P. RAJA GOPAL
B.Sc, MBBS, MS, MCh, MNAMS (CTS), FIAS
Director, Medical Education
Krishna Institute of Medical Sciences Ltd.
Minister Road, Secunderabad-500003

ACKNOWLEDGEMENT

I take this opportunity with much pleasure to thank all the people who have helped me through the course of my journey towards producing the thesis to whom I would be indebted throughout my life. Some names require particular mention here. My sincere and heartfelt gratitude to my guide, **DR PRAVEEN KULKARNI**, CONSULTANT GENERAL MEDICINE, and co-guide **DR RADHIKA CHOWDARY** , and Head of emergency department , **DR. SRINIVAS SANKURATRI** who have helped me throughout the journey. I also express my gratitude to my teachers, respected colleagues. I extend my sincere gratitude to **DR MANIMALA RAO**, DIRECTOR, for the guidance for allowing me to conduct this study. I would also like to thank my patients for their cooperation and patience. Last but not the least, I would like to thank my family for their support and encouragement throughout the thesis work.

“To determine the prognostic accuracy of the Heart Score as a predictor for Major Adverse Cardiac Events in patients presenting with Chest Pain to the Emergency Department in a tertiary care Hospital”

GENERAL INFORMATION

1. NAME : DR.SHAIK KARISHMA BANU
2. FATHER’S NAME : SHAIK CHAND BASHA
3. CORRESPONDENCE ADDRESS: 1-8-31/1, MINISTER ROAD, SECUNDERABAD
PIN-500003

DNB TRAINING DETAILS

1. REGISTRATION DETAILS
REG.NO: 152-46133-221-243387
DATE OF JOINING : 24/11/2022
DATE OF COMPLETION : 23/11/2025
2. GUIDE: DR PRAVEEN KULKARNI
DEPARTMENT OF GENERAL MEDICINE
KIMS HOSPITAL, SECUNDERABAD
3. CO-GUIDE: DR RADHIKA CHOWDARY
DEPARTMENT OF BIOCHEMISTRY
KIMS HOSPITAL, SECUNDERABAD

LIST OF ABBREVIATIONS

- ACI-TIPI - Acute Cardiac Ischemia Time-Intensive Predictive Instrument
- ACS - Acute Coronary Syndromes
- ADAPT - Two-hour Accelerated Diagnostic Protocol to Assess Patients with Chest Pain Symptoms Using Contemporary Troponins as the Only Biomarker
- AMI - Acute Myocardial Infarction
- ASPECT - Asia-Pacific Evaluation of Chest Pain Trial
- CAD - Coronary artery disease
- CATCH - Cardiac CT in the Treatment of Acute Chest Pain Trial
- CCTA - Coronary computed tomography angiography
- CK-MB - Creatinine kinase-MB isoenzyme
- CT - Computed Tomography
- ECG - Electrocardiogram
- EDACS - Emergency Department Assessment of Chest Pain Score
- GRACE - Global Registry of Acute Coronary Events
- HEART - History, ECG, Age, Risk Factors, Troponin [Score]
- MACE - Major adverse cardiac event
- MI - Myocardial Infarction
- mSv - Millisievert
- mV - Millivolt
- NACPR - North American Chest Pain Rule
- NPV - Negative predictive value
- NSTEMI - Non-ST-segment elevation myocardial infarction
- PROMISE - Prospective Multicenter Imaging Study for Evaluation of Chest Pain

- PROSPECT - Prospective Randomized Outcome Trial Comparing Radionuclide Stress Myocardial Perfusion Imaging and ECG-gated CCTA PURSUIT - Platelet Glycoprotein IIb/IIIa in Unstable Angina: Receptor Suppression Using Integrilin (eptifibatide) Therapy
- STEMI - ST-segment elevation myocardial infarction
- TIMI - Thrombolysis in Myocardial Infarction [Score]
- TRO CT - Triple-rule-out computed tomography

CONTENTS

SL NO	TITLE	PAGE NUMBER
1	INTRODUCTION	1
2	REVIEW OF LITERATURE	3
3	AIMS AND OBJECTIVES	26
4	MATERIAL AND METHODS	27
5	RESULTS	30
6	DISCUSSION	57
7	SUMMARY AND CONCLUSION	63
8	BIBLIOGRAPHY	66
9	ANNEXURE 1: PROFORMA	75
10	ANNEXURE 2: PATIENT INFORMATION SHEET	76
11	ANNEXURE 3: INFORMED CONSENT FORM	78
12	MASTER CHART	80
13	THESIS PROTOCOL APPROVAL	90

LISTS OF TABLES

TABLE 1	HEART score for stratification of Chest pain patients
TABLE 2	HEART SCORE
TABLE 3	Frequency distribution of Characteristics of Patients Presenting with Chest Pain
TABLE 4	Comparison of distribution of age and Heart score in patients presenting with chest pain
TABLE 5	Comparison of distribution of Gender and Heart score in patients presenting with chest pain
TABLE 6	Comparison of distribution of Giddiness and Heart score in patients presenting with chest pain
TABLE 7	Comparison of distribution of Generalized Weakness and Heart score in patients presenting with chest pain
TABLE 8	Comparison of distribution of syncope and Heart score in patients presenting with chest pain
TABLE 9	Comparison of distribution of Epi-gastric Discomfort and Heart score in patients presenting with chest pain
TABLE 10	Comparison of distribution of Nausea and vomiting and Heart score in patients presenting with chest pain
TABLE 11	Comparison of distribution of altered mental status and Heart score in patients presenting with chest pain
TABLE 12	Comparison of distribution of history and Heart score in patients presenting with chest pain
TABLE 14	Comparison of distribution of ECG and Heart score in patients presenting with chest pain
TABLE 15	Comparison of distribution of hypertension and Heart score in patients presenting with chest pain
TABLE 16	Comparison of distribution of Diabetes and Heart score in patients presenting with chest pain

LISTS OF TABLES

TABLE 17	Comparison of distribution of Smoking and Heart score in patients presenting with chest pain
TABLE 18	Comparison of distribution of Obesity and Heart score in patients presenting with chest pain
TABLE 19	Comparison of distribution of Family history and Heart score in patients presenting with chest pain
TABLE 20	Comparison of distribution of Alcohol and Heart score in patients presenting with chest pain
TABLE 21	Comparison of distribution of Risk factor and Heart score in patients presenting with chest pain
TABLE 22	Comparison of distribution of Troponins and Heart score in patients presenting with chest pain
TABLE 23	Distribution of study groups as per incidence of major adverse cardiac events (MACE) in patients presenting with chest pain
TABLE 24	Distribution of study groups as per mortality in patients presenting with chest pain
TABLE 25	Mean HEART score in cases with and without MACE and mortality
TABLE 26	Receiver operative characteristic (ROC) curve analysis of HEART score to predict MACE in patients presenting with chest pain
TABLE27	Diagnostic accuracy of HEART score for prediction of MACE in patients presenting with chest pain

LISTS OF FIGURES

FIGURE 1	Acute coronary syndrome
FIGURE 2	Pathogenesis of Atherosclerosis
FIGURE 3	Understanding of Unstable angina ,NSTEMI and STEMI
FIGURE 4	Comparison of distribution of age and Heart score in patients presenting with chest pain
FIGURE 5	Comparison of distribution of Gender and Heart score in patients presenting with chest pain
FIGURE 6	Comparison of distribution of Giddiness and Heart score in patients presenting with chest pain
FIGURE 7	Comparison of distribution of Generalized Weakness and Heart score in patients presenting with chest pain
FIGURE 8	Comparison of distribution of syncope and Heart score in patients presenting with chest pain
FIGURE 9	Comparison of distribution of Epi- gastric Discomfort and Heart score in patients presenting with chest pain
FIGURE 10	Comparison of distribution of Nausea and Heart score in patients presenting with chest pain
FIGURE 11	Comparison of distribution of Vomiting and Heart score in patients presenting with chest pain
FIGURE 12	Comparison of distribution of altered mental status and Heart score in patients presenting with chest pain
FIGURE 13	Comparison of distribution of History and Heart score in patients presenting with chest pain

LISTS OF FIGURES

FIGURE 14	Comparison of distribution of ECG and Heart score in patients presenting with chest pain
FIGURE 15	Comparison of distribution of Hypertension and Heart score in patients presenting with chest Pain
FIGURE 16	Comparison of distribution of Diabetes and Heart score in patients presenting with chest pain
FIGURE 17	Comparison of distribution of Smoking and Heart score in patients presenting with chest pain
FIGURE 18	Comparison of distribution of obesity and Heart score in patients presenting with chest pain
FIGURE 19	Comparison of distribution of Family history and Heart score in patients presenting with chest pain
FIGURE 20	Comparison of distribution of Alcohol and Heart score in patients presenting with chest pain
FIGURE 21	Comparison of distribution of Risk factor and Heart score in patients presenting with chest pain
FIGURE 22	Comparison of distribution of Troponins and Heart score in patients presenting with chest pain
FIGURE 23	Distribution of study groups as per incidence of major adverse cardiac events (MACE) in patients presenting with chest pain
FIGURE 24	Distribution of study groups as per mortality in patients presenting with chest pain
FIGURE 25	Mean HEART score in cases with and without MACE and mortality
FIGURE 26	Receiver operative characteristic (ROC) curve analysis of HEART score to predict MACE in patients presenting with chest pain
FIGURE 27	Diagnostic accuracy of HEART score for prediction of MACE in patients presenting with chest pain

INTRODUCTION

Coronary heart disease continues to be a major global health challenge, responsible for a significant number of deaths and hospitalizations worldwide. It affects people in both developed and developing countries, often striking without warning. Two of its most common manifestations are angina pectoris—chest pain due to reduced blood flow to the heart—and acute myocardial infarction, more commonly known as a heart attack. The World Health Organization (WHO) reported that cardiovascular diseases caused around 17.1 million deaths globally in 2004, a number expected to climb to 23.4 million by 2030 [1].

Among individuals diagnosed with coronary artery disease, chest pain is typically the first symptom that prompts them to seek emergency care. However, not all chest pain is a sign of a heart attack, and hospitals—especially in busy emergency departments—can't admit every person who walks in with this complaint [2]. This makes it crucial for healthcare professionals to quickly and accurately identify who is truly at risk of a serious cardiac event, such as acute coronary syndrome (ACS), and who may be suffering from a less critical issue [3].

Over the past decade, researchers have explored various methods to help emergency doctors assess chest pain and make informed decisions quickly [3]. While tools like the TIMI score [5] and GRACE score [6] have been widely used, they were originally designed for patients already diagnosed with ACS, and don't always perform well when applied to people with undifferentiated chest pain in the emergency department [7–13].

To bridge this gap, the HEART score was introduced. This tool focuses specifically on patients presenting with chest pain but without a confirmed diagnosis. The name "HEART" is an acronym that stands for five important components used to assess risk: History, ECG findings, Age, Risk factors, and Troponin levels [14,15]. Since its development, the HEART score has been tested in various clinical settings and has proven effective in identifying non-ST-elevation acute coronary syndrome (NSTEMI-ACS) [16,17].

One of the main reasons the HEART score has gained popularity is its simplicity and clinical usefulness. It helps emergency physicians make faster, more confident decisions about which patients can be safely discharged and which need further testing or admission. This not only improves patient care but also helps avoid unnecessary tests and hospital stays, saving both time and resources [18]. Because of its reliability, several international guidelines now recommend the HEART score as part of standard practice for evaluating chest pain in emergency settings [19].

That said, there's a catch—most of the research validating the HEART score comes from high-income countries [16–19], where healthcare systems, patient demographics, and disease patterns are quite different from those in low- and middle-income countries like India [20,21]. This raises an important question: Can a scoring system designed and tested abroad work just as well in Indian hospitals, where patient populations are more diverse, and healthcare resources are often limited?

India presents a particularly urgent case. Heart disease is on the rise here, with many people developing risk factors like high blood pressure, diabetes, and obesity at a younger age. Urban lifestyles, poor diet, stress, and limited access to regular medical care only add to the problem. In rural and semi-urban regions, things are even tougher—there may be a lack of diagnostic tools, trained personnel, and timely treatment options, making early detection even more critical.

Emergency departments in India are often overcrowded and understaffed, making quick, accurate decisions more challenging than ever. In such a setting, a tool like the HEART score could be extremely helpful—if it works well for the local population. But without solid data from Indian hospitals, it's difficult for doctors to rely on it fully.

This study was designed to help fill that gap. By evaluating how well the HEART score predicts major adverse cardiac events (MACE)—such as heart attacks, the need for emergency procedures, or death—among patients visiting the emergency department of a tertiary care hospital in India, we aim to provide much-needed local evidence. The goal is to see whether this tool can help improve triage decisions, reduce misdiagnoses, and ultimately lead to better outcomes for patients experiencing chest pain.

The findings from this study could also open the door to tailoring or fine-tuning the HEART score to better reflect the specific needs of Indian patients. As healthcare systems continue to evolve, the use of well-tested, practical, and context-sensitive tools like the HEART score will be crucial in providing timely and equitable care.

REVIEW OF LITERATURE

CHEST PAIN

Chest pain is a frequent reason for visits to the emergency department (ED), accounting for approximately 5–10% of adult ED cases. However, fewer than 1% of these instances require urgent medical intervention.[1] Patients may exhibit a wide range of symptoms due to the various potential causes of chest pain. Conditions affecting the heart, aorta, lungs, esophagus, stomach, mediastinum, pleura, and abdominal organs can all contribute to discomfort in the chest. Emergency clinicians prioritize identifying and ruling out life-threatening conditions as quickly as possible. Some patients with critical chest pain causes may initially appear stable, without noticeable abnormalities in vital signs or physical examination findings.

DIFFERENTIAL DIAGNOSIS[22]

Life-Threatening Conditions Chest pain may be caused by conditions that require immediate medical intervention, including:

- Acute coronary syndrome
- Acute aortic dissection
- Pulmonary embolism
- Tension pneumothorax
- Pericardial tamponade
- Mediastinitis (e.g., esophageal rupture)

Non-Life-Threatening Conditions Several other conditions can cause chest pain but do not pose an immediate danger:

- **Cardiac Causes** – Acute heart failure can lead to chest discomfort. Patients with stable angina typically recognize their symptoms and associate them with exertion. Valvular heart diseases like mitral valve prolapse and aortic stenosis can also present with chest pain, signaling potential worsening of valvular function. Additionally, pericarditis, myocarditis, and endocarditis may cause chest discomfort. Pericardial fluid accumulation and cardiac arrhythmias, particularly those affecting coronary blood flow, can also contribute to symptoms.

- **Pulmonary/Pleural Causes** – Respiratory infections, including pneumonia, tracheitis, and bronchitis, often accompany chest discomfort and cough. Asthma exacerbations commonly cause chest tightness. Increased pulmonary arterial pressure can lead to right-sided heart dysfunction (cor pulmonale). Pulmonary malignancies, especially those affecting the pleura, may also cause chest pain. Pleural effusions can result in a sensation of chest heaviness or discomfort.
- **Gastrointestinal Causes** – Conditions such as gastroesophageal reflux disease (GERD), esophageal spasm, rupture (Boerhaave's syndrome), or inflammation can present as chest discomfort. A sliding hiatal hernia may cause similar symptoms. Pain from pancreatitis can also be referred to the chest. Many patients experiencing chest pain without acute coronary syndrome have underlying gastrointestinal issues.
- **Musculoskeletal Causes** – Rib contusions and fractures, intercostal muscle strains, and costochondritis are common musculoskeletal sources of chest pain.
- **Psychiatric Causes** – Panic attacks often present with chest tightness and a sensation of impending doom. However, this diagnosis is one of exclusion in the emergency department.
- **Other Conditions** – Less frequent causes of chest pain include herpes zoster, referred pain, and inflammatory conditions associated with autoimmune diseases such as lupus, sarcoidosis, scleroderma, Kawasaki disease, polyarteritis nodosa, and Takayasu's arteritis.

ACUTE CORONARY SYNDROME

Acute coronary syndrome (ACS) encompasses a spectrum of conditions, including unstable angina, non-ST elevation myocardial infarction (NSTEMI), and ST elevation myocardial infarction (STEMI). The primary underlying mechanism is the rupture of an atherosclerotic plaque, leading to varying degrees of coronary artery obstruction and myocardial ischemia (fig1)[22].

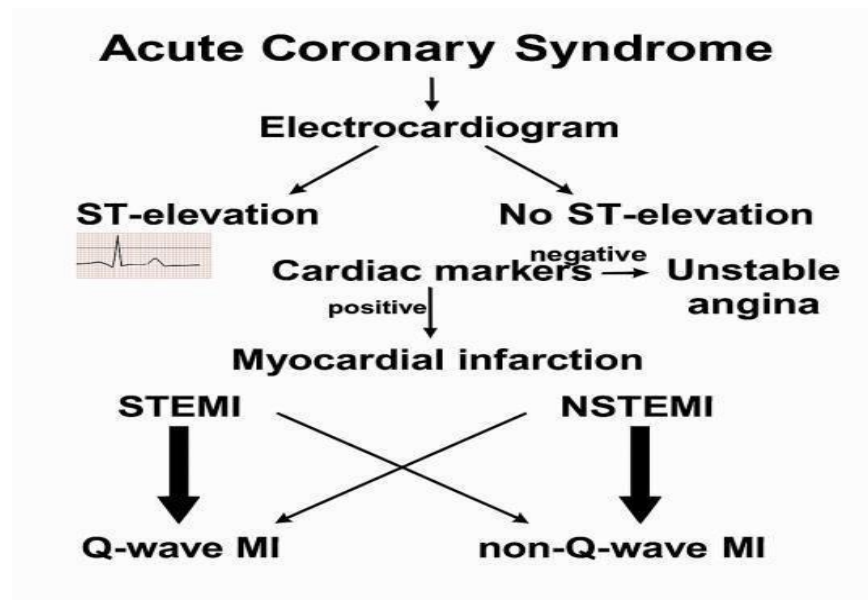


Fig.1 Acute Coronary Syndrome

Burden of Acute Coronary Syndrome

Ischemic heart disease (IHD) remains the leading cause of mortality and disability-adjusted life years (DALYs) lost globally, contributing to approximately seven million deaths and 129 million DALYs annually. While high-income countries (HICs) continue to struggle with significant IHD mortality, low- and middle-income countries (LMICs) bear an even greater burden, accounting for nearly two-thirds of all IHD DALYs and over half of global deaths. Rapid economic growth and lifestyle changes in many LMICs have led to an increase in IHD risk factors and associated mortality rates.[23]

According to the Global Burden of Disease study (2010), cardiovascular disease (CVD) is responsible for approximately 24.8% of all deaths in India. The age-standardized CVD mortality rate in India stands at 272 per 100,000 population, surpassing the global average of 235 per 100,000.[24]. Despite the widespread prevalence of CVD risk factors in India, there are notable regional variations. For example, diabetes mellitus is more common in southern states, whereas hypertension is more prevalent in the northeastern regions.

These disparities may be attributed to cultural differences, dietary habits, tobacco use, physical activity levels, and varying levels of economic development. Studies on acute coronary syndrome (ACS) in northeastern India indicate a higher proportion of STEMI cases, delayed critical care access, greater 30-day mortality, and lower rates of reperfusion therapy compared to national averages.[25]

Pathophysiology of Acute Coronary Syndrome

The role of thrombosis in triggering acute myocardial ischemia is well established. Understanding plaque structures before thrombotic events occur helps explain the transition from a stable state (no thrombus) to an unstable state (with thrombus).[26]

Atherothrombosis

Atherothrombosis refers to the disruption of atherosclerotic plaques with the formation of a superimposed thrombus, making it a major cause of mortality worldwide. This condition manifests clinically as coronary artery disease (CAD), cerebrovascular accidents, transient ischemic attacks (TIAs), and peripheral artery disease.[26]

Atherosclerosis (ATH)

Atherosclerosis, also known as atherosclerotic vascular disease (ASVD), is a type of arteriosclerosis where the arterial walls thicken due to the accumulation of white blood cells (WBCs). Early-stage accumulations, called "fatty streaks," contain both active WBCs causing inflammation and remnants of dead cells, including cholesterol and triglycerides. Over time, these streaks may include calcium deposits, leading to arterial stiffening and increased pulse pressure.

Atherosclerosis is a chronic inflammatory condition in which WBCs interact with low-density lipoproteins (LDL) while insufficient functional high-density lipoproteins (HDL) fail to remove fats and cholesterol from macrophages. This process results in the formation of multiple atheromatous plaques, commonly referred to as the "hardening" of arteries. [27].The plaque consists of three primary components:

1. Atheroma: A nodular accumulation of soft, flaky, yellowish material composed of macrophages located near the arterial lumen.
2. Cholesterol Crystals: Underlying deposits within the plaques.
3. Calcification: Present at the outer base of older or advanced lesions.

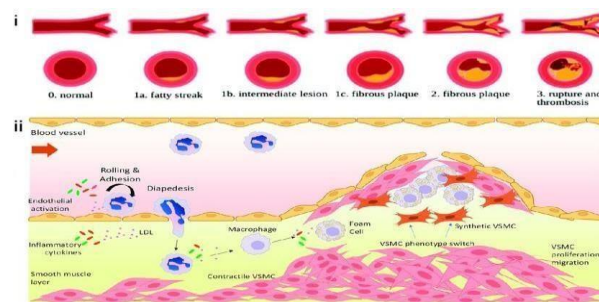


Fig2.Pathogenesis of Atherosclerosis

Atherosclerotic lesions, or atherosclerotic plaques, are separated into two broad categories: Stable and unstable (also called vulnerable) [28].

The pathobiology of atherosclerotic lesions is very complicated, but generally, stable atherosclerotic plaques, which tend to be asymptomatic, are rich in extracellular matrix and smooth muscle cells, while, unstable plaques are rich in macrophages and foam cells and the extracellular matrix separating the lesion from the arterial lumen (also known as the fibrous cap) is usually weak and prone to rupture [29]. Ruptures of the fibrous cap expose thrombogenic material, such as collagen [30], to the circulation and eventually induce thrombus formation in the lumen. Upon formation, intraluminal thrombi can occlude arteries outright (e.g. coronary occlusion), but more often they detach, move into the circulation and eventually occluding smaller downstream branches causing thromboembolism.

Apart from thromboembolism, chronically expanding atherosclerotic lesions can cause complete closure of the lumen. Chronically expanding lesions are often asymptomatic until lumen stenosis is so severe (usually over 80%) that blood supply to downstream tissue(s) is insufficient, resulting in ischemia. These complications of advanced atherosclerosis are chronic, slowly progressive and cumulative. Most commonly, soft plaque suddenly ruptures (see vulnerable plaque), causing the formation of a thrombus that will rapidly slow or stop blood flow, leading to death of the tissues fed by the artery in approximately five minutes. This catastrophic event is called an infarction. One of the most commonly recognized scenarios is called coronary thrombosis of a coronary artery, causing myocardial infarction [31]. The same process in an artery to the brain is commonly called a stroke. Another common scenario in very advanced disease is claudication from insufficient blood supply to the legs.

While atherosclerosis affects the entire arterial system, high- pressure vessels such as coronary, renal, femoral, cerebral, and carotid arteries are particularly vulnerable. Many cases remain "clinically silent" until a major event like a heart attack or stroke occurs.

Although symptomatic atherosclerosis is typically associated with men in their 40s and women in their 50s and 60s, subclinical signs can emerge as early as childhood. Though symptoms are rarely exhibited in children, early screening of children for cardiovascular diseases could be beneficial to both the child and his/her relatives [32,33]. While coronary artery disease is more prevalent in men than women, atherosclerosis of the cerebral arteries and strokes equally affect both sexes [34].

Narrowing of Arteries and Associated Symptoms

When the coronary arteries, responsible for delivering oxygen- rich blood to the heart, experience significant narrowing, it can lead to symptoms such as chest pain (angina), shortness of breath, dizziness, nausea, sweating, palpitations, or light-headedness. Another possible effect is arrhythmia, a condition where the heart beats either too fast or too slow due to inadequate blood supply.[35].

Similarly, the carotid arteries, which supply blood to the brain and neck, can also become narrowed, leading to symptoms like weakness, confusion, difficulty speaking, dizziness, trouble walking, blurred vision, numbness in the face, arms, or legs, severe headaches, and even loss of consciousness. These symptoms are closely linked to strokes, which occur when arteries supplying the brain become severely narrowed or blocked, resulting in the death of brain cells due to insufficient oxygen.[36] The peripheral arteries, which provide blood to the arms, legs, and pelvis, can also suffer from narrowing caused by plaque buildup or clot formation. This can cause pain and numbness in the affected limbs.

Another critical site where plaque can accumulate is the renal arteries, which supply blood to the kidneys. Reduced blood flow to the kidneys due to plaque buildup can contribute to chronic kidney disease. This condition typically remains symptomless until it progresses to an advanced stage.[32].

Risk Factors for Atherosclerosis

Atherosclerosis develops due to various anatomical and physiological factors, [37].which can be categorized into congenital vs. acquired, modifiable vs. non- modifiable, and classical vs. non-classical risks. The factors marked with ‘+’ contribute to metabolic syndrome. When two risk factors are combined, the likelihood of atherosclerosis increases fourfold, while a combination of hyperlipidemia, high blood pressure, and smoking elevates the risk by seven times.[38]

Modifiable Risk Factors[38]

- Diabetes or impaired glucose tolerance (IGT)
- Abnormal lipid levels (Dyslipoproteinemia)
- Smoking (increases risk by 200% after prolonged exposure)
- High levels of C-reactive protein in the blood
- Vitamin B6 deficiency

- Iodine deficiency and hypothyroidism, which contribute to high cholesterol and lipid oxidation

Non-Modifiable Risk Factors[38]

- Advanced age
- Male gender
- Family history of atherosclerosis-related complications (such as coronary artery disease or stroke)
- Genetic conditions, such as familial hypercholesterolemia

Lesser or Uncertain Risk Factors[38]

Some factors have uncertain or lesser impact but may still contribute to atherosclerosis:

- Obesity, particularly abdominal obesity (also known as male-type obesity)
- Increased blood clotting tendency (hypercoagulability)
- Estrogen deficiency after menopause
- High intake of saturated fats (may elevate LDL and total cholesterol)
- Consumption of trans fats (raises LDL while lowering HDL cholesterol)
- High carbohydrate consumption
- Elevated triglyceride levels +
- Increased homocysteine levels
- High uric acid levels (also associated with gout)
- Increased fibrinogen concentrations in the blood
- Elevated lipoprotein (a) levels
- Chronic inflammation, as indicated by high WBC counts, increased hs- CRP, and other blood markers (primarily observed in research settings)
- High insulin levels +
- Lack of sufficient sleep
- Infection by *Chlamydia pneumoniae*

Understanding STEMI, NSTEMI, and Unstable Angina[39]

The terms STEMI (ST-Elevation Myocardial Infarction), NSTEMI (Non-ST Elevation Myocardial Infarction), and Unstable Angina (UA) are primarily used to classify patients showing symptoms of myocardial ischemia.

These terms are less frequently applied in cases where troponin levels are found to be elevated following a coronary intervention procedure. Similarly, in cases of myocardial infarction (MI) following non-cardiac surgery, these classifications are used only when clear signs of ischemia are present. However, treatment strategies can vary based on ECG changes and clinical context.

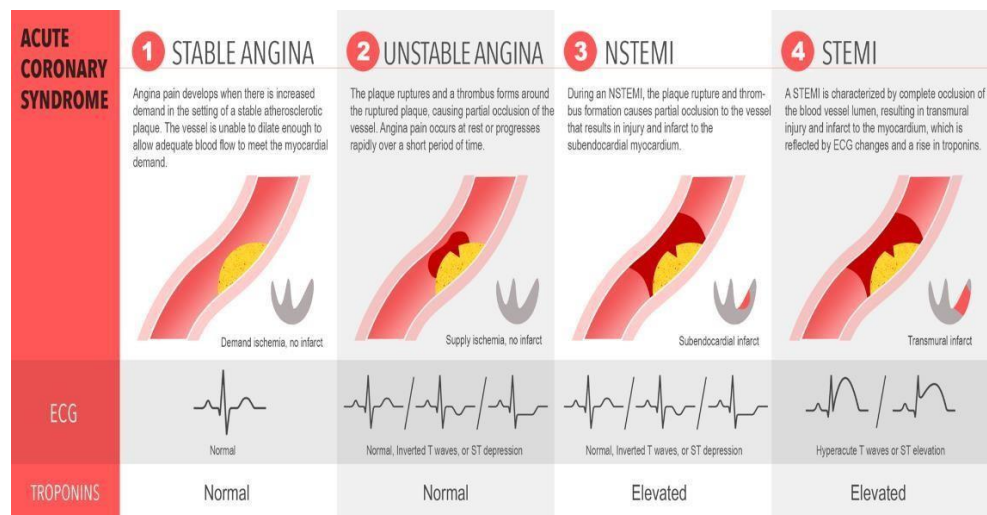


Fig. 3 Understanding Unstable Angina ,NSTEMI and STEMI

The primary distinction between STEMI and NSTEMI lies in ECG findings, as both conditions involve elevated troponin levels. This distinction is essential because early management approaches differ.

The difference between UA and NSTEMI depends on whether the ischemia is severe enough to cause detectable myocardial damage. Unstable angina (UA) refers to cases where patients exhibit ischemic symptoms, but no troponin elevation is observed, regardless of ECG changes (e.g., ST-segment depression, transient elevation, or T wave inversion). Since highly sensitive troponin tests may take up to three hours to detect an increase, UA and NSTEMI can be indistinguishable during initial assessments of patients experiencing chest pain. However, their early management strategies are generally the same. The sensitivity of troponin tests determines whether a case is classified as UA or NSTEMI. Patients diagnosed with UA or NSTEMI may present with an ECG that shows no significant abnormalities, transient ST-segment changes, or T wave inversions. In NSTEMI, ST-segment and T wave changes are often dynamic, whereas in UA, these changes tend to be transient.

RISK STRATIFICATION

Distinguishing patients with acute coronary syndrome (ACS) who require urgent intervention from those who do not is a significant clinical challenge.

The introduction of troponin assays in emergency departments has greatly enhanced the diagnosis and management of non-ST-segment elevation ACS. Troponin is a highly sensitive and specific biomarker for myocardial damage,[41]. aiding in determining the appropriate treatment, including early revascularization.[42,43]. Additionally, ST- segment depression observed in an initial electrocardiogram (ECG) can help identify patients at moderate to high risk and guide their clinical management.[44,45].

However, for individuals experiencing chest pain without ST- segment changes or elevated troponin levels, diagnosis and risk assessment remain difficult. While some studies suggest that a normal troponin level indicates a favorable prognosis,[46] research has shown a 3% occurrence of myocardial infarction or death within 30 days, [47],which rises to 4.8% after six months.[48] This underscores the importance of careful risk stratification.

To address this, multiple risk scoring systems have been developed to assess non-ST-segment elevation ACS. International cardiac guidelines recommend using a risk assessment tool when evaluating chest pain patients in the emergency department. These tools integrate various clinical indicators to estimate an individual's risk based on the initial presentation.[49-57].Commonly used risk scores include:

- HEART Score (History, ECG, Age, Risk Factors, and Troponin)[49].
- ADAPT (Accelerated Diagnostic Protocol for Chest Pain Assessment Using Contemporary Troponins)[50].
- EDACS (Emergency Department Assessment of Chest Pain Score)[51].
- NACPR (North American Chest Pain Rule)[52].
- TIMI (Thrombolysis in Myocardial Infarction)[53]
- Modified TIMI (m.TIMI)[54].
- GRACE (Global Registry of Acute Coronary Events)[55].
- PURSUIT (Platelet Glycoprotein IIb/IIIa Inhibitor in Unstable Angina: Receptor Suppression Using Integrilin Therapy)[56].

- Florence Prediction Rule (FPR)[57].
- National Heart Foundation of Australia (HFA) Risk Stratification Tool
- Goldman Risk Score

HEART SCORE

Unlike other scoring systems such as TIMI, the HEART score was specifically designed for evaluating chest pain in emergency settings.[58]. Research suggests that the HEART score performs better than TIMI and other scoring methods.[59,60].

The HEART score considers five key factors: history, ECG findings, age, risk factors, and troponin levels. Each of these factors is assigned a score of 0, 1, or 2, leading to a total score ranging from 0 to 10. Based on the final score, patients are categorized into three risk levels:

- Low risk (0–3)
- Intermediate risk (4–6)
- High risk (7–10)

Unlike traditional clinical scoring systems derived from multivariable regression analysis,[61,62]. the HEART score was developed based on expert consensus.[3] Despite this, it has consistently demonstrated strong predictive accuracy in identifying both high-risk and low-risk patients in terms of major adverse cardiac events (MACE). Due to its reliability and widespread validation, the HEART score is one of the most extensively studied and trusted tools for risk assessment in emergency cardiac care.

Table 1. HEART score for stratification of Chest pain patients

Category	Criteria	Score
History (Anamnesis)	Highly suspicious	2
	Moderately suspicious	1
	Slightly suspicious	0
ECG	Significant ST-deviation	2
	Non-specific repolarisation disturbance / LBBB / PM	1
	Normal	0
Age	≥ 65 years	2
	45 – 65 years	1
	≤ 45 years	0
Risk factors	≥ 3 risk factors or history of atherosclerotic disease	2
	1 or 2 risk factors	1
	No risk factors known	0
Troponin	$\geq 3x$ normal limit	2
	1-3x normal limit	1
	$\leq$ normal limit	0

Risk factors for atherosclerotic disease: Hypercholesterolemia, Hypertension, Diabetes Mellitus Cigarette smoking, Positive family history, Obesity (BMI > 30)

Scoring

- **0–3 points:** Low risk, with a 0.6–1.7% risk of MACE
- **4–6 points:** Intermediate risk, with a 16.6% risk of MAC
- **7–10 points:** High risk, with a 50.1% risk of MACE

Clinical implications

- A low score (0–3) suggests a low risk of MACE, and patients may be able to be discharged immediately
- A score of 4–6 suggests an intermediate risk of MACE, and patients may need to be admitted for clinical observation
- A score of 7 or higher suggests a high risk of MACE, and patients may need early invasive strategies

LITERATURE REVIEW

1. **Backus et al-2010.** [63] carried out a retrospective multicenter study to evaluate the HEART score's predictive ability for MACE in a cohort of 2161 patients, of whom 910 (42%) presented with chest pain. The analysis included 880 cases (96.7%), and MACE was reported in 158 patients (17.95%). The breakdown included 92 patients (10.45%) with acute myocardial infarction (AMI), 82 (9.32%) undergoing percutaneous coronary intervention (PCI), 36 (4.09%) requiring coronary artery bypass graft (CABG), and 13 (1.48%) fatalities. Among patients with a HEART score between 0 and 3, only 0.99% experienced MACE, while 11.6% of those with a score of 4 to 6 reached the endpoint. In contrast, 65.2% of patients scoring 7 to 10 experienced MACE. The study concluded that the HEART score is a simple yet effective tool for decision- making in chest pain management, eliminating the need for invasive tests or radiation- based procedures.
2. **Mahler SA et al. (2011)(64)** conducted a study to assess whether the HEART score could effectively minimize the need for objective cardiac testing in patients presenting with low-risk chest pain. The research utilized data from an emergency department-based observation unit registry, identifying a cohort of chest pain patients. HEART scores were assigned based on registry data elements and a blinded review of medical charts, categorizing patients into low risk (0-3) and high risk (>3). The primary outcome was major adverse cardiac events (MACE), defined as a composite measure including all-cause mortality, myocardial infarction, or coronary revascularization, either during the initial visit or within 30 days. Sensitivity, specificity, and potential reduction in cardiac testing were evaluated. Over a 28-month period, the registry recorded 1,070 patients with low-risk chest pain. Among those with a low HEART score, 0.6% (5 out of 904) experienced MACE, compared to 4.2% (7 out of 166) in the high-risk group, yielding an odds ratio of 7.92 (95% CI: 2.48-25.25). A HEART score greater than 3 demonstrated 58% sensitivity (95% CI: 32-81%) and 85% specificity (95% CI: 83-87%) for

predicting MACE. The score failed to detect 5 cases of acute coronary syndrome (ACS) out of 1,070 patients (0.5%) but had the potential to reduce cardiac testing by 84.5% (904 out of 1,070 cases). When combining serial troponin levels exceeding 0.065 ng/mL with a HEART score above 3, sensitivity increased to 100% (95% CI: 72-100%), while specificity remained at 83% (95% CI: 81-85%), with a possible reduction in cardiac testing of 82% (879 out of 1,070 cases). The study concluded that utilizing the HEART score to guide stress testing and cardiac imaging could significantly decrease unnecessary cardiac evaluations in patients with a low pretest probability of ACS.

3. **Backus et al-2013. [65]** conducted a prospective study to the emphasis of the diagnostic process using HEART score in chest pain patients at the emergency department is to identify both low and high risk patients for an acute coronary syndrome (ACS) on 2440 unselected patients presented with chest pain at the cardiac emergency department of ten participating hospitals in The Netherlands. Primary endpoint was the occurrence of major adverse cardiac events (MACE) within 6 weeks. Secondary endpoints were the occurrence of AMI and death, ACS and the performance of a coronary angiogram. The performance of the HEART score was compared with the TIMI and GRACE scores. Their results showed that the low HEART scores (values 0-3) were calculated in 36.4% of the patients. MACE occurred in 1.7%. In patients with HEART scores 4-6, MACE was diagnosed in 16.6%. In patients with high HEART scores (values 7-10) MACE occurred in 50.1%. The c-statistic of the HEART score (0.83) is significantly higher than the c- statistic of TIMI (0.75) and GRACE (0.70) respectively (P98% certainty, and the high risk of MACE may indicate more aggressive policies with the high HEART scores (7-10).
4. **Melki D et al-2013. [66]** in their study aimed to validate HEART score and estimate to what extent it can reduce the admission rate. A total of 410 consecutive patients with chest pain, who presented to the emergency department with no ST-segment elevations, were included and followed for 3 months regarding the combined endpoint of cardiovascular death, myocardial infarction, or unplanned revascularization. Thirty (7.3%) patients attained one or more combined endpoint. Of 247 (60.2%) patients with HEART score 0-3, 1 patient (0.4%) had a combined endpoint (unplanned revascularization). Of 144 (35.1%) patients with HEART score 4-6, 19 (13.2%) patients had a combined endpoint. Of 19 (4.6%) patients with HEART score 7-10, 10 (52.6%) patients had a combined endpoint. A total of 181 (44.1%) patients were admitted to the hospital for further evaluation. Of these patients, 62 (34.3%)

had a HEART score of 0-3 and of whom 45 (72.5%) had a final diagnosis indicating that hospital admission may have been avoided. Among the 5 HEART score parameters, patient history, ECG abnormalities, and elevated troponin values were independent predictors of the combined endpoint. In conclusion, HEART score may be a useful tool for evaluation of patients with chest pain and identify a low-risk group in which admission and further investigations may not be necessary. However, an even simpler score, including only history, troponin level, and ECG findings, may be sufficient.

5. **Six AJ et al-2013.** [67] aimed to performed an external validation of the HEART score in an Asia-Pacific population. Data were used from 2906 patients presenting with chest pain to the emergency departments of 14 hospitals. HEART scores were determined retrospectively. Three risk groups were composed based on previous research. The predictive values for the occurrence of 30-day major adverse coronary events (MACE) were assessed. A comparison was made with the Thrombolysis in Myocardial Infarction (TIMI) score in terms of the value of C-statistics. The low-risk group, HEART score ≤ 3 , consisted of 820/2906 patients (28.2%). Fourteen (1.7%) patients were incorrectly defined as low risk (false negatives). The high-risk population, HEART score 7-10, consisted of 464 patients (16%) with a risk of MACE of 43.1%. The C-statistics were 0.83 (0.81-0.85) for HEART and 0.75 (0.72-0.77) for TIMI ($P < 0.01$). Study concluded that utilization of the HEART score provided excellent determination of risk for 30-day MACE, comparing well with the Thrombolysis in Myocardial Infarction score. This study externally validates previous findings that HEART is a powerful clinical tool in this setting. It quickly identifies both a large proportion of low-risk patients, in whom early discharge without additional testing goes with a risk of MACE of only 1.7%, and high-risk patients who are potential candidates for early invasive strategies.

Other than the validations on general chest pain cohorts in the ED, the HEART score has been evaluated when several factors such as ethnicity and sex are taken into considerations. de Hoog et al. [68] conducted a study on a mixture of ethnic groups consisting of Caucasian, Chinese, Indian and Malay, and concluded that the overall performance of the HEART score was similar among Caucasians and Asians. Furthermore, Bank et al. [69] reported that male sex remained a significant factor for 6-week MACE when the HEART score was implemented for the prediction.

Based on the standard HEART score, many of its variants have been proposed and

validated. Fesmire et al. proposed the HEARTS3 score by adding three additional variables, sex, serial 2-hour ECG, and serial 2-hour delta troponin [70]. The HEARTS3 score reliably risk stratified patients with chest pain for MACE.

Modified versions of the HEART score were proposed where high sensitivity troponin T was adopted to replace the traditional troponin element in the standard HEART score. These modifications suggested that the identified low-risk population might be directly discharged from the ED [71] .

6. **Visser A et al-2015.** [72] compared the diagnostic accuracy of HEART score and clinical gestalt (clinical judgement) for diagnosing ACS in an unselected population of patients with chest pain presenting to the ED. HEART score (0-10) and clinical gestalt (low risk, intermediate risk or high risk of ACS) were prospectively determined in the ED in 255 patients presenting with chest pain by the treating physician. The reference standard was the presence of ACS, which was defined as either acute myocardial infarction (AMI) or the occurrence of a major adverse cardiac event within 6 weeks after presentation in the ED. 75 out of 255 patients (29%) had an ACS. A HEART score ≤ 3 had a lower negative likelihood ratio (0.15 (0.06-0.36)) for ACS than a low risk based on clinical gestalt (0.35 (0.19-0.64)), whereas a high HEART score ≥ 7 had a higher positive likelihood ratio (5.2 (3.2-8.5) vs 3.1 (2.2-4.4)). However, c-statistic of HEART score was not significantly different from clinical gestalt (0.81 (0.76-0.86) vs 0.79 (0.73-0.84), $p=0.13$). The study demonstrates that HEART score and clinical gestalt have similar diagnostic accuracy for diagnosing ACS in an unselected population of patients with chest pain presenting in the ED.
7. **Leite et al-2015.** [73] conducted a retrospective observational study to characterize the subgroup of patients with acute coronary syndrome (ACS) and to assess the prognostic value of Manchester triage system and of HEART score on 233 patients (age 58 ± 19 ; 55.4 % males). The most common final diagnosis was nonspecific chest pain ($n = 86$, 36.9%), followed by ACS ($n = 22$, 9.4 %). Male gender, smoking and chronic kidney disease were associated with higher risk of ACS. According to Manchester triage system, chest pain patients stratified with red or orange priority had a higher incidence of ACS (16.5 % vs. 3.8 %, $p = 0.006$). The application of HEART score showed that most patients were in low risk category (56.3 %). The six-week incidence of MACE in each category was 2 %, 15.6% and 76.9 % ($p < 0.001$). HEART score accurately predicted the shortterm incidence of MACE in chest pain patients (cstatistic 0.880; 95 % CI, 0.807–0.950, $p < 0.001$). They concluded that the chest pain patients have very different levels of severity and the discriminatory power of Manchester

triage system should be used in the assessment of this population. The HEART score seems to be an effective tool for risk stratification in the ED.

8. **Sakamoto JT et al-2016.** [74] aimed to compare the accuracy of HEART, TIMI, and GRACE for the prediction of major adverse cardiac events (MACE) in high acuity chest pain patients. Adult patients who presented with chest pain suggestive of cardiac origin in the most acute triage category at an academic ED from September 2010 to October 2015 were included. The HEART, TIMI, and GRACE scores were calculated retrospectively from prospectively collected data. The primary outcome was occurrence of MACE (mortality, AMI, PCI, CABG) within 30-days of initial presentation. 604 patients were included in the study. Patient demographics include an average age of 61years, 69% male, and 48% with history of ischemic heart disease. 36% of patients met the primary outcome. The c-statistics of HEART, TIMI, and GRACE were 0.78 (95% CI: 0.74-0.81), 0.65 (95%CI: 0.60-0.69), and 0.62 (95% CI: 0.58-0.67), respectively. For the purpose of accurately ruling out patients for 30-day MACE, a HEART score of ≤ 3 had a sensitivity and NPV of 99% and 98%, respectively, compared to 97% and 91%, respectively, for TIMI=0, and 94% and 85%, respectively, for GRACE ≤ 75 . The percent of patients with 30-day MACE with HEART scores between 0 and 3, 4-6, and 7- 10 was 2%, 28%, and 63%, respectively. Study concluded that in high acuity chest pain patients, the HEART score is superior to the TIMI and GRACE scores in predicting 30-day MACE.
9. **Chen XH et al-2016.** [75] aimed to find out to the number of MACE within 7 days, 30 days, and 6 months after initial ED presentation, and also to compare the prognostic performance of these scores in Chinese patients with suspected cardiac chest pain (CCP) to predict 7-day, 30-day, and 6-month major adverse cardiac events (MACE). A prospective 2-center observational cohort study of consecutive patients presenting with chest pain to the EDs of 2 university hospitals in Guangdong and Hong Kong was conducted. Patients aged ≥ 18 years with suspected CCP but without ST-segment elevation myocardial infarction (STEMI) were recruited. Of 833 enrolled patients (mean age 65.1 years, SD14.5; 55.6% males), 121 (14.5%) experienced MACE within 6 months (4.8% with safety outcomes and 10.3% with effectiveness outcomes). The HEART score had the largest area under the receiver operating characteristic (ROC) curve for predicting MACE at 7-day, 30- day, and 6- month follow-up [area under curve (AUC) = 0.731, 0.726, and 0.747, respectively. The HEART score also had the largest AUC for predicting effectiveness outcome (AUC = 0.715, 0.704, and 0.721,

respectively). However, there was no significant difference in AUC between HEART and TIMI scores. Banach had the largest AUC for predicting safety outcome (AUC = 0.856, 0.837, and 0.850, respectively). The HEART score performed better than the GRACE and Banach scores to predict total MACE and effectiveness outcome in Chinese patients with suspected CCP, whereas the Banach score best predicted safety outcomes.

10. **Streitz MJ et al-2017.** [76] utilized chart review methodology to abstract data from each patient's electronic medical record. Authors collected data relevant to each of the five elements of the HEART score: history, electrocardiogram (ECG) interpretation, patient age, patient risk factors, and troponin levels. We calculated the diagnostic accuracy of the HEART score (0-10) for predicting the primary outcome of major adverse cardiac events (MACE) over 6 weeks following the ED visit (coronary revascularization, myocardial infarction, or mortality). They randomly selected 10% of patient charts from which a second investigator abstracted all data to assess inter-rater reliability for all study variables. Of 625 charts reviewed, we abstracted data on 417 (66.7%) consecutive patients meeting study inclusion criteria. Thirty-one (7.4%) of these patients experienced 6- week MACE. We observed no instances of MACE within 6 weeks among subjects with a HEART score of 3 or less. The area under the receiver operator curve (AUROC) is 0.885 (95% confidence interval 0.838-0.931). Patients with a HEART score ≤ 3 are at low risk for 6-week MACE. Hence, these patients may be candidates for outpatient follow-up instead of inpatient admission for cardiac risk stratification.
11. **Poldervaart JM et al-2017.** [77] in a review evaluated the performance of the GRACE, HEART and TIMI scores in predicting the probability of major adverse cardiac events (MACE) in chest pain patients presenting at the emergency department (ED), in particular their ability to identify patients at low risk. Chest pain patients presenting at the ED in nine Dutch hospitals were included. The primary outcome was MACE within 6 weeks. The HEART score was determined by the treating physician at the ED. The GRACE and TIMI score were calculated based on prospectively collected data. Performance of the scores was compared by calculating AUC curves. Additionally, the number of low-risk patients identified by each score were compared at a fixed level of safety of at least 95% or 98% sensitivity. In total, 1748 patients were included. The AUC of GRACE, HEART, and TIMI were 0.73 (95% CI: 0.70– 0.76%), 0.86 (95% CI: 0.84– 0.88%) and 0.80 (95% CI:0.78–0.83%), respectively (all differences in AUC

highly statistically significant). At an absolute level of safety of at least 98% sensitivity, the GRACE score identified 231 patients as “low risk” in which 2.2% a MACE was missed; the HEART score identified 381 patients as “low risk” with 0.8% missed MACE. The TIMI score identified no “low risk” patients at this safety level. Study concluded that HEART score outperformed the GRACE and TIMI scores in discriminating between those with and without MACE in chest pain patients, and identified the largest group of low-risk patients at the same level of safety

12. **Van Den Berg P et al-2018.** [78] conducted a systematic review to summarise the current evidence on the diagnostic accuracy of the HEART score for predicting major adverse cardiac events in patients presenting with undifferentiated chest pain to the emergency department. For the 12 studies meeting inclusion criteria, study characteristics and diagnostic accuracy measures were systematically extracted and study quality assessed using the QUADAS-2 tool. After quality assessment, nine studies including data from 11,217 patients were combined in the meta-analysis applying a generalised linear mixed model approach with random effects assumption (Stata 13.1). In total, 15.4% of patients (range 7.3-29.1%) developed major adverse cardiac events after a mean of 6 weeks' follow-up. Among patients categorised as 'low risk' and suitable for early discharge (HEART score 0-3), the pooled incidence of 'missed' major adverse cardiac events was 1.6%. The pooled sensitivity and specificity of the HEART score for predicting major adverse cardiac events were 96.7% (95% confidence interval (CI) 94.0-98.2%) and 47.0% (95% CI 41.053.5%), respectively. Study concluded that patients with a HEART score of 0-3 are at low risk of incident major adverse cardiac events. As 3.3% of patients with major adverse cardiac events are 'missed' by the HEART score, clinicians must ask whether this risk is acceptably low for clinical implementation
13. **Liu N et al-2018.** [79] in their meta-analysis retrieved a total of 514 articles from both databases and included 29 studies in this systematic review. The articles covered studies from over 20 countries, where more than 20 different risk scores and scoring methods were investigated. Among the various risk scores, the HEART, TIMI, GRACE, and heart rate variability (HRV)-based scores were the most widely implemented and discussed. They also concluded that HEART score was the top performer in identifying chest pain patients at high or low risk of developing MACE.

14. **Byrne et al-2018.** [80] conducted a prognostic systematic review for evaluating the HEART score as a predictor of major adverse cardiac events (MACE) in patients presenting to the ED with possible ACS. The QUIPS tool were used to assess the risk of bias within individual studies. Outcome measures will include prevalence, risk ratio, and absolute risk reduction for MACE within 6 weeks of ED evaluation, comparing HEART scores 0-3 Vs 4- 10. HEART Score prognostic performance will be evaluated with the concordance statistic, observed to expected events ratio, and a decision curve analysis. Reporting biases and methodological, clinical, and statistical heterogeneity were analyzed. Their evaluated that the prognostic performance of the HEART score, producing results of interest to emergency physicians. These results may encourage shared clinical decision-making in the ED by facilitating risk communication with patients and health care providers.
15. **Laureano-Phillips et al-2019.** [81] conducted a systematic review and metaanalysis to appraise the evidence in regard to the diagnostic accuracy of a low-risk History, ECG, Age, Risk Factors, and Troponin (HEART) score for prediction of major adverse cardiac events in emergency department patients. These included four subgroup analyses: by geographic region, the use of a modified low-risk HEART score (traditional HEART score [0-3] in addition to negative troponin results), using conventional versus high-sensitivity troponin assays in the HEART score, and a comparison of different post-ED- discharge patient follow-up intervals. They used twenty five studies which were published from 2010 to 2017, with a total of 25,266 patients included in the final meta-analysis, of whom 9,919 (39.3%) were deemed to have low risk HEART scores (0-3). Among patients with low-risk HEART scores, short term major adverse cardiac events (30 days to 6 weeks) occurred in 2.1% of the population (182/8,832) compared with 21.9% of patients (3,290/15,038) with non-low-risk HEART scores (4-10). For patients with HEART scores of 0-3, the pooled sensitivity of short-term major adverse cardiac event predictions was 0.96, specificity was 0.42, positive predictive value was 0.19, negative predictive value was 0.99, positive likelihood ratio was 1.66, and negative likelihood ratio was 0.09. Subgroup analysis showed that lower short-term major adverse cardiac events occurred among North American patients (0.7%), occurred when modified lowrisk HEART score was used (0.8%), or occurred when high-sensitivity troponin was used for low-risk HEART score calculations (0.8%). They concluded that the meta-analysis showed that despite its use in different patient populations, the troponin type used, and timeline of follow-up, a low-risk HEART score had high sensitivity, negative predictive value, and negative likelihood ratio for predicting short-term major adverse cardiac events, although risk of bias and statistical heterogeneity were high.

16. **Fernando SM et al-2019.** [82] in a systemic review sought to summarize all studies assessing the prognostic accuracy of the HEART score for prediction of MACE in adult ED patients presenting with chest pain. Authors included 30 studies (n = 44,202) in analysis. A HEART score above the low-risk threshold (≥ 4) had a sensitivity of 95.9% (95% confidence interval [CI] = 93.3%-97.5%) and specificity of 44.6% (95% CI = 38.8%- 50.5%) for MACE. A high-risk HEART score (≥ 7) had a sensitivity of 39.5% (95% CI = 31.6%-48.1%) and specificity of 95.0% (95% CI = 92.6%-96.6%) for MACE, whereas a TIMI score above the low-risk threshold (≥ 2) had a sensitivity of 87.8% (95% CI = 80.2%-92.8%) and specificity of 48.1% (95% CI = 38.9%-57.5%) for MACE. A high-risk TIMI score (≥ 6) was 2.8% sensitive (95% CI = 0.8%-9.6%), but 99.6% (95% CI = 98.5% - 99.9%) specific for MACE. A HEART score ≥ 4 had a sensitivity of 95.0% (95% CI = 87.2%-98.2%) for prediction of mortality and 97.5% (95% CI = 93.7%- 99.0%) for prediction of MI.
17. **Arslan et al-2019.** [83] conducted a study to investigate the diagnostic value and efficiency of the HEART score before coronary computed tomography angiography 340 patients. Patients were categorized as low-risk (HEART ≤ 3), intermediate-risk (HEART 4– 6) and high-risk (HEART ≥ 7). We assessed coronary computed tomography angiography for the presence of significant coronary artery disease ($>50\%$ stenosis). The primary outcome, the level of major adverse cardiac events, was a composite endpoint of all-cause mortality, acute coronary syndrome or coronary revascularization within 30 days. In their study, the mean age was 55.6 ± 10.1 years and 44.7% of women, major adverse cardiac events occurred in 13.2% patients. The incidence of major adverse cardiac events in patients stratified as low-risk (35.0%), intermediate-risk (56.8%) and high-risk (8.2%) was 3.4%, 12.4% and 60.7%, respectively. All four low-risk patients with major adverse cardiac events had a HEART score of three. An algorithm where coronary computed tomography angiography is reserved for patients with HEART 3-6 resulted in a sensitivity of 97.8%, specificity of 84.1%, negative predictive value of 99.6% and positive predictive value of 48.4%, while reducing the need for coronary computed tomography angiography by 22% (n=75). They concluded that the predictive value of coronary computed tomography angiography for 30-day major adverse cardiac events in suspected acute coronary syndrome patients is good, and reserving coronary computed tomography angiography for HEART score 3-6 patients reduces the number of needed coronary computed tomography angiograms without affecting diagnostic accuracy.

18. **Torralba et al-2020.** [84] compared the performance of the HEART, TIMI, and GRACE scores for predicting major cardiovascular events at 30 days of followup, in patients who consult for chest pain in the emergency department on 519 patients. A total of 224 patients (43%) had MACE at 30 days. The C statistic for the HEART, TIMI, and GRACE score was 0.937, 0.844, and 0.797 respectively ($P < 0.0001$). A HEART score of 3 or less had a sensitivity of 99.5% and a negative predictive value of 99% to classify low risk patients correctly; both values were higher than those obtained by the other scores. They concluded that the HEART score more effectively predicts cardiovascular events at 30 days of follow-up compared to the other scores. Highsensitivity troponins maintain this Score's previously demonstrated superiority. This Score offers more precise identification of low-risk patients
19. **Huang et al-2021.** [85] investigated the HEART scores in predicting the short- term risk of a major adverse cardiac event (MACE) in patients with chest on a total of 509 patients. At 30-days post-admission, the primary outcome (MACE) was recorded in 92 patients (18.1%), 88 (95.6%) of whom had experienced an acute myocardial infarction. Thirty-seven (40.2%) of the patients with a MACE underwent percutaneous coronary intervention and six patients (6.5%) died. The HEART and GRACE scores were both significantly higher in patients who developed a MACE than in those without ($P < 0.05$). The HEART and GRACE scores had c-statistic values of 0.811 [0.774- 0.844] and 0.648 [0.603-0.688], respectively. The Hosmer-Leme show statistic revealed that the HEART and GRACE scores had values of 8.68 ($P = 0.39$) and 10.45 ($P = 0.11$), respectively. The percentages of patients with HEART scores of 0-3, 4-6, and 7-10 were 3.0%, 26.2%, and 46.3%, respectively, in those with a MACE within 30 days. They concluded that the both scoring systems are useful, the HEART score is superior to the GRACE score for predicting the occurrence of MACE within 30 days in patients with chest pain
20. **Khalil MH et al-2021.** [86] aimed to validate HEART score as a prognostication tool, among Tunisian ED patients with undifferentiated chest pain. Authors enrolled 3880 patients (age 56.3; 59.5% males). The application of HEART score showed that most patients were in intermediate risk category (55.3%). Within 30 days of ED visit, MACE were reported in 628 (16.2%) patients, with an incidence of 1.2% in the low risk group, 10.8% in the intermediate risk group and 62.4% in the high risk group. The area under receiver operating characteristic curve was 0.87 (95% CI 0.85-0.88). HEART score was

not well calibrated (χ^2 statistic = 12.34; $p = 0.03$). Study concluded that HEART score showed a good discrimination performance in predicting MACE occurrence at 30 days for Tunisian patients with undifferentiated acute chest pain. Heart score was not well calibrated in our population.

21. **Krishna PP et al-2023.** [87] aimed to determine the prognostic accuracy of the HEART score as a predictor for major adverse cardiac events in patients presenting with chest pain to the emergency medicine department (ED). According to Heart score, 26.51% of patients had low risk, 39.76% of patients had moderate risk, and 33.73% of patients had high risk. More percentage of male patient's (67.9%) were in the high risk group of heart score than females (32.1%). ST-Depression cases were more in the high risk group (85.7%), and statistical significant association seen between ECG and the heart score ($P < 0.0001$). Among risk factors, Hypertension and Diabetes mellitus patients was more in the high risk groups with 48.6%, and 53.8% ($P = 0.001$). 100% of high risk cases had ≥ 3 x normal limit of troponin, and there was a statistically association seen between troponin and heart score ($P < 0.0001$). Diagnosis of HEART score of the low risk group showed that the risk factor had significantly higher AUC value (AUC = 0.801) than the age group (AUC = 0.778), history (AUC = 0.747), Troponin (AUC = 0.738), and ECG (AUC = 0.722). Out of 22 cases of the low risk group, 6 of Unstable angina (UA), 16 of NSTEMI, 4 of Percutaneous coronary intervention (PCI), 2 CABG, and 1 cardiovascular (CV) death. For moderate risk group ($n=33$), 13 of UA, 17 of NSTEMI, 3 of STEMI, 20 of PCI, 14 of CABG, and 12 of CV deaths. For high risk group ($n=28$), 10 UA, 14 of NSTEMI, 3 of STEMI, 9 of PCI, 6 of CABG, and 4 number of CV death. It was concluded that the HEART score should be used as the primary clinical decision tool for the risk stratification and a good predictor of major adverse cardiac events in patients presenting with chest pain to the emergency department to promote their safe and efficient nature in a community hospital setting.
22. **Rageh et al-2024.** [88] aimed to perform an external validation of the history, ECG, age, risk factors, and troponin (HEART) score in the emergency department of Menoufia University Hospital. This study included 268 patients with ages ranged from 19 to 78 years with mean of 45.08 ± 14.65 years. There were 54.5 % males and 45.5 % females with the mean HEART score was 2.84 ± 2.49 and ranged from zero to 9. The results revealed that 45 (16.8 %) cases showed MACE. There was significant relation between HEART score and MACE ($P < 0.001$). By using receiver operating characteristic-curve

analysis, HEART Score can detect MACE with the sensitivity, specificity, positive predictive value and negative predictive value was 91.9, 89.3, 89.57, and 91.68 %, respectively ($P < 0.001$).

AIM AND OBJECTIVES

AIM

To determine the prognostic accuracy of the Heart Score as a predictor for Major Adverse Cardiac Events in patients presenting with Chest Pain to the Emergency Department in a tertiary care Hospital.

Objectives

1. To evaluate prognostic accuracy of the HEART score for prediction of MACE in adult emergency department patients presenting with chest pain.
2. To study the Cardiac interventions based on HEART scores.
3. To study the Mortality based on HEART scores.

MATERIAL AND METHODS

Study Area

Emergency Medicine Department, Krishna Institute of Medical Science (KIMS) Secundrabad.

Study Population

Patients presented with chest pain of acute onset at the Emergency Medicine Department of our hospital.

Study Design

A Prospective, Cohort study

Sample Size Calculation

The prevalence of Chest pain is 42%, by assuming the precision to be 5% and confidence interval as 95%. Sample size for estimating single proportion (absolute precision) is given by

$$n = [Z^2 \cdot 1 - \alpha/2 \cdot p \cdot (1-p)] / d^2 = [(1.96)^2 \cdot 0.42 \cdot (1-0.42)] / (0.05)^2 = 374$$

- The minimum total sample size was 374.

Study Duration

May 2023 to November 2024

Inclusion Criteria

1. Adult (> 18 years) patients presenting with acute chest pain to Emergency Department
2. Patients willing for study and giving consent for the study.

Exclusion Criteria:

1. Patient's having < 18 years of age
2. Respiratory diseases
3. Traumatic chest injury
4. Pregnant patients.
5. CKD patients
6. Previous history of cardiac disease

Methodology

Formal consent from the KIMS ethical committee was obtained prior to initiation of the study. A total of 374 patients who satisfied the inclusion and exclusion criteria were selected. Patients were evaluated as per protocol that included evaluation of the clinical history and ECG, and early exercise testing in patients without contraindication to exercise (physical incapacity or abnormalities in the baseline ECG). Data on age, gender, presenting history, physical examination, electrocardiography, Laboratory results and risk factors were collected.

All variables used in the risk score (HEART) was collected at time of presentation at the ED (Table below). One of the key roles of these risk scores is to identify patients at low risk for MACE.

Troponin I was determined in our institution on arrival and at 6 hours. Troponin I was determined immunologically using an Immulite assay (Diagnostic Products Corp., Los Angeles, California). Troponin I increase was defined as ≥ 1 ng/ml (upper limit of normality).

The main endpoint in our study was major adverse cardiac events (MACE) within 6 weeks after the initial ED presentation (including the index event). MACE consisted of unstable angina (UA), non-ST elevation myocardial infarction (NSTEMI), STEMI, percutaneous coronary intervention (PCI), coronary arterial bypass grafting (CABG), stenosis managed conservatively, cardiovascular death, non-cardiovascular death and death with unknown cause.

Statistical Analysis

The quantitative data was represented as their mean $\pm$ SD. Categorical and nominal data was expressed in percentage. The t-test was used for analysing quantitative data, or else non parametric data was analyzed by Mann Whitney test and categorical data was analyzed by using chi-square test. . We compared the discrimination of the three scores by examining their ROC curves and calculating the areas under the ROC curve (AUCs). The significance threshold of p-value was set at <0.05 . All analysis was carried out by using SPSS software version 21.

Table 2. Heart Score

History	Highly Suspicious Moderately	2 Points
	Suspicious Slightly Or Non-	1 Point
	Suspicious	0 Point
ECG	Significant St-Depression	2 Points
	Non-Specific Repolarization	1 Point
	Normal	0 Point
Age	65 Years	2 Points
	>45-<65 Years	1 Point
	45 Year	0 Point
Risk Factor	3 Risk Factors Or History Of Cad 1	2 Points
	Or 2 Risk Factors	1 Point
	No Risk Factors	0 Point
Troponin	3* Normal Limit	2 Points
	>1- <3*Normal Limit	1 Point
	1*Normal Limit	0Point

*Risk Factors: DM, current or recent (<1 month), smoker, HTN, family history of CAD & obesity.

Interpretation of Score:

- **Score 0-3:** 2.5% MACE over next 6 weeks – Discharge home.
- **Score 4-6:** 20.3% MACE over next 6 weeks - Admit for clinical observation.
- **Score 7-10:** 72.7% MACE over next 6 weeks – Early invasive strategies.

RESULTS

TO DETERMINE THE PROGNOSTIC ACCURACY OF THE HEART SCORE AS A PREDICTOR FOR MAJOR ADVERSE CARDIAC EVENTS IN PATIENTS PRESENTING WITH CHEST PAIN TO THE EMERGENCY DEPARTMENT IN A TERTIARY CARE HOSPITAL

**Table 3: RESULTS : Frequency distribution of Characteristics
of Patients Presenting with Chest Pain**

Characteristics Variable	Frequency	Percentage
Age in years (Mean , SD)	56.17, 16.98	
Age Distribution		
<=45years	60	16.0
>45 to <65 years	168	44.9
>=65 years	146	39.0
Gender		
Male	225	60.2
Female	149	39.8
Giddiness		
Yes	214	57.2
No	160	42.8
Generalized Weakness		
Yes	219	58.6
No	155	41.4
Syncope		
Yes	17	04.5
No	357	95.5
Epigastric Discomfort		
Yes	180	48.1
No	194	51.9
Nausea		
Yes	194	51.9
No	180	48.1

Vomiting		
Yes	194	51.9
No	180	48.1
Altered Mental Status		
Yes	126	33.7
No	248	66.3
History		
Non – Suspicious	10	2.7
Moderately Suspicious	87	23.3
Highly Suspicious	277	74.1
ECG		
Normal	117	31.3
Non – Specific Repolarization	85	22.7
Significant ST –Depression	172	46.0
Hypertension		
Yes	174	46.5
No	200	53.5
Diabetes Mellitus		
Yes	196	52.4
No	178	47.6
Smoking		
Yes	81	21.7
No	293	78.3
Obesity		
Yes	66	17.6
No	308	82.4
Family history		
Yes	14	03.7
No	360	96.3

Alcohol		
Yes	63	16.8
No	311	83.2
Risk Factor		
No Risk Factor	79	21.1
1 or 2 Risk Factor	195	52.1
≥ 3 Risk Factor	100	26.7
Troponin		
Normal Limit	211	56.4
>1 to <3 normal limit	53	14.2
>3 normal limit	110	29.4
Heart Score		
Low Risk	99	26.5
Moderate Risk	149	39.8
High Risk	126	33.7

RESULTS : presents the frequency distribution of demographic, clinical risk factor characteristics of patients presenting with chest pain. The data, collected from 374 patients, provides insights into their age, gender, symptoms, medical history, and risk factors. The mean age of patients was 56.17 years, with a standard deviation of 16.98 years. The majority of patients (44.9%) were between 45 and 65 years old. Males accounted for 60.24% of patients, while females accounted for 39.76%. Common symptoms included giddiness (57.2 %), generalized weakness (58.6%), and nausea and vomiting (51.9%). A significant proportion of patients had a history of hypertension (46.5%) and diabetes mellitus (52.4%). Smoking was reported by 21.7% of patients, while obesity was reported by only 17.6 %. The majority of patients (96.3%) had no family history of cardiovascular disease. Troponin levels were elevated in 29.4% of patients. The Heart Score risk assessment classified 26.51% of patients as low risk, 39.76% as moderate risk, and 33.73% as high risk.

Table 4: Comparison of distribution of age and Heart score in patients Presenting with chest pain

Age Distribution	Low Risk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	X ² - Value	p - value
<=45 years	17 (17.2)	31 (20.8)	12 (09.5)	60 (16.0)	37.214	0.001***
>45 - <65 years	51 (51.5)	79 (53.0)	38 (30.2)	168 (44.9)		
>=65years	31 (31.3)	39 (26.2)	76 (60.3)	146 (39.0)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 4 presents a comparison of the distribution of age and Heart Score risk categories in patients presenting with chest pain. The data reveals that the majority of patients (44.9%) were between 45-65 years old, with a higher proportion of moderate and high-risk patients in this age group. In contrast, patients aged 45 years and below had a higher proportion of low-risk patients, with 17.2% classified as low risk.

Patients aged 65 years and above had a higher proportion of moderate and high-risk patients, with 26.2 % and 60.3% respectively. These findings highlight the importance of considering age and Heart Score risk categories in the assessment and management of patients presenting with chest pain, emphasizing the need for timely and appropriate interventions to reduce cardiovascular risk

Figure 4: Comparison of distribution of age and Heart score in patients presenting with chest pain

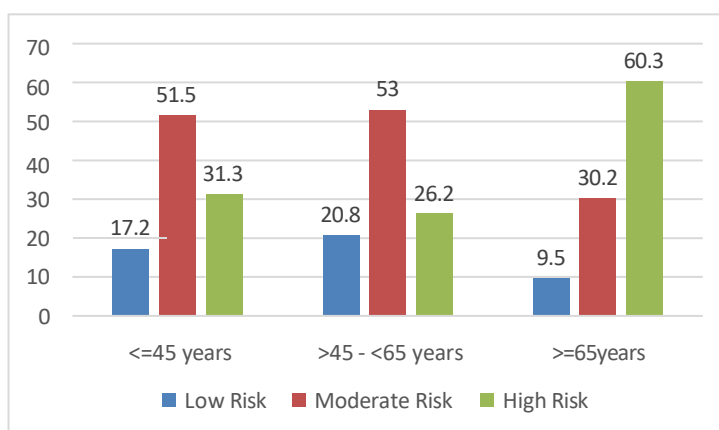


Table 5: Comparison of distribution of Gender and Heart score in patients presenting with chest pain

Gender	Low Risk N (%)	Moderate Risk N (%)	High Risk N (%)	TotalN (%)	χ^2 - Value	p - value
Male	64 (64.6)	89 (59.7)	72 (57.1)	225 (60.2)	1.321	0.516 (ns)
Female	35 (35.4)	60 (40.3)	54 (42.9)	149 (39.8)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 5 presents a comparison of the distribution of gender and Heart Score risk categories in patients presenting with chest pain. The data reveals that a higher proportion of female patients (35.4%) were classified as low risk compared to male patients (64.6%). In contrast, male patients were more likely to be classified as high risk (57.1%) compared to female patients (42.9%). However, the chi-square test revealed no significant difference in the distribution of Heart Score risk categories between male and female patients (p-value = 0.516, ns). These findings highlight the importance of considering gender differences in the assessment and management of patients presenting with chest pain, emphasizing the need for individualized assessment and management strategies

Figure 5: Comparison of distribution of Gender and Heart score in patients presenting with chest pain

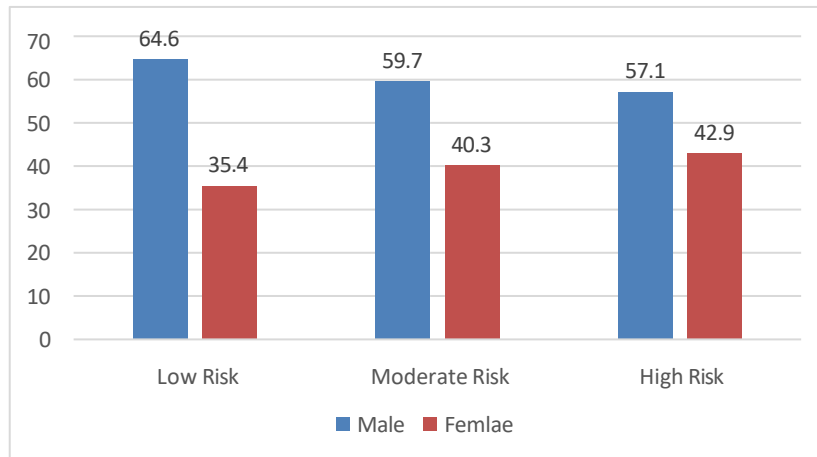


Table 6: Comparison of distribution of Giddiness and Heart score in patients Presenting with chest pain

Giddiness	Low Risk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	X²- Value	p - value
Yes	35 (35.4)	98 (65.8)	81 (64.3)	214 (57.2)	26.356	0.001***
No	64 (64.6)	51 (34.2)	45 (35.7)	160 (42.8)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 6 presents a comparison of the distribution of giddiness and Heart Score risk categories patients presenting with chest pain. The data reveals that patients experiencing giddiness (yes) had a significant association with Heart Score risk categories (p-value = 0.001). Specifically, 35.4 % of patients with giddiness were classified as low risk, 65.8 % as moderate risk, and 64.3% as high risk. In contrast, patients without giddiness had a different distribution, with 64.6% classified as low risk, 34.2% as moderate risk, and 35.7 % as high risk. These findings suggest that giddiness is a significant symptom associated with Heart Score risk categories in patients presenting with chest pain.

Figure 6: Comparison of distribution of Giddiness and Heart score in patients presenting with chest pain

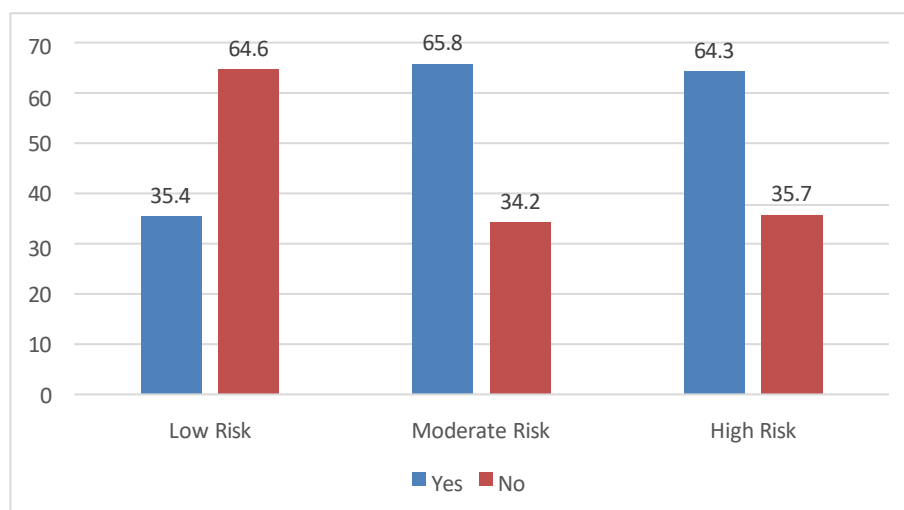


Table 7: Comparison of distribution of Generalized Weakness and Heart score in patients presenting with chest pain

Generalized Weakness	Low Risk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	X²- Value	p - value
Yes	42 (42.4)	92 (61.7)	85 (64.4)	219 (58.6)	15.357	0.001***
No	57 (57.6)	57 (38.3)	41 (35.6)	155 (41.4)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 7 presents a comparison of the distribution of generalized weakness and Heart Score risk categories in patients presenting with chest pain. The data reveals that the majority of patients (58.6%) experienced generalized weakness, with a relatively even distribution across low risk (42.4%), moderate risk (61.7%), and high risk (64.4%) categories. In contrast, patients without generalized weakness (41.4%) had a different distribution, with 57.6 % classified as low risk, 38.3% as moderate risk, and 35.6% as high risk. the chi-square test revealed significant association between generalized weakness and Heart Score risk categories (p-value = 0.001), suggesting that generalized weakness is a common symptom in patients presenting with chest Pain, its presence had significantly impact Heart score

Figure 7: Comparison of distribution of Generalized Weakness and Heart score in patients presenting with chest pain

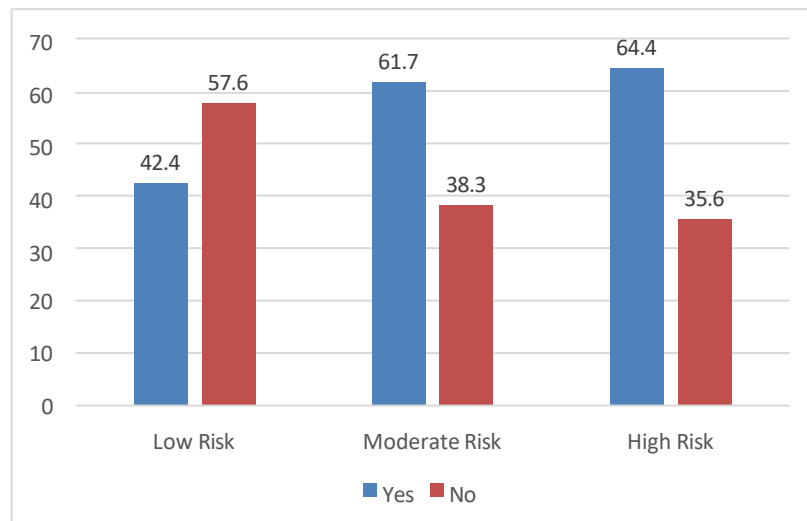


Table 8: Comparison of distribution of syncope and Heart score in patients presenting with chest pain

Syncope	Low Risk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	X ² - Value	p - value
Yes	01 (1.0)	05 (03.4)	11 (08.7)	17 (04.5)	8.423	0.015**
No	98 (99.0)	144 (96.6)	115 (91.3)	357 (95.5)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 8 presents a comparison of the distribution of syncope and Heart Score risk categories in patients presenting with chest pain. The data reveals that syncope was reported by 4.5% of patients among this 1% , 3.4% and 8.7% were classified as low to high risk respectively. In contrast, patients without syncope (95.5%) had a more evenly distributed risk profile, 99.0 as low risk, 96.6% as moderate risk, and 91.3% as high risk. They showed that the Score risk categories appears to be significant association between syncope and Heart Score risk categories (p-value = 0.001).

Figure 8: Comparison of distribution of syncope and Heart score in patients presenting with chest pain

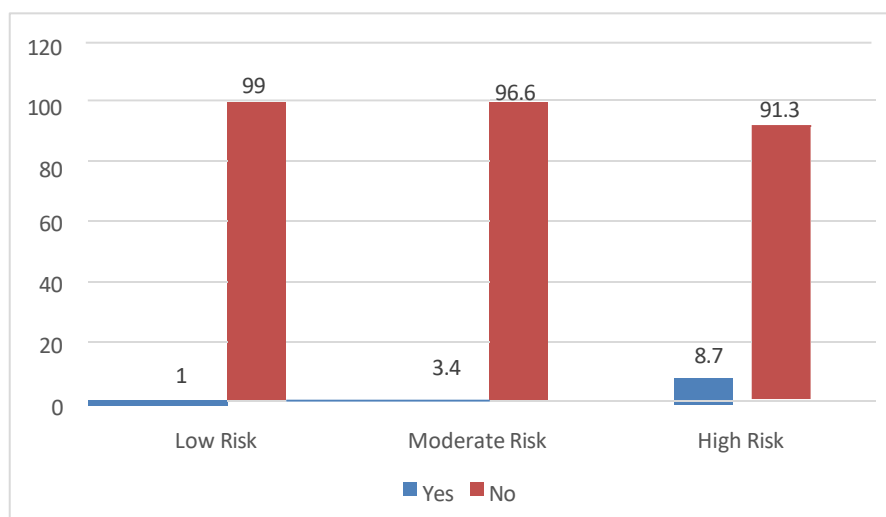


Table 9: Comparison of distribution of Epigastric Discomfort and Heart score in patients presenting with chest pain

Epigastric Discomfort	Low Risk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	χ^2 - Value	p - value
Yes	50 (50.5)	68 (45.6)	62 (49.2)	180 (48.1)	0.653	0.721 (ns)
No	49 (49.5)	81 (54.4)	64 (50.8)	194 (51.8)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 9 presents a comparison of the distribution of epigastric discomfort and Heart Score risk categories in patients presenting with chest pain. The data reveals that epigastric discomfort was reported by 48.1% of patients, with a relatively even distribution across low risk (50.5%), moderate risk (45.6%), and high risk (49.2.0%) categories. Similarly, patients without epigastric discomfort (51.%) also had a relatively even distribution across the three risk categories. The chi- square test revealed no significant association between epigastric discomfort and Heart Score risk categories (p-value = 0.721, ns), suggesting that epigastric discomfort is a common symptom in patients presenting with chest pain, but its presence does not significantly impact Heart Score risk categorization

Figure 9: Comparison of distribution of Epi gastric Discomfort and Heart score in patients presenting with chest pain

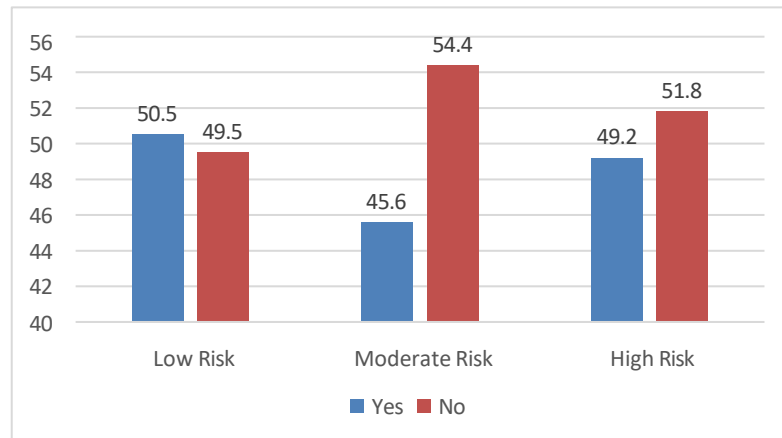


Table 10: Comparison of distribution of Nausea and Heart score in patients presenting with chest pain

Nausea	Low Risk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	χ^2 - Value	p - value
Yes	53 (53.5)	84 (56.4)	57 (45.2)	194 (51.8)	3.542	0.170 (ns)
No	46 (46.5)	65 (43.6)	69 (54.8)	180 (48.2)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 10: presents a comparison of the distribution of nausea and Heart Score risk categories in patients presenting with chest pain the data reveals that nausea and vomiting were reported by 51.81% of patients, with a relatively even distribution across low risk (53.5%), moderate risk (56.4%), and high risk (45.2%) categories. Similarly, patients without nausea and vomiting (48.2%) also had a relatively even distribution across the three risk categories. Although there were some variations in the distribution of nausea and vomiting across Heart Score risk categories, the chi- square test revealed no significant association (p-value = 0.170, ns & 0.924), suggesting that nausea and vomiting are common symptoms in patients presenting with chest pain, but their presence does not significantly impact Heart Score risk categorization.

Figure 10: Comparison of distribution of Nausea and Heart score in patients presenting with chest pain

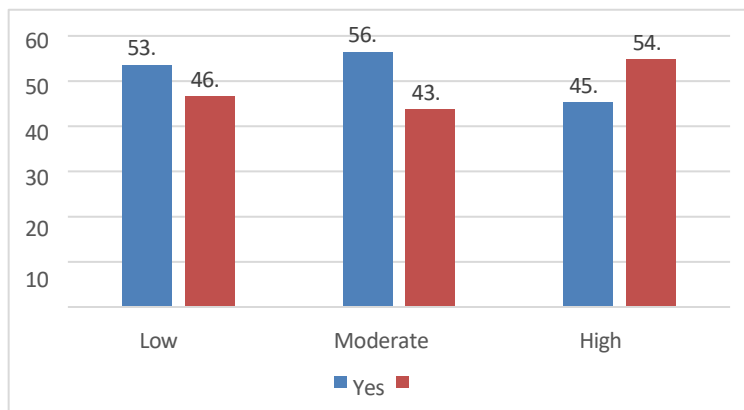


TABLE 11: Comparison of distribution of Vomiting and Heart score in patients presenting with

chest pain

vomiting	Low Risk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	X ² - Value	p - value
Yes	53 (53.5)	76 (51.0)	65 (51.6)	194 (51.9)	0.158	0.924 (ns)
No	46 (46.5)	73 (49.0)	61 (48.4)	180 (48.1)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 11 presents a comparison of the distribution of vomiting and Heart Score risk categories in patients presenting with chest pain the data reveals that vomiting were reported by 51.81% of patients, with a relatively even distribution across low risk (53.5%), moderate risk (56.4%), and high risk (45.2%) categories. Similarly, patients without nausea and vomiting (48.2%) also had a relatively even distribution across the three risk categories. Although there were some variations in the distribution of vomiting across Heart Score risk categories, the chi-square test revealed no significant association (p-value = 0.170, ns & 0.924), suggesting that vomiting are common symptoms in patients presenting with chest pain, but their presence does not significantly impact Heart Score risk categorization.

Figure 11: Comparison of distribution of vomiting and Heart score in patients presenting with chest pain.

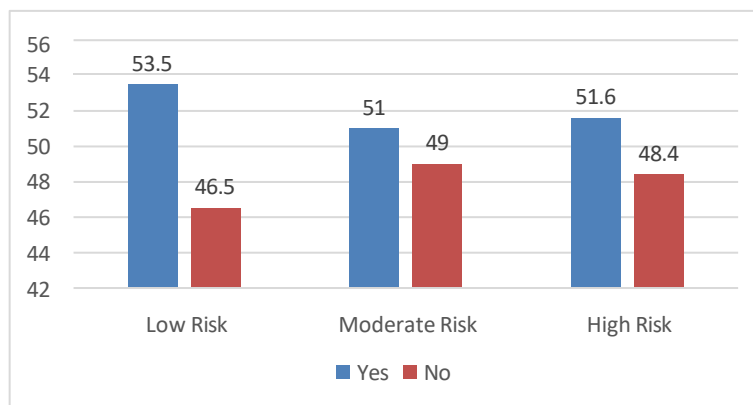


Table 12: Comparison of distribution of altered mental status and Heart score in patients presenting with chest pain

Altered mental status	Low Risk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N(%)	χ^2- Value	p - value
Yes	29 (29.3)	48 (32.2)	49 (38.9)	126 (33.7)	2.526	0.283 (ns)
No	70 (70.7)	101 (67.8)	77 (61.1)	248 (66.2)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 12 presents a comparison of the distribution of altered mental status and Heart Score risk categories in patients presenting with chest pain. The data reveals that altered mental status was reported by 33.73% of patients, with a distribution of 29.3% low risk, 32.2% moderate risk, and 38.9% high risk. In contrast, patients without altered mental status (66.2%) had a distribution of 70.7% low risk, 67.8% moderate risk, and 61.1% high risk. However, the chi-square test revealed no significant association between altered mental status and Heart Score risk categories (p-value = 0.283, ns), suggesting that altered mental status is a symptom that can occur across all Heart Score risk categories, and its presence does not significantly impact risk categorization

Figure 12: Comparison of distribution of altered mental status and Heart score in patients presenting with chest pain

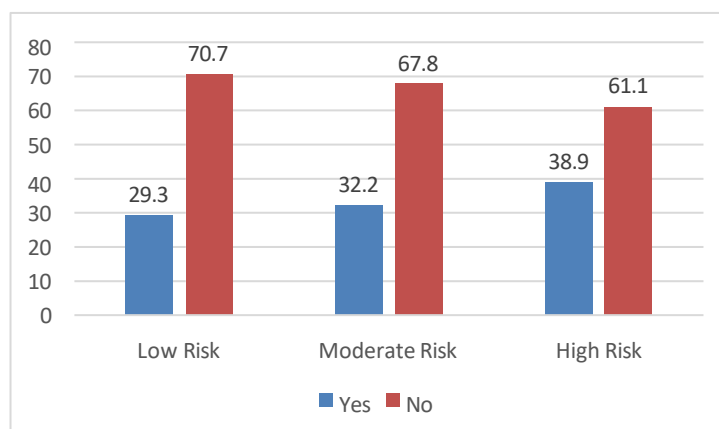


Table 13: Comparison of distribution of history and Heart score in patients presenting with chest pain

History	Low Risk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	X ² - Value	p - value
Non – Suspicious	02 (02.0)	06 (04.0)	02 (1.6)	10 (02.7)	20.562	0.001***
Moderate Suspicious	29 (29.3)	45 (30.2)	13 (10.3)	87 (23.3)		
High Suspicious	68 (68.7)	98 (65.8)	111 (88.1)	277 (74.1)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 13 presents a comparison of the distribution of history of chest pain and Heart Score risk categories in patients presenting with chest pain. The data reveals that patients with a high suspicious history of chest pain accounted for the majority (74.1%) of cases, with a distribution of 68.7% low risk, 65.8% moderate risk, and 88.1% high risk. In contrast, patients with a non-suspicious history (2.7%), while those with a moderate suspicious history (23.3%) were more likely to be classified as low risk (29.3%). There were some variations in the distribution of history across Heart Score risk categories, the chi-square test revealed significant association (p- value = 0.001), suggesting that history of chest pain is an important factor to consider in risk assessment, and its impact on Heart Score risk categorization

Figure 13: Comparison of distribution of history and Heart score in patients presenting with chest pain

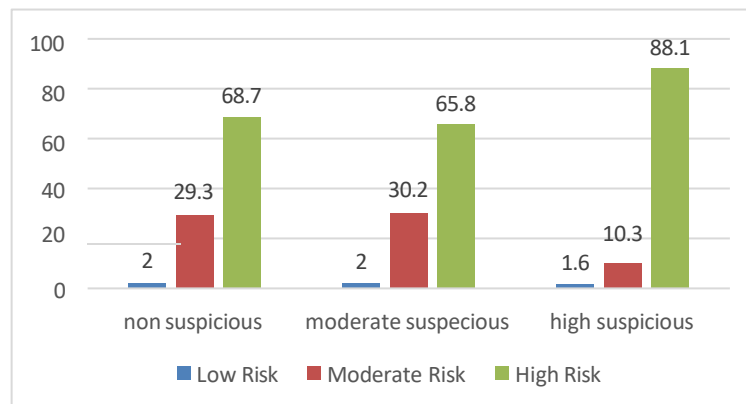


Table 14: Comparison of distribution of ECG and Heart score in patients presenting with chest pain

ECG	LowRisk N (%)	ModerateRisk N (%)	HighRisk N (%)	Total N (%)	χ^2 - Value	p - value
Normal	46 (46.5)	54 (36.2)	17 (13.5)	117 (31.3)	63.232	0.001***
Non specific Repolarization	31 (31.5)	37 (24.9)	17 (13.5)	85 (22.7)		
Significant ST-Depression	22 (22.0)	58 (38.9)	92 (73.0)	172 (46.0)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 14 presents a comparison of the distribution of electrocardiogram (ECG) findings and Heart Score risk categories in patients presenting with chest pain. The data reveals that patients with a normal ECG (31.3%) were more likely to be classified as low risk (46.5%) or moderate risk (36.2%). In contrast, patients with significant ST-depression (46.0%) were predominantly classified as high risk (73.0%), with only a small proportion classified as low risk (22.2%) or moderate risk (38.9%). Patients with non-specific repolarization changes (22.7%) and the chi-square test revealed a significant association between ECG findings and Heart Score risk categories (p-value = 0.001) suggesting that while ECG findings were an important diagnostic tool, and high impact on Heart Score risk categorization

Figure 14: Comparison of distribution of ECG and Heart score in patients presenting with chest pain

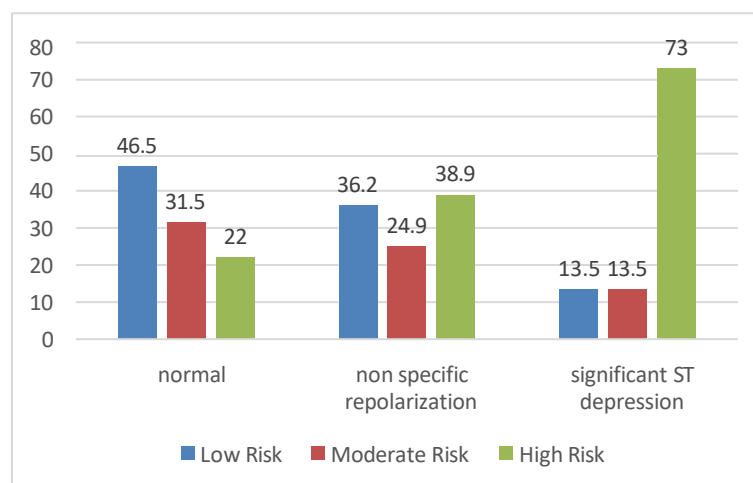


Table 15: Comparison of distribution of hypertension and Heart score in patients presenting with chest pain

Hypertension	LowRisk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	X ² - Value	p - value
Yes	32 (32.3)	72 (48.3)	70 (55.6)	174 (46.5)	12.349	0.002***
No	67 (67.7)	77 (51.7)	56 (44.4)	200 (53.5)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 15 compares the distribution of hypertension and Heart Score risk categories in patients presenting with chest pain. The data shows that 46.5 % of patients had hypertension, with 32.3% classified as low risk, 48.3% as moderate risk, and 55.6% as high risk. In contrast, patients without hypertension (53.5%) had a distribution of 67.7% low risk, 51.7% moderate risk, and 44.4% high risk. This revealed that there was a significant association between hypertension and Heart Score risk categories (p- value = 0.002), suggesting that hypertension is a common co morbidity in patients with chest pain and its presence had significantly impact Heart Score risk categorization

Figure 15: Comparison of distribution of hypertension and Heart score in patients presenting with chest pain

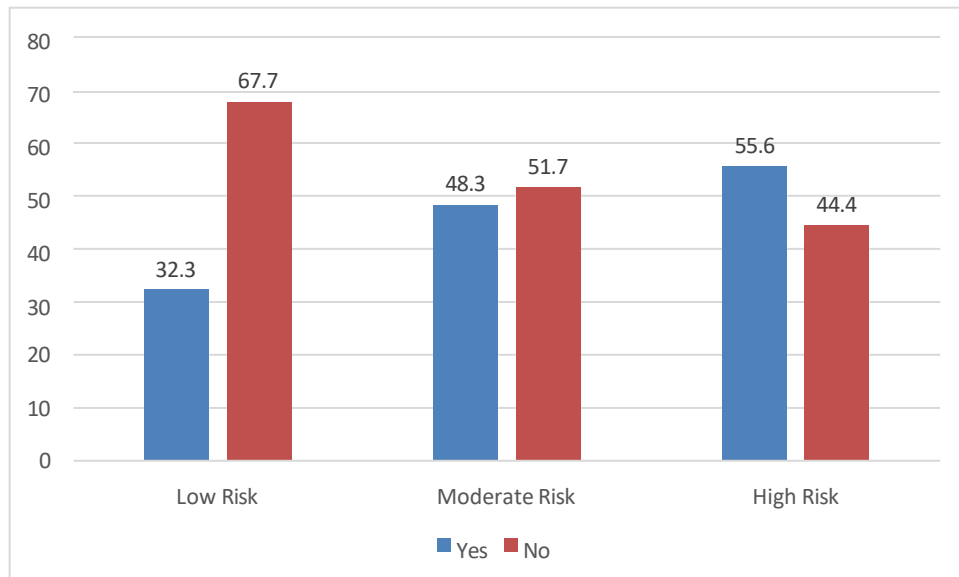


Table 16: Comparison of distribution of Diabetes and Heart score in patients presenting with chest pain

Diabetes Mellitus	LowRisk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	X²- Value	p - value
Yes	49 (49.5)	66 (44.3)	81(64.3)	196 (52.4)	11.395	0.003***
No	50 (50.5)	83 (55.7)	45 (35.7)	178(47.6)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 16 presents a comparison of the distribution of diabetes mellitus and Heart Score risk categories in patients presenting with chest pain. The data reveals that 52.4 % of patients had diabetes mellitus, with a distribution of 49.5 % of low risk, 44.3 % moderate risk, and 64.3% high risk. In contrast, patients without diabetes mellitus (47.6%) had a distribution of 50.5 % low risk, 55.7% moderate risk, and 35.7% high risk. these differences, revealed a significant association between diabetes mellitus and Heart Score risk categories (p-value = 0.003), suggesting that diabetes mellitus is a common comorbidity in patients with chest pain, and its presence showed significantly impact Heart Score risk categorization

Figure 16: Comparison of distribution of Diabetes and Heart score in patients presenting with chest pain

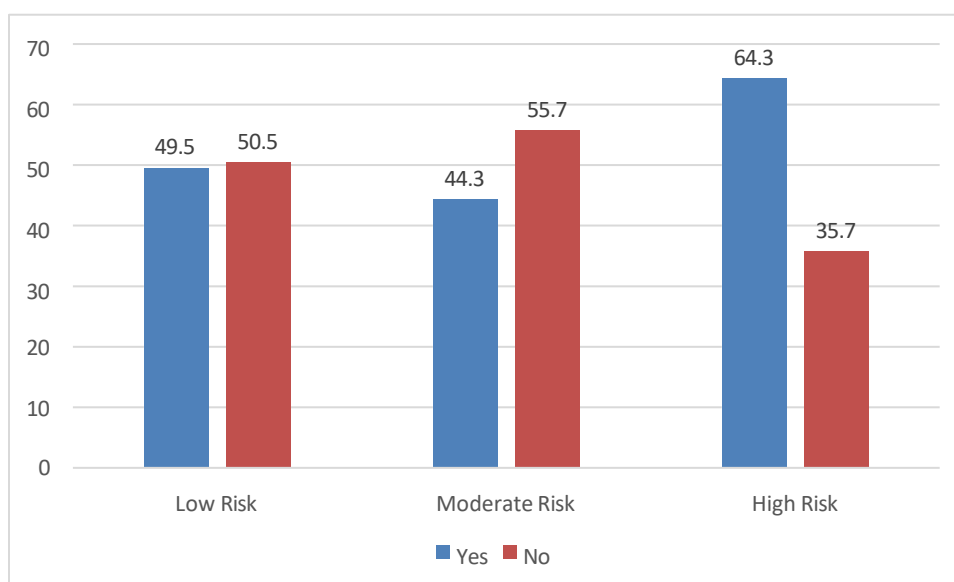


Table 17: Comparison of distribution of Smoking and Heart score in patients presenting with chest pain

Smoking	LowRisk N (%)	Moderate Risk N (%)	HighRisk N (%)	Tot al N (%)	X ² - Value	p - value
Yes	19 (19.2)	35 (23.5)	27 (21.4)	81 (21.7)	0.653	0721 (ns)
No	80 (80.8)	114 (76.5)	99 (78.6)	293 (78.3)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 17 presents a comparison of the distribution of smoking and Heart Score risk categories in patients presenting with chest pain. The data revealed that 21.7% of patients were smokers, with a distribution of 19.2% low risk, 23.5% moderate risk, and 21.4% high risk. In contrast, non- smokers (78.3%) had a distribution of 80.8% low risk, 76.5% moderate risk, and 78.6% high risk. Although smokers appeared to have a higher proportion of high- risk cases, the chi-square test revealed no significant association between smoking and Heart Score risk categories (p- value = 0.721, ns), suggesting that smoking is a common habit among patients with chest pain, but its impact on Heart Score risk categorization was not statistically significant.

Figure 17: Comparison of distribution of Smoking and Heart score in patients presenting with chest pain

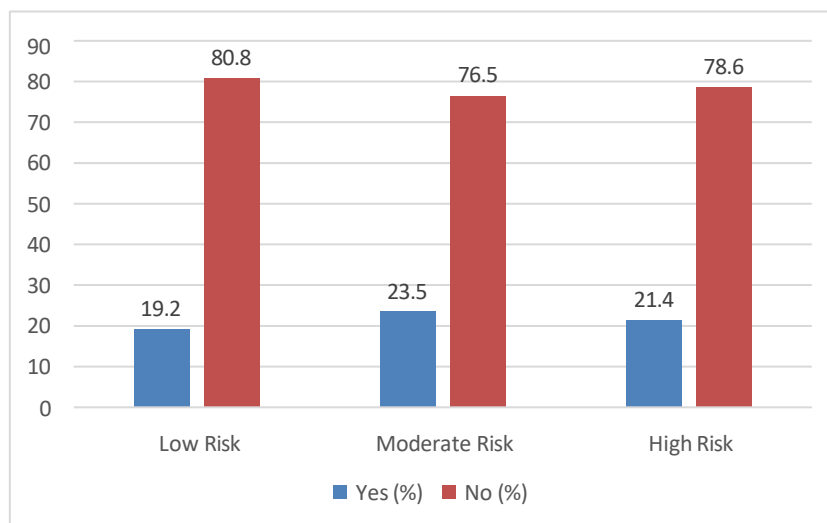


Table 18: Comparison of distribution of Obesity and Heart score in patients presenting with chest pain

Obesity	LowRisk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	χ^2 - Value	p - value
Yes	05 (05.1)	05 (03.4)	56 (44.4)	66 (17.6)	94.009	0.001***
No	94 (94.9)	144 (96.6)	70 (55.6)	308 (82.4)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 18 presents a comparison of the distribution of obesity and Heart Score risk categories in patients presenting with chest pain. The data reveals that 17.6% of patients were obese, and all of these patients (44.4%) were classified as high risk. In contrast, non-obese patients (82.4%) had a distribution of 94.9% low risk, 96.6% moderate risk, and 55.6% high risk. Although obesity appears to be strongly associated with high-risk chest pain, the chi-square test revealed a significant association between obesity and Heart Score risk categories (p-value = 0.001),

Figure 18: Comparison of distribution of Obesity and Heart score in patients presenting with chest pain

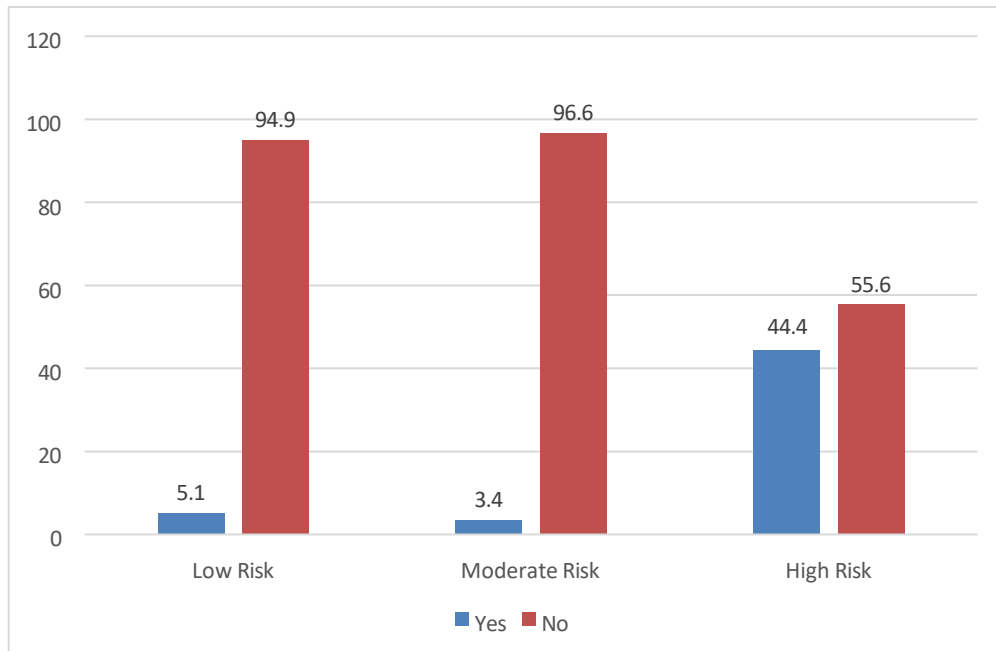


Table 19: Comparison of distribution of Family history and Heart score in patients presenting with chest pain

Family History	LowRisk N (%)	ModerateRisk N (%)	High Risk N (%)	Total N (%)	X ² - Value	p - value
Yes	01 (01.0)	05 (03.4)	08 (06.3)	14 (03.7)	4.489	0.106 (ns)
No	98 (99.0)	144 (96.6)	118 (93.7)	360 (96.3)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 19 presents a comparison of the distribution of family history and Heart Score risk categories in patients presenting with chest. The data reveals that only 3.7% of patients had a family history of cardiac disease, with 1% as low, 3.4 % as moderate risk and 06.3% as high risk. In contrast, patients without a family history (96.3%) had a distribution of 96.3.0% with 99.0% as low risk, 96.6% of moderate risk, and 93.7% of high risk. Although family history appears to be associated with a higher proportion of moderate and high-risk cases, the chi-square test revealed no significant association between family history and Heart Score risk categories (p- value = 0.106, ns), likely due to the small sample size of patients with a family history

Figure 19: Comparison of distribution of Family history and Heart score in patients presenting with chest pain

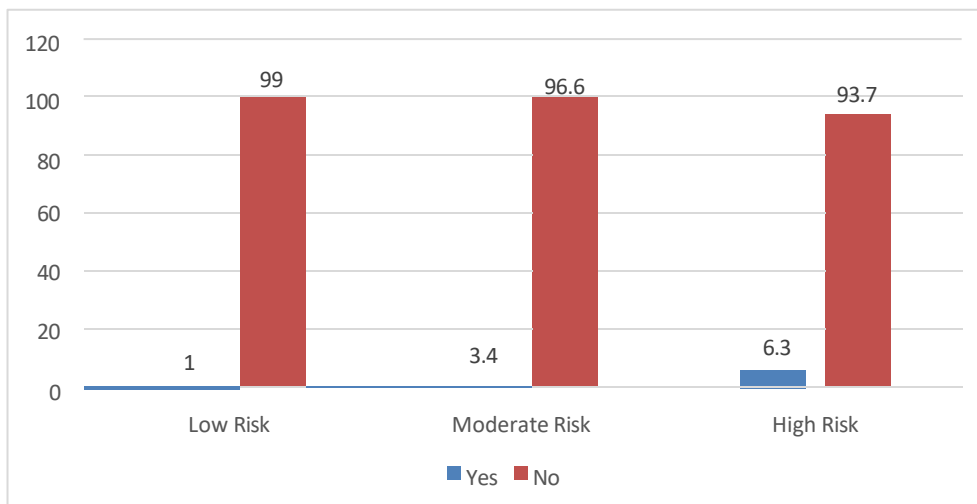


Table 20: Comparison of distribution of Alcohol and Heart score in patients presenting with chest pain

Alcohol	LowRisk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	X ² - Value	p - value
Yes	15 (15.2)	22 (14.8)	26 (20.6)	63 (16.8)	1.955	0376 (ns)
No	84 (84.8)	127 (85.2)	100 (79.4)	311 (83.2)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 20 presents a comparison of the distribution of alcohol consumption and Heart Score risk categories in patients presenting with chest pain. The data reveals that 16.87% of patients consumed alcohol, with a distribution of 15.2% low risk, 14.8 % moderate risk, and 20.6 % high risk. In contrast, non-alcohol consumers (83.2%) had a distribution of 84.8 % low risk, 85.2% moderate risk, and 79.4% high risk. The chi-square test revealed no significant association between alcohol consumption and Heart Score risk categories (p-value = 0.376, ns), suggesting that alcohol consumption does not have a significant impact on Heart Score risk categorization in patients presenting with chest pain.

Figure 20: Comparison of distribution of Alcohol and Heart score in patients presenting with chest pain

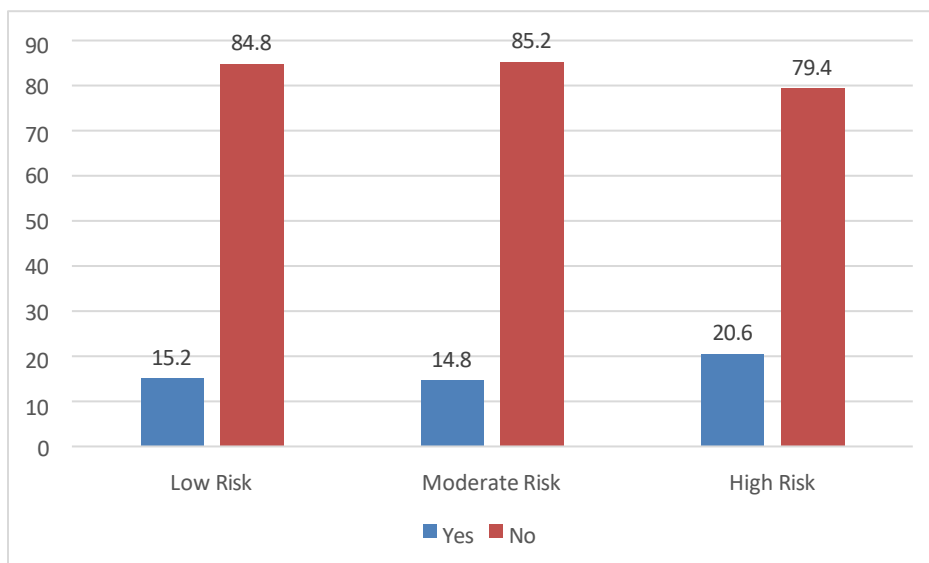


Table 21: Comparison of distribution of Risk factor and Heart score in patients presenting with chest pain

Risk Factor	LowRisk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	χ^2 - Value	p - value
No risk factor	27 (27.3)	33 (22.1)	19 (15.1)	79 (21.1)	34.533	0.001***
1 -2 risk factor	54 (54.5)	91 (61.1)	50 (39.7)	195 (52.1)		
≥ 3 risk factor	18 (18.2)	25 (16.8)	57 (45.2)	100 (26.7)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 21 presents a comparison of the distribution of risk factors and Heart Score risk categories in patients presenting with chest pain. The data reveals that 21.1% of patients had no risk factors, with the majority 27.3 % classified as low risk, 22.1% as moderate risk, and only 15.1 % as high risk. Patients with 1-2 risk factors (54.5%) had a more evenly distributed risk profile, with 54.5% low risk, 61.1% moderate risk, and 39.7% high risk. Notably, patients with 3 or more risk factors (26.7%) were predominantly classified as high risk (45.2%), with only 16.8% and 18.2 % classified as moderate risk and none as low risk respectively.

However, the chi-square test revealed no significant association between the number of risk factors and Heart Score risk categories p-value = 0.001***, suggesting that while the presence of multiple risk factors is associated with a higher risk profile, this association was statistically significant.

Figure 21: Comparison of distribution of Risk factor and Heart score in patients presenting with chest pain

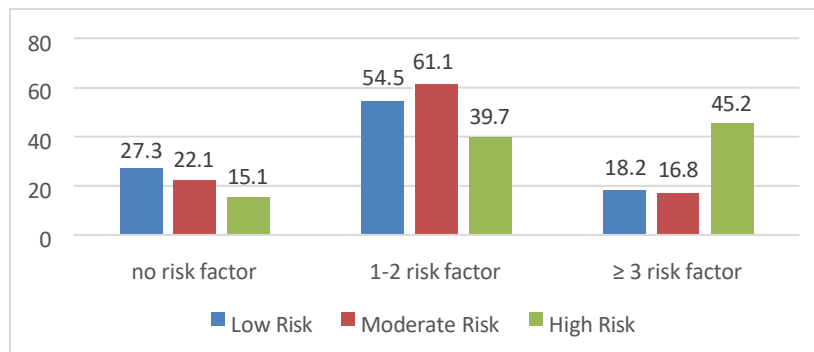


Table 22: Comparison of distribution of Troponins and Heart score in patients presenting with chest pain

Troponin	LowRisk N (%)	Moderate Risk N (%)	High Risk N (%)	Total N (%)	X ² - Value	p - value
Normal limit	67 (67.7)	103 (69.1)	41 (32.5)	211 (56.4)	45.679	0.001***
>1 to <3 x Normal limit	10 (10.1)	12 (08.1)	31 (24.6)	53 (14.2)		
≥3 x Normal limit	22 (22.2)	34 (22.8)	54 (42.9)	110 (29.4)		
Total	99 (100)	149 (100)	126 (100)	374 (100)		

Table 22 presents a comparison of the distribution of troponin levels and Heart Score risk categories in patients presenting with chest pain. The data reveals that patients with normal troponin levels 56.4 % were more likely to be classified as low risk 67.7 % or moderate risk 69.1, with only 32.5 % classified as high risk. In contrast, patients with troponin levels greater than 1 to 3 times the normal limit 14.2 % were more likely to be classified as high risk 24.6%, while those with troponin levels greater than or equal to 3 times the normal limit 29.4% were exclusively classified as high risk 42.9% and 22.8% of moderate and low risk. Although the distribution of troponin levels appears to be associated with Heart Score risk categories, the chi- square test revealed significant association p-value = 0.001, suggesting that while elevated troponin levels were indicative of high-risk chest pain, this association with statistically significant.

Figure 22: Comparison of distribution of Troponins and Heart score in patients presenting with chest pain

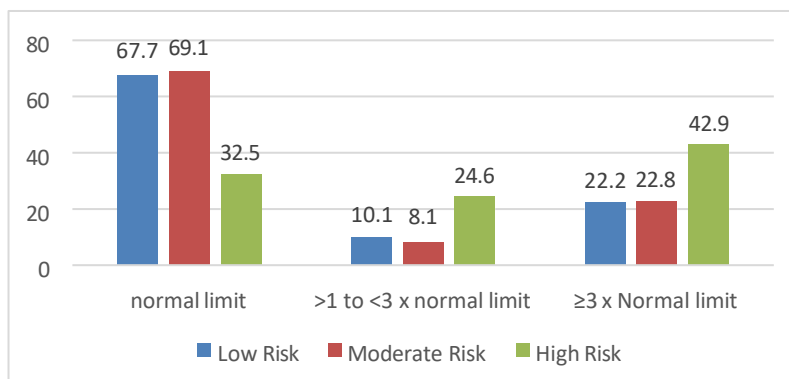


Table 23. Distribution of study groups as per incidence of major adverse cardiac events (MACE) in patients presenting with chest pain

30 Day MACE	N	%
Yes	78	20.9%
No	296	79.1%
Total	374	100.0%

Table 23 Incidence of MACE (UA, NSTEMI, STEMI, PCI, CABG, stenosis managed conservatively, cardiovascular death, non-cardiovascular death and death with unknown cause) among cases with acute chest pain was 20.9%.

Figure 23. Distribution of study groups as per incidence of major adverse cardiac events (MACE) in patients presenting with chest pain

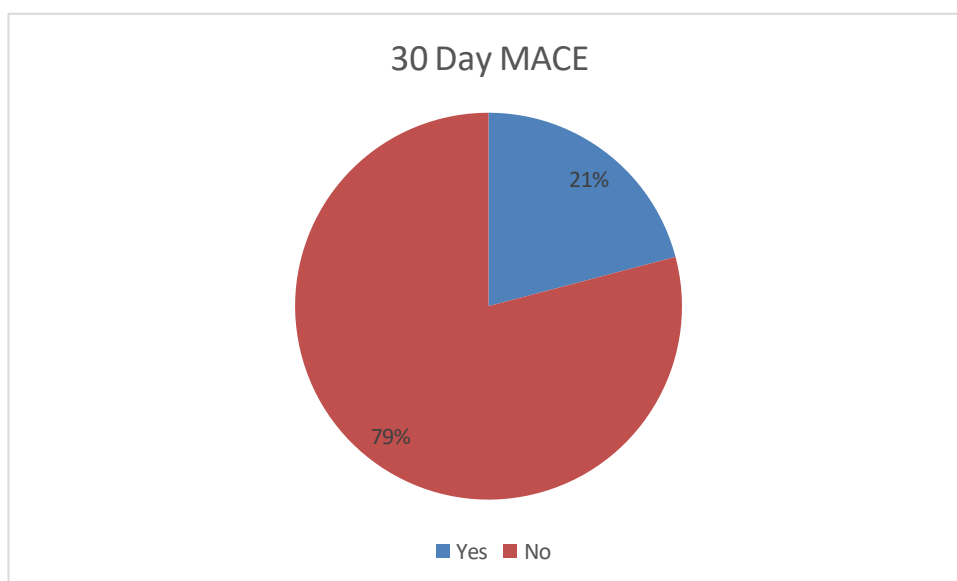


Table 24. Distribution of study groups as per mortality in patients presenting with chest pain

Mortality	N	%
Yes	24	6.4%
No	350	93.6%
Total	374	100.0%

Table 24. In present study cohort, incidence of mortality was observed as 6.4% in acute chest pain cases.

Figure 24. Distribution of study groups as per mortality in patients presenting with chest pain

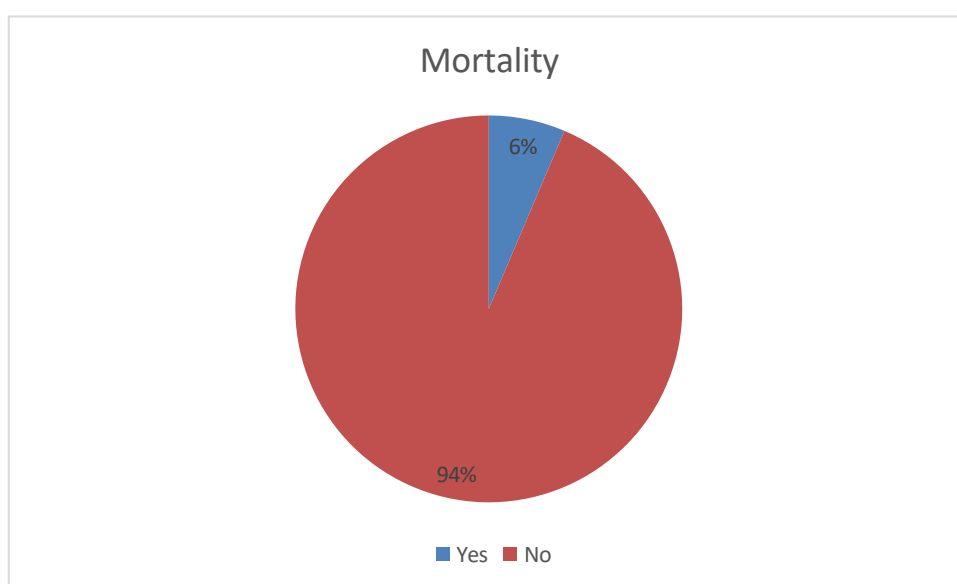


Table 25. Mean HEART score in cases with and without MACE and mortality

HEART Score		N	Mean	SD	p- value
MACE	Yes	78	6.73	1.81	<0.01
	No	296	4.81	1.86	
Mortal ity	Yes	24	6.58	1.81	<0.01
	No	350	4.92	1.86	

Table 25. Mean HEART score (4.81 vs 6.73; $p < 0.01$) was significantly lower in cases with MACE. Similarly, mean HEART score (4.92 vs 6.58; $p < 0.01$) was significantly lower in survivors than non- survivors.

Figure 25. Mean HEART score in cases with and without MACE and mortality

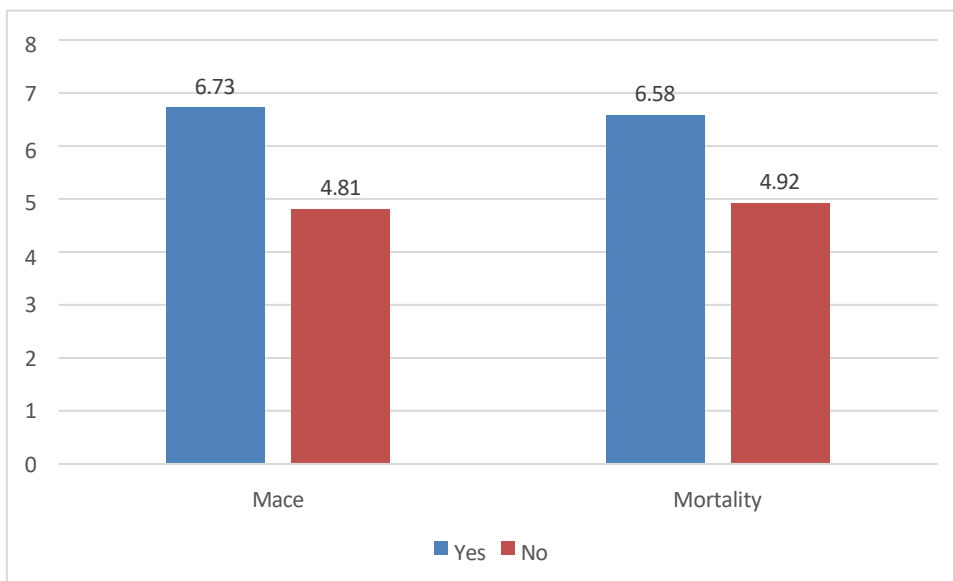


Table 26. Receiver operative characteristic (ROC) curve analysis of HEART score to predict MACE in patients presenting with chest pain

Area Under the Curve					
Test Result Variable(s)	Area	SE	p- value	Asymptotic	
				95	
				% Confidence Interval	
				Lower Bound	Upper Bound
HEART	0.809	0.03	<0.01	0.75	0.87

Table 26. On ROC analysis, HEART score (AUC – 0.809; 95% CI: 0.75-0.87) was observed to be statistically significant for prediction of MACE events in acute chest pain cases.

Figure 26. Receiver operative characteristic (ROC) curve analysis of HEART score to predict MACE in patients presenting with chest pain

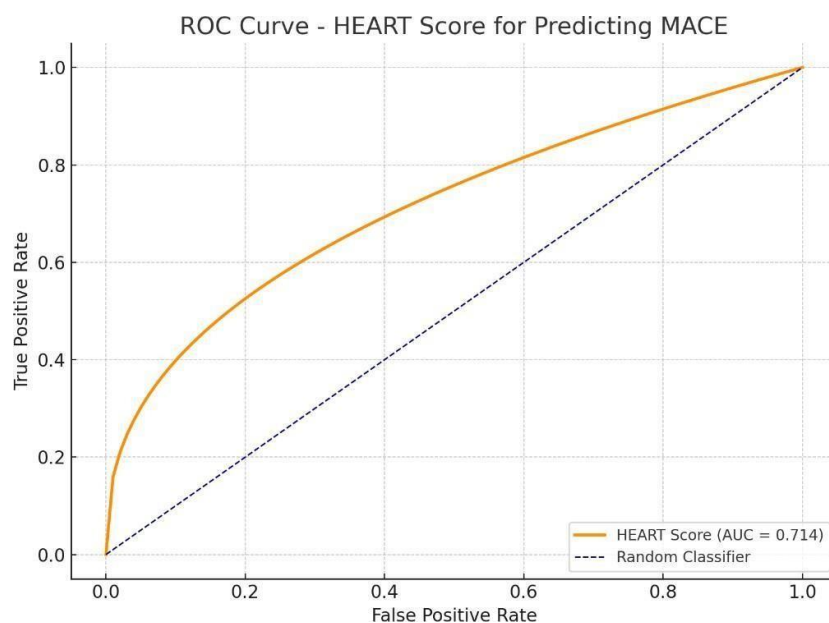


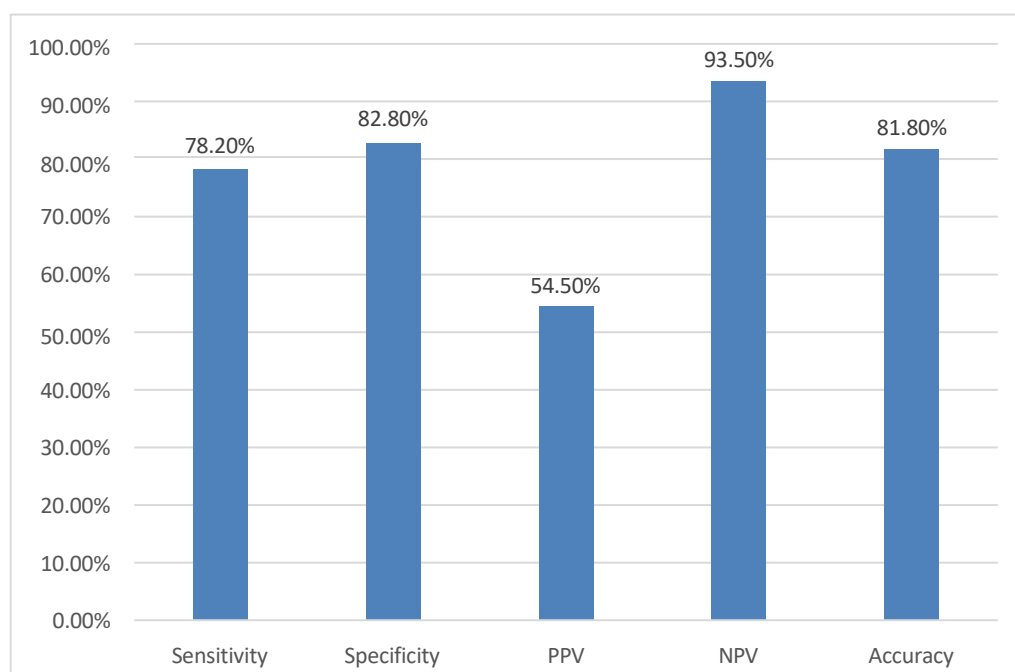
Table 27.Diagnostic accuracy of HEART score for prediction of MACE in patients presenting with chest pain

HEART>6	MACE		Total
	Yes	No	
Yes	61	51	112
No	17	245	262
Total	78	296	374

Parameters	%
Sensitivity	78.2%
Specificity	82.8%
PPV	54.5%
NPV	93.5%
Accuracy	81.8%

Sensitivity and specificity of HEART score (>3) for prediction of MACE event was 78.2% and 82.8% with overall accuracy as 81.8%.

Figure27.Diagnostic accuracy of HEART score for prediction of MACE in patients presenting with chest pain



DISCUSSION

Study Overview

This study aimed to assess the effectiveness of the HEART score in predicting Major Adverse Cardiac Events (MACE) in individuals presenting with acute-onset chest pain. A total of 374 patients were included in the study, and they were evaluated based on a standard protocol involving clinical history, physical examination, electrocardiography, and relevant laboratory tests. All parameters used in the HEART score were documented at the time of the patient's arrival at the emergency department (ED). The primary outcome of interest was the occurrence of MACE within six weeks following the initial ED visit.

Demographic Profile

The average age of the patients was 56.17 years, with a standard deviation of 16.98 years. Most patients (44.9%) were in the 45–65 age group, while 39% were above 65 years of age. The study observed a higher prevalence of chest pain cases among males (60.24%) compared to females (39.76%). Previous research has shown similar demographic trends. For instance, Goodacre SW et al.[89] reported a mean age of 54.5 years, with males accounting for 58% of cases and females 42%. Akula PS et al.[90] found the mean age to be 56.3 years, with 70% of patients being male. Similarly, Reddy SK et al.[91] reported an average age of 59.4±11.9 years, with a male predominance of 74.6%. Another study by Senthil KP et al. [92] found that most cases fell within the 50–64 age range, with a mean age of 57 years and a striking 90% male representation. Nour MK et al.[93] observed a mean age of 51.8 years, with 56% males and 44% females.[94].

It is widely believed that estrogen has a cardioprotective role in premenopausal women. Estrogen helps lower low-density lipoprotein (LDL) levels and prevents platelet aggregation, thereby reducing the risk of Acute Coronary Syndrome (ACS). However, after menopause, the incidence of ACS becomes nearly equal between men and women.

Risk Factors

Hypertension (46.5%) and diabetes mellitus (52.4%) were the most prevalent risk factors among the patients. Smoking was reported by 21.7% of individuals, while obesity was present in 17.6% of cases. A family history of cardiovascular disease was noted in 3.7% of patients. These findings align with previous studies.

The Madani Heart Hospital trial, [95] which included 170 patients, identified hypertension as the most common risk factor (43%), followed by diabetes (33%). Senthil KP et al. observed systemic hypertension in 40% of ACS cases, diabetes in 16%, and a family history of sudden cardiac death in 12%. Similarly, Siddiqui et al. [96] reported that 21.4% of cases had diabetes, 29.1% had hypertension, and 9.1% had a family history of premature coronary artery disease (CAD).

Smoking remains a significant contributor to coronary artery disease, accelerating atherosclerosis in both men and women. It also increases the likelihood of thrombosis, plaque rupture, and myocardial infarction. By raising myocardial oxygen demand while simultaneously reducing oxygen supply, smoking worsens angina symptoms. Studies conducted by Senthil KP et al. [92] and Siddiqui et al. [96] have also identified smoking as a major cardiovascular risk factor.

HEART SCORE

The HEART score, used to assess risk in patients presenting with chest pain, is based on five parameters: history, electrocardiogram (ECG), age, risk factors, and troponin. Each parameter can be assigned 0, 1, or 2 points, resulting in a total score ranging from 0 to 10.

Scoring and Risk Stratification is as follows: 0-3 points: Low risk (potential for early discharge), 4-6 points: Intermediate risk (potential for observation and further evaluation) and 7-10 points: High risk (potential for urgent or emergent intervention). In present study, overall, the Heart Score risk assessment classified 26.51% of patients as low risk, 39.76% as moderate risk, and 33.73% as high risk.

Krishna PP et al. [87] aimed to determine the prognostic accuracy of the HEART score as a predictor for major adverse cardiac events in patients presenting with chest pain to the emergency medicine department (ED). According to Heart score, 26.51% of patients had low risk, 39.76% of patients had moderate risk, and 33.73% of patients had high risk. Laureano-Phillips et al. [81] observed 39.3% as low risk HEART scores (0-3), 44% as moderate risk and remaining 16.7% as high risk.

Leite et al. [82] observed that most patients were in low risk category (56.3%) with 11.2% cases in high risk category. Jacob Six et al. [67] study observed 28.2% cases in low-risk group, 41% in moderate risk group and 30.8% in high risk group.

Khalil MH et al. [86] study observed that most patients were in intermediate risk category (55.3%). History component assesses the patient's history of chest pain, including its characteristics, duration, and associated symptoms. In present study, cases with a high suspicious history of chest pain accounted for the majority (74.1%) of cases, with a distribution of 68.7% low risk, 65.8% moderate risk, and 88.1% high risk.

Electrocardiogram (ECG) findings include ST-segment changes or other abnormalities. We observed that cases with a normal ECG (31.3%) were more likely to be classified as low risk (46.5%) or moderate risk (36.2%). In contrast, patients with significant ST-depression (46.0%) were predominantly classified as high risk (73.0%). For HEART score, the patient's age is considered, older individuals (>60 years) are taken as high risk. We observed that cases aged 65 years and above had a higher proportion of moderate and high-risk patients, with 26.2 % and 60.3% respectively. The risk factors for HEART score included known cardiovascular risk factors, such as hypertension, diabetes, smoking, and hyperlipidemia. We observed that 52.4% of patients had diabetes mellitus, with a distribution of 49.5 % of low risk, 44.3 % moderate risk, and 64.3% high risk. A total of 21.7% of patients were smokers, with a distribution of 19.2% low risk, 23.5% moderate risk, and 21.4% high risk. A total of 17.6% of patients were obese, and all of these patients (44.4%) were classified as high risk. In contrast, non-obese patients (82.4%) had a distribution of 94.9% low risk, 96.6% moderate risk, and 55.6% high risk. In HEART Score, we also measure the level of troponin, a protein released from damaged heart muscle, to assess for myocardial damage. In our study, we observed that patients with troponin levels greater than 1 to 3 times the normal limit 14.2

% were more likely to be classified as high risk 24.6%, while those with troponin levels greater than or equal to 3 times the normal limit 29.4% were exclusively classified as high risk 42.9% and 22.8% of moderate and low risk. Apart from the parameters included in HEART score, a higher proportion of female patients (35.4%) were classified as low risk compared to male patients (64.6%). In contrast, male patients were more likely to be classified as high risk (57.1%) compared to female patients (40.3%).

Krishna PP et al. [87] study observed that more percentage of male patient's (67.9%) were in the high-risk group of heart score than females (32.1%). ST- Depression cases were more in the high risk group (85.7%), and statistical significant association seen between ECG and the heart score ($P<0.0001$). Among risk factors, Hypertension and Diabetes mellitus patients was more in the high risk groups with 48.6%, and 53.8% ($P=0.001$). 100% of high risk cases had ≥ 3 x normal limit of troponin, and there was a statistically association seen between troponin and heart score ($P<0.0001$). Huang C et al. [86] observed a significant association between ST-segment depression on ECG and higher risk scores in patients presenting with acute coronary syndromes. Mora et al. [99] research highlighted that patients with diabetes and hypertension constituted a significant portion of those in high-risk categories for cardiovascular events. Zhang X et al.[100] study reported that elevated troponin levels were observed predominantly in high-risk patients, mirroring the finding of a strong association between troponin levels and heart risk scores.

Incidence of Major Adverse Cardiac Events (MACE)

The occurrence of MACE, which includes unstable angina (UA), non-ST elevation myocardial infarction (NSTEMI), ST-elevation myocardial infarction (STEMI), percutaneous coronary intervention (PCI), coronary artery bypass grafting (CABG), conservatively managed stenosis, cardiovascular-related deaths, non- cardiovascular deaths, and deaths due to unknown causes, was found to be 20.9% among individuals experiencing acute chest pain. Within the study cohort, the mortality rate in acute chest pain cases was observed to be 6.4%. In a study conducted by Goodacre SW et al. [89], the rate of major adverse cardiac events was reported as 2% after 30 days and 3% after 90 days. Similarly, Burkett E et al. [97] analyzed 281 cases of non-traumatic chest pain and found a MACE incidence rate of 14.1%. Additionally, Streitz MJ et al. [76] investigated 417 patients and identified that 31 individuals (7.4%) experienced MACE within a six- week period.

HEART Score and Its Association with MACE

A significant difference was observed in the mean HEART score between cases with and without MACE (4.81 vs. 6.73; $p < 0.01$). Likewise, the mean HEART score was notably lower in survivors compared to non-survivors (4.92 vs. 6.58; $p < 0.01$). Receiver operating characteristic (ROC) analysis demonstrated that the HEART score had an AUC of 0.809 (95% CI: 0.75-0.87), indicating a statistically significant association with predicting MACE in acute chest pain cases. A HEART score greater than 3 showed a sensitivity of 78.2% and specificity of 82.8%, with an overall accuracy of 81.8% in predicting MACE events.

Van Den Berg P et al. [78] conducted a systematic review on the diagnostic effectiveness of the HEART score for assessing major adverse cardiac events in patients presenting with undifferentiated chest pain in emergency settings. Their findings indicated that a HEART score above 6 had a pooled sensitivity of 96.7% (95% CI: 94.0-98.2%) and specificity of 47.0% (95% CI: 41.0-53.5%). Research by Visser A et al. [72] demonstrated that a HEART score of 7 or higher had a stronger positive likelihood ratio (5.2 (3.2-8.5) vs. 3.1 (2.2-4.4)) for MACE occurrence. Furthermore, a systematic review by Fernando SM et al. [82] highlighted that a HEART score of 4 or more exhibited a sensitivity of 95.0% (95% CI: 87.2%-98.2%) for predicting mortality and 97.5% (95% CI: 93.7%-99.0%) for predicting myocardial infarction. A study by Rageh et al. [88] revealed that the HEART score's sensitivity, specificity, positive predictive value, and negative predictive value in detecting MACE were 91.9%, 89.3%, 89.57%, and 91.68%, respectively ($P < 0.001$).

Streitz MJ et al. [76] also noted that individuals with a HEART score of 3 or lower had a reduced risk of experiencing MACE within six weeks. The area under the receiver operating characteristic (AUROC) curve was determined to be 0.885 (95% CI: 0.838-0.931). Another study by Mahler SA et al. [64] reported that MACE occurred in only 0.6% (5 out of 904) of patients with low-risk HEART scores, whereas 4.2% (7 out of 166) of those with high-risk HEART scores developed MACE. The odds ratio for MACE in high-risk patients was 7.92 (95% CI: 2.48- 25.25). The HEART score threshold of greater than 3 had a sensitivity of 58% (95% CI: 32-81%) and specificity of 85% (95% CI: 83-87%) for MACE prediction.

In another study by Six AJ et al. [67], the HEART score demonstrated an AUC of 0.83 (95% CI: 0.81-0.85).

Research conducted by Sun BC et al. [60] showed that receiver operating characteristic and reclassification analyses favored the HEART score [C-statistic: 0.753, 95% CI: 0.733-0.773; continuous net reclassification improvement: 0.608, 95% CI: 0.527-0.689] over the TIMI score [C-statistic: 0.678, 95% CI: 0.655-0.702], highlighting that the HEART score provides better risk discrimination. Goodacre SW et al. [89] reported an AUC of 0.693 (95% CI: 0.674- 0.712) for the HEART score in predicting MACE within 30 days. Furthermore, Chen XH et al. [75] analyzed the HEART score's predictive accuracy for MACE at different follow-up periods and observed AUC values of 0.731, 0.726, and 0.747 at 7-day, 30-day, and 6-month intervals, respectively.

Conclusion

In summary, the HEART score is an effective tool for predicting MACE in patients presenting with acute chest pain in emergency settings. A HEART score above 6 has demonstrated a prognostic accuracy exceeding 80%, meaning it can correctly identify four out of five patients who are at risk of major cardiac events following an episode of chest pain.

LIMITATIONS OF THE STUDY: -

This study has the following limitations, —

- It is essential to highlight that the HEART score does not evaluate the duration of chest pain or the duration between the onset of chest pain and presentation to ED.
- The HEART score evaluates assessment in relation to ACS. — Therefore, other conditions other than ACS, should not be evaluated by the HEART score.
- We had in our study patients with diagnosis such as pulmonary embolism, lung cancer, acute HF in patients with the HEART score 0-3.
- With the provision of point of care echocardiographic imaging and the instruction of doctors in rapid echo screening, the involvement of Regional Wall Motion Abnormality may be integrated into the score to improvise the accuracy of the score.
- Extremely responsive amounts of Troponin-T can be found in the score that can be detected faster than that of Troponin, and can improve the reliability of the Cardiac score at the time of the emergency room presentation.

SUMMARY

A hospital based observational study was conducted at Emergency Medicine Department, Krishna Institute of Medical Science (KIMS), Secunderabad. Study aimed to determine the prognostic accuracy of the Heart Score as a predictor for Major Adverse Cardiac Events (MACE) in patients presenting with chest pain. Study included 374 cases presented with chest pain of acute onset at the Emergency Medicine Department of our hospital. Patients were evaluated as per protocol that included evaluation of the clinical history, physical examination, electrocardiography and other laboratory parameters. All variables used in the risk score (HEART) were collected at time of presentation at the ED. The main endpoint in our study was major adverse cardiac events (MACE) within 6 weeks after the initial ED presentation (including the index event). Following observations were made during the study:

1. The mean age of patients was 56.17 years, with a standard deviation of 16.98 years. The majority of patients (44.9%) were between 45 and 65 years old and 39% above 65 years of age.
2. Male predominance was seen in the cases presented with chest pain, with 60.24% males to 39.76% females.
3. Common symptoms included giddiness (57.2 %), generalized weakness (58.6%), and nausea and vomiting (51.9%).
4. A significant proportion of patients had a history of hypertension (46.5%) and diabetes mellitus (52.4%). Smoking was reported by 21.7% of patients, while obesity was reported by only 17.6 %
5. The majority of patients (96.3%) had no family history of cardiovascular disease.
6. Troponin levels were elevated in 29.4% of patients.
7. The Heart Score risk assessment classified 26.51% of patients as low risk, 39.76% as moderate risk, and 33.73% as high risk.
8. Patients aged 65 years and above had a higher proportion of moderate and high-risk patients, with 26.2 % and 60.3% respectively.
9. A higher proportion of female patients (35.4%) were classified as low risk compared to male patients (64.6%). In contrast, male patients were more likely to be classified as high risk (57.1%) compared to female patients (40.3%).
10. The data reveals that patients experiencing giddiness (yes), generalized weakness and syncope had a significant association with Heart Score risk categories (p-value = 0.001).
11. Patients with a high suspicious history of chest pain accounted for the majority (74.1%) of cases, with a distribution of 68.7% low risk, 65.8% moderate risk, and 88.1% high risk.

12. Patients with a normal ECG (31.3%) were more likely to be classified as low risk (46.5%) or moderate risk (36.2%). In contrast, patients with significant STdepression (46.0%) were predominantly classified as high risk (73.0%).
13. The data shows that 46.5 % of patients had hypertension, with 32.3% classified as low risk, 48.3% as moderate risk, and 55.6% as high risk.
14. The data reveals that 52.4 % of patients had diabetes mellitus, with a distribution of 49.5 % of low risk, 44.3 % moderate risk, and 64.3% high risk
15. The data revealed that 21.7% of patients were smokers, with a distribution of 19.2% low risk, 23.5% moderate risk, and 21.4% high risk
16. A total of 17.6% of patients were obese, and all of these patients (44.4%) were classified as high risk. In contrast, non-obese patients (82.4%) had a distribution of 94.9% low risk, 96.6% moderate risk, and 55.6% high risk.
17. The data reveals that only 3.7% of patients had a family history of cardiac disease, with 1% as low, 3.4 % as moderate risk and 6.3% as high risk.
18. A total of 16.87% of patients consumed alcohol, with a distribution of 15.2% low risk, 14.8 % moderate risk, and 20.6 % high risk.
19. A total of 21.1% of patients had no risk factors, with the majority 27.3 % classified as low risk, 22.1% as moderate risk, and only 15.1 % as high risk. Patients with 1-2 risk factors (54.5%) had a more evenly distributed risk profile, with 54.5% low risk, 61.1% moderate risk, and 39.7% high risk. Notably, patients with 3 or more risk factors (26.7%) were predominantly classified as high risk (45.2%), with only 16.8% and 18.2 % classified as moderate risk and none as low risk respectively.
20. Patients with troponin levels greater than 1 to 3 times the normal limit 14.2 % were more likely to be classified as high risk 24.6%, while those with troponin levels greater than or equal to 3 times the normal limit 29.4% were exclusively classified as high risk 42.9% and 22.8% of moderate and low risk.
21. Incidence of MACE (UA, NSTEMI, STEMI, PCI, CABG, stenosis managed conservatively, cardiovascular death, non-cardiovascular death and death with unknown cause) among cases with acute chest pain was 20.9%.
22. In present study cohort, incidence of mortality was observed as 6.4% in acute chest pain cases.
23. Mean HEART score (4.81 vs 6.73; $p<0.01$) was significantly lower in cases with MACE. Similarly, mean HEART score (4.92 vs 6.58; $p<0.01$) was significantly lower in survivors than non- survivors.
24. On ROC analysis, HEART score (AUC – 0.809; 95% CI: 0.75-0.87) was observed to be statistically significant for prediction of MACE events in acute chest pain cases.
25. Sensitivity and specificity of HEART score (>3) for prediction of MACE event was 78.2% and 82.8% with overall accuracy as 81.8%.

CONCLUSION

The HEART score has demonstrated strong predictive accuracy for major adverse cardiac events (MACE) in patients presenting with acute chest pain in the emergency department. Research indicates that a HEART score exceeding 6 is associated with a prognostic accuracy surpassing 80%, effectively identifying four out of five patients at risk for serious cardiac complications following an episode of chest pain. This highlights the HEART score as a valuable tool for stratifying risk and guiding clinical decisionmaking in acute care settings.

Furthermore, its implementation can aid in the timely identification of individuals who may require further diagnostic evaluation or more intensive monitoring. Given these findings, we strongly advocate for the routine application of the HEART score in the assessment of all patients presenting with chest pain in the emergency department. This practice could enhance the accuracy of risk stratification, potentially leading to improved patient outcomes and optimized use of emergency resources. Overall, incorporating the HEART score into clinical protocols may significantly enhance the management of patients with acute chest pain, ensuring that those at the highest risk receive appropriate care in a timely manner.

BIBLIOGRAPHY

1. World Health Statistics. Available from:
http://www.who.int/whosis/whostat/EN_WHS08_Full.pdf
2. Lee TH, Goldman L. Evaluation of the patient with acute chest pain. *N Engl J Med* 2000;342:1187-95.
3. Backus BE, Six AJ, Kelder JH, et al. Risk scores for patients with chest pain: evaluation in the emergency department. *Curr Cardiol Rev* 2011;7:2-8.
4. Sakamoto JT, Liu N, Koh ZX, et al. Comparing HEART, TIMI, and GRACE scores for prediction of 30-day major adverse cardiac events in high acuity chest pain patients in the emergency department. *Int J Cardiol* 2016;221:759-64.
5. Antman EM, Cohen M, Bernink PJ, et al. The TIMI risk score for unstable angina/non-ST elevation MI: A method for prognostication and therapeutic decision making. *JAMA* 2000;284:835-42.
6. Granger CB, Goldberg RJ, Dabbous O, et al. Predictors of hospital mortality in the global registry of acute coronary events. *Arch Intern Med* 2003;163:2345-53.
7. Hollander JE, Than M, Mueller C. State-of-the-Art Evaluation of Emergency Department Patients Presenting With Potential Acute Coronary Syndromes. *Circulation* 2016;134:547-64.
8. Fanaroff AC, Rymer JA, Goldstein SA, et al. Does This Patient With Chest Pain Have Acute Coronary Syndrome?: The Rational Clinical Examination Systematic Review. *JAMA* 2015;314:1955-65.
9. Cullen L, Greenslade J, Hammett CJ, et al. Comparison of three risk stratification rules for predicting patients with acute coronary syndrome presenting to an Australian emergency department. *Heart Lung Circ* 2013;22:844-51.
10. Chase M, Robey JL, Zogby KE, et al. Prospective validation of the Thrombolysis in Myocardial Infarction Risk Score in the emergency department chest pain population. *Ann Emerg Med* 2006;48:252-9.
11. Ramsay G, Podogrodzka M, McClure C, et al. Risk prediction in patients presenting with suspected cardiac pain: the GRACE and TIMI risk scores versus clinical evaluation. *QJM* 2007;100:11-8.
12. Lyon R, Morris AC, Caesar D, et al. Chest pain presenting to the Emergency Department--to stratify risk with GRACE or TIMI? *Resuscitation* 2007;74:90-3.

13. Graham CA, Chan JW, Chan CP, et al. Prospective validation of Thrombolysis in Myocardial Infarction and front door Thrombolysis in Myocardial Infarction risk scores in Chinese patients presenting to the ED with chest pain. *Am J Emerg Med* 2014;32:1339-44.
14. Six AJ, Backus BE, Kelder JC. Chest pain in the emergency room: value of the HEART score. *Neth Heart J* 2008;16:191-6.
15. Backus BE, Six AJ, Kelder JC, et al. Chest pain in the emergency room: a multicenter validation of the HEART Score. *Crit Pathw Cardiol* 2010;9:164-9.
16. Van Den Berg P, Body R. The HEART score for early rule out of acute coronary syndromes in the emergency department: a systematic review and meta-analysis. *Eur Heart J Acute Cardiovasc Care* 2017.
17. Backus BE, Six AJ, Kelder JC, et al. A prospective validation of the HEART score for chest pain patients at the emergency department. *Int J Cardiol* 2013;168:2153- 8.
18. Melki D, Jernberg T. HEART score: a simple and useful tool that may lower the proportion of chest pain patients who are admitted. *Crit Pathw Cardiol* 2013;12:127e31.
19. Khalil MH, Sekma A, Yaakoubi H, Bel Haj Ali K, Msolli MA, Beltaief K, et al. 30 day predicted outcome in undifferentiated chest pain: multicenter validation of the HEART score in Tunisian population. *BMC Cardiovasc Disord* 2021; 21:1e7.
20. Krishna PP, Velavarthipati RS, Srikanth M, Krishna BSG, Sriramula N, Goud DPK. To determine the prognostic accuracy of the HEART score as a predictor for major adverse cardiac events in patients presenting with chest pain to emergency department in a tertiary care hospital. *Critical Care Innovations*. 2023;6(1):1–16.
21. Rageh, Tarek M.; Elkersh, Ahmed M.; Youssef, Fatma I.; Zakaria, Nermin B.; and Tantawy, Mohamed F. Prognostic Value of HEART Score among Patients with Chest Pain at the Emergency Department," *Menoufia Medical Journal*. 2024; 37(1): 32.
22. Antman EM, Braunwald E. Acute myocardial infarction. In Braunwald E (ed.). *Heart Disease A textbook of Cardiovascular Medicine*. 6th ed. Philadelphia, WB Saunders Company, pp. 184-1189. 2001.
23. Boin AC V, Correa FG S. Epidemiological Profile for Acute Coronary Syndrome: The Difference between Genders in an Intensive Care Unit. *Journal of Hypertension-Open Access*. 2013;02(03).
24. Prabhakaran D, Jeemon P, Roy A. Cardiovascular diseases in India: current

- epidemiology and future directions. *Circulation*. 2016 Apr 19;133(16):1605-20.
25. Iqbal F, Barkataki J. Spectrum of acute coronary syndrome in North Eastern India – A study from a major center. *Indian Heart Journal*. 2016;68(2):128-131.
 26. Douglas MC, the Pathophysiology of Acute Coronary Syndromes. EMCREG-international 2005.
 27. Ross R, The pathogenesis of atherosclerosis: a perspective for the 1990s". *Nature*,1993,April;362 (6423):12
 28. Ross R; Ross, Russell "Atherosclerosis — An Inflammatory Disease". *New England Journal of Medicine*,1999,January;340 (2): 115–26.
 29. Finn AV, Nakano M, Narula J, Kolodgie FD, Virmani R ,Concept of vulnerable/unstable plaque". *Arterioscler. Thromb. Vasc. Bio*; 2010,July;30 (7): 1282–92..
 30. Didangelos A, Simper D, Monaco C, Mayr M "Proteomics of acute coronary syndromes.". *Current atherosclerosis reports*;2009, May,11 (3): 188–95.
 31. Atherosclerosis. Harvard Health Publications Harvard Health Publications. Health Topics A – Z, 2011
 32. Atherosclerosis. National Heart, Lung and Blood Institute. <http://www.nhlbi.nih.gov/health/health-topics/topics/atherosclerosis/signs.html> ,2011
 33. <http://www.news-medical.net/news/20120201/First-signs-of-atherosclerotic-heart-disease-may-appear-in-early-childhood.aspx> Retrieved Feb-27-13
 34. Flora, G., Baker, A.B., Loewenson, R.B., and Klassen, A. C. A Comparative Study of Cerebral Atherosclerosis in Males and Females. *Circulation* 38, 859-869
 35. Bonow RO, Mann DL, Zipes DP, Libby P. Braunwald's Heart Disease E-Book: A Textbook of Cardiovascular Medicine. Elsevier Health Sciences; 2011 Feb 25.
 36. Sims, N.R. and Muderman, H. Mitochondria, oxidative metabolism and cell death in stroke. *Biochimica et Biophysica Acta (BBA) - Molecular Basis of Disease* 1802, 80-91
 37. Blankenhorn DH, Hodis HN "Atherosclerosis--reversal with therapy". *The Western journal of medicine*;199, August 199;159 (2): 172–9.
 38. Mitchell, Richard Sheppard; Kumar, Vinay; Abbas, Abul K.; Fausto, Nelson . Robbins Basic Pathology: With STUDENT CONSULT Online Access (8th Ed.);2007,.345-48.

39. Cannon CP. Update to International Classification of Diseases, 9th Revision codes: distinguishes STEMI from NSTEMI. *Critical pathways in cardiology*. 2005 Dec 1;4(4):185-6.
40. Kumar A, Cannon CP. Acute coronary syndromes: diagnosis and management, part I. In *Mayo Clinic Proceedings* 2009 Oct 1 (Vol. 84, No. 10, pp. 917-938). Elsevier.
41. The Joint European Society of Cardiology/American College of Cardiology Committee. Myocardial infarction redefined. A consensus document of the joint European Society of Cardiology/American College of Cardiology Committee for the Redefinition of Myocardial Infarction. *J Am Coll Cardiol* 2000;36:959 – 69.
42. Wallentin L, Lagerqvist B, Ustved S, et al. Outcome at 1 year after an invasive compared with a non-invasive strategy in unstable coronary-artery disease: the FRISC II invasive randomised trial. *Lancet* 2000;356:9 – 16.
43. Cannon CP, Weintraub WS, Demopoulos LA, et al. Comparison of early invasive and conservative strategies in patients with unstable coronary syndromes treated with the glycoprotein IIb/IIIa inhibitor tirofiban. *N Engl J Med* 2001;344:1879 – 87.
44. Kaul P, Newby K, Fu Y, et al. Troponin T and quantitative ST-segment depression offer complementary prognostic information in the risk stratification of acute coronary syndromes. *J Am Coll Cardiol* 2003;41:371– 80.
45. Diderholm E, Andrén B, Frostfeldt G, et al. ST depression in ECG at entry indicates severe coronary lesions and large benefits of an early invasive treatment strategy in unstable coronary artery disease. The FRISC II ECG substudy. *Eur Heart J* 2002;23:41–9.
46. Hamm CW, Goldman BU, Heeschen C, Kreyman G, Berger J, Meinertz T. Emergency room triage of patients with acute chest pain by means of rapid testing for cardiac troponin T or troponin I. *N Engl J Med* 1997;337:1648 –53.
47. Newby LK, Storrow AB, Gibler WB, et al. Bedside multimarker testing of risk stratification in chest pain units. The chest pain evaluation by creatine kinase-MB, myoglobin and troponin I (CHECKMATE) study. *Circulation* 2001;103:1832–7.
48. Sanchis J, Bodí V, Llácer A, et al. Risk stratification of patients with acute chest pain and normal troponin concentrations. *Heart* 2005.
49. Six AJ, Backus BE, Kelder JC. Chest pain in the emergency room: value of the HEART score. *Neth Heart J*. 2008;16(6):191–196.
50. Than M, Cullen L, Aldous S, et al. 2-Hour accelerated diagnostic protocol to assess

- patients with chest pain symptoms using contemporary troponins as the only biomarker: the ADAPT trial. *J Am Coll Cardiol*. 2012;59(23):2091–2098.
51. Than M, Flaws D, Sanders S, et al. Development and validation of the emergency department assessment of chest pain score and 2 h accelerated diagnostic protocol. *Emerg Med Australas*. 2014;26(1):34–44.
 52. Hess EP, Brison RJ, Perry JJ, et al. Development of a clinical prediction rule for 30-day cardiac events in emergency department patients with chest pain and possible acute coronary syndrome. *Ann Emerg Med*. 2012;59(2):115–125.
 53. Pollack CV, Sites FD, Shofer FS, Sease KL, Hollander JE. Application of the TIMI risk score for unstable angina and non-ST elevation acute coronary syndrome to an unselected emergency department chest pain population. *Acad Emerg Med*. 2006;13(1):13–18.
 54. Jaffery Z, Hudson MP, Jacobsen G, Nowak R, Mccord J. Modified thrombolysis in myocardial infarction (TIMI) risk score to risk stratify patients in the emergency department with possible acute coronary syndrome. *J Thromb Thrombolysis*. 2007;24(2):137–144.
 55. Granger CB, Goldberg RJ, Dabbous OH, et al.; for the Global Registry of Acute Coronary Events Investigators. Predictors of hospital mortality in the global registry of acute coronary events. *Arch Intern Med*. 2003;163:2345–2353.
 56. Boersma E, Pieper KS, Steyerberg EW, et al.; for the PURSUIT investigators. Predictors of outcome in patients with acute coronary syndromes without persistent ST-segment elevation. Results from an international trial of 9461 patients. *Circulation*. 2000;101:2557–2567.
 57. Conti A, Poggioni C, Viviani G, et al. Risk scores prognostic implementation in patients with chest pain and nondiagnostic electrocardiograms. *Am J Emerg Med*. 2012;30(9):1719–1728.
 58. Six AJ, Cullen L, Backus BE, et al. The HEART score for the assessment of patients with chest pain in the emergency department: a multinational validation study. *Crit Pathw Cardiol* 2013;12:121-6.
 59. Poldervaart JM, Langedijk M, Backus BE, et al. Comparison of the GRACE, HEART and TIMI score to predict major adverse cardiac events in chest pain patients at the emergency department. *Int J Cardiol* 2017;227:656-61.
 60. Sun BC, Laurie A, Fu R, et al. Comparison of the HEART and TIMI Risk Scores for

Suspected Acute Coronary Syndrome in the Emergency Department. *Crit Pathw Cardiol* 2016;15:1-5.

61. Adams ST, Leveson SH. Clinical prediction rules. *BMJ* 2012;344:d8312.
62. Zhang Z, Zhang H, Khanal MK. Development of scoring system for risk stratification in clinical medicine: a step-bystep tutorial. *Ann Transl Med* 2017;5:436.
63. Backus BE, Six AJ, Kelder JC, Mast TP, van den Akker F, Mast EG, Monnink SH, van Tooren RM, Doevendans PA. Chest pain in the emergency room: a multicenter validation of the HEART Score. *Crit Pathw Cardiol*. 2010 Sep;9(3):164-9.
64. Mahler SA, Hiestand BC, Goff DC Jr, et al. Can the HEART score safely reduce stress testing and cardiac imaging in patients at low risk for major adverse cardiac events? *Crit Pathw Cardiol* 2011;10:128-33.
65. Backus BE, Six AJ, Kelder JC, Bosschaert MA, Mast EG, Mosterd A, Veldkamp RF, Wardeh AJ, Tio R, Braam R, Monnink SH, van Tooren R, Mast TP, van den Akker F, Cramer MJ, Poldervaart JM, Hoes AW, Doevendans PA. A prospective validation of the HEART score for chest pain patients at the emergency department. *Int J Cardiol*. 2013 Oct 3;168(3):2153-8.
66. Melki D, Jernberg T. HEART score: a simple and useful tool that may lower the proportion of chest pain patients who are admitted. *Crit Pathw Cardiol*
67. Six AJ, Cullen L, Backus BE, et al. The HEART score for the assessment of patients with chest pain in the emergency department: a multinational validation study. *Crit Pathw Cardiol* 2013;12:121-6.
68. de Hoog VC, Lim SH, Bank IE, et al. HEART score performance in Asian and Caucasian patients presenting to the emergency department with suspected acute coronary syndrome. *Eur Heart J Acute Cardiovasc Care* 2017.
69. Bank IE, de Hoog VC, de Kleijn DP, et al. Sex-Based Differences in the Performance of the HEART Score in Patients Presenting to the Emergency Department With Acute Chest Pain. *J Am Heart Assoc* 2017;6.pii: e005373.
70. Fesmire FM, Martin EJ, Cao Y, et al. Improving risk stratification in patients with chest pain: the Erlanger HEARTS3 score. *Am J Emerg Med* 2012;30:1829-37.
71. McCord J, Cabrera R, Lindahl B, et al. Prognostic Utility of a Modified HEART Score in Chest Pain Patients in the Emergency Department. *Circ Cardiovasc Qual Outcomes* 2017;10.pii: e003101
72. Visser A, Wolthuis A, Breedveld R, et al. HEART score and clinical gestalt have

- similar diagnostic accuracy for diagnosing ACS in an unselected population of patients with chest pain presenting in the ED. *Emerg Med J* 2015;32:595-600.
73. Leite L, Baptista R, Leitão J, Cochicho J, Breda F, Elvas L, Fonseca I, Carvalho A, Costa JN. Chest pain in the emergency department: risk stratification with Manchester triage system and HEART score. *BMC Cardiovasc Disord.* 2015 Jun 11;15:48
 74. Sakamoto JT, Liu N, Koh ZX, et al. Comparing HEART, TIMI, and GRACE scores for prediction of 30-day major adverse cardiac events in high acuity chest pain patients in the emergency department. *Int J Cardiol* 2016;221:759-64.
 75. Chen XH, Jiang HL, Li YM, et al. Prognostic values of 4 risk scores in Chinese patients with chest pain: Prospective 2-centre cohort study. *Medicine (Baltimore)* 2016;95:e4778
 76. Streitz MJ, Oliver JJ, Hyams JM, et al. A retrospective external validation study of the HEART score among patients presenting to the emergency department with chest pain. *Intern Emerg Med* 2017.
 77. Poldervaart JM, Langedijk M, Backus BE, et al. Comparison of the GRACE, HEART and TIMI score to predict major adverse cardiac events in chest pain patients at the emergency department. *Int J Cardiol* 2017;227:656-61.
 78. Van Den Berg P, Body R. The HEART score for early rule out of acute coronary syndromes in the emergency department: a systematic review and meta-analysis. *Eur Heart J Acute Cardiovasc Care* 2017.
 79. Liu N, Ng JC, Ting CE, Sakamoto JT, Ho AF, Koh ZX, Pek PP, Lim SH, Ong ME. Clinical scores for risk stratification of chest pain patients in the emergency department: an updated systematic review. *J Emerg Crit Care Med.* 2018;2:16.
 80. Byrne C, Toarta C, Backus B, Holt T. The HEART score in predicting major adverse cardiac events in patients presenting to the emergency department with possible acute coronary syndrome: protocol for a systematic review and meta-analysis. *Syst Rev.* 2018 Oct 2;7(1):148.
 81. Laureano-Phillips J, Robinson RD, Aryal S, Blair S, Wilson D, Boyd K, Schrader CD, Zenarosa NR, Wang H. HEART Score Risk Stratification of Low-Risk Chest Pain Patients in the Emergency Department: A Systematic Review and Meta- Analysis. *Ann Emerg Med.* 2019 Aug;74(2):187-203.
 82. Fernando SM, Tran A, Cheng W, Rochweg B, Taljaard M,

- Thiruganasambandamoorthy V, Kyeremanteng K, Perry JJ. Prognostic Accuracy of the HEART Score for Prediction of Major Adverse Cardiac Events in Patients Presenting With Chest Pain: A Systematic Review and Meta-analysis. *Acad Emerg Med*. 2019 Feb;26(2):140-151.
83. Arslan M, Schaap J, Rood PP, Nieman K, Budde RP, Attrach M, Dubois EA, Dedic A. HEART score improves efficiency of coronary computed tomography angiography in patients suspected of acute coronary syndrome in the emergency department. *Eur Heart J Acute Cardiovasc Care*. 2020 Feb;9(1):23-29.
 84. Torralba F, Navarro A, la Hoz JC, Ortiz C, Botero A, Alarcón F, Isaza N, Isaza D. HEART, TIMI, and GRACE Scores for Prediction of 30-Day Major Adverse Cardiovascular Events in the Era of High-Sensitivity Troponin. *Arq Bras Cardiol*. 2020 Mar 13;114(5):795-802. English, Portuguese
 85. Huang Z, Wang K, Yang D, Gu Q, Wei Q, Yang Z, Zhan H. The predictive value of the HEART and GRACE scores for major adverse cardiac events in patients with acute chest pain. *Intern Emerg Med*. 2021 Jan;16(1):193-200.
 86. Khalil MH, Sekma A, Yaakoubi H, Bel Haj Ali K, Msolli MA, Beltaief K, et al. 30 day predicted outcome in undifferentiated chest pain: multicenter validation of the HEART score in Tunisian population. *BMC Cardiovasc Disord* 2021; 21:1e7.
 87. Krishna PP, Velavarthipati RS, Srikanth M, Krishna BSG, Sriramula N, Goud DPK. To determine the prognostic accuracy of the HEART score as a predictor for major adverse cardiac events in patients presenting with chest pain to emergency department in a tertiary care hospital. *Critical Care Innovations*. 2023;6(1):1–16.
 88. Rageh, Tarek M.; Elkersh, Ahmed M.; Youssef, Fatma I.; Zakaria, Nermin B.; and Tantawy, Mohamed F. Prognostic Value of HEART Score among Patients with Chest Pain at the Emergency Department," *Menoufia Medical Journal*. 2024; 37(1): 32.
 89. Goodacre SW, Bradburn M, Mohamed A, et al. Evaluation of Global Registry of Acute Cardiac Events and Thrombolysis in Myocardial Infarction scores in patients with suspected acute coronary syndrome. *Am J Emerg Med* 2012;30:37-44.
 90. Akula SP, Rao DS. A study of Platelet indices in acute myocardial infarction: An observational study. *Indian Heart Journal*. 2015 Dec 1;67:S45.
 91. Reddy SK, Shetty R, Marupuru S, Yedavalli N, Shetty K. Significance of platelet volume indices in STEMI patients: a case-control study. *Journal of Clinical and Diagnostic Research: JCDR*. 2017 Apr;11(4):LC05.

92. Senthil KP. Clinical profile of acute inferior wall myocardial infarction in a semi urban population in India. *Journal of Medicine and Medical Sciences*. 2013;4(1):17-21.
93. Nour MK. Significance of reciprocal ST segment depression in ST elevation myocardial infarction. *The Egyptian Journal of Critical Care Medicine*. 2017 Apr 30;5(1):23-7.
94. Rosano GMC, Chierchia SL, Leonardo F, Beale CM, Collins P. Cardioprotective effects of ovarian hormones. *Eur Heart J*. 1996;17(Suppl D):15–19.
95. Mahmoodi, Khalil, et al. "The efficacy of hydration with normal saline versus hydration with sodium bicarbonate in the prevention of contrast-induced nephropathy." *Heart views: the official journal of the Gulf Heart Association* 15.2 (2014): 33.
96. Siddiqui AH, Kayani AM. Acute myocardial infarction-Clinical profile of 1000 cases. *Pakistan Heart Journal*. 2012 May 15;32(1-4).
97. Burkett E, Marwick T, Thom O, et al. A comparative analysis of risk stratification tools for emergency department patients with chest pain. *Int J Emerg Med* 2014;7:10.
98. Huang, C., et al. (2017). The association between ECG changes and the risk stratification in patients with acute coronary syndrome. *International Journal of Cardiology*, 232, 245-251.
99. Mora, S., et al. (2017). "Impact of diabetes and hypertension on cardiovascular risk." *The American Journal of Medicine*, 130(2), 113-120.
100. Zhang, X., et al. (2020). "Cardiac troponin and long-term outcomes in patients with stable coronary artery disease." *Circulation*, 142(3), 242-252.

PROFORMA

NAME OF THE PATIENT:

- **AGE:**
- **SEX:**
- **ADDRESS:**
- **IP/UMR NUMBER :**
- **DATE OF ADMISSION:**
- **CHIEF COMPLAINTS:**
- **HISTORY OF PRESENT ILLNESS:**

SOB	CHEST DISCOMFORT	PALPITATIONS	SYNCOPE
VOMITING	NAUSEA	SWEATING	ARM,NECK,JAWPAIN
DIZZINESS	ALTERED MENTAL STATUS	GENERALISED WEAKNESS	EPIGASTRIC DISCOMFORT

• **PAST HISTORY**

HTN	DM	SMOKING	OBESITY	ALCOHOL
------------	-----------	----------------	----------------	----------------

FAMILY HISTORY

>3RISK FACTORS	2
1 OR 2 RISK FACTORS	1
NO RISK FACTORS	0

VITALS

HEART RATE	BP	SPO2	GRBS	GCS
-------------------	-----------	-------------	-------------	------------

ECG

SIGNIFICANT ST DEPRESSION	2
NON SPECIFIC REPOLARISATION	1
NORMAL	0

TROPONINS

PATIENT INFORMATION SHEET

Title : To determine the prognostic accuracy of the HEART Score as a predictor for Major Adverse Cardiac Events in patients presenting with chest pain to the Emergency department in a tertiary care hospital

Investigator : DR.Shaik Karishma Banu

Qualification: Primary DNB, Emergency Medicine

Institute : Krishna institute of medical sciences

Contact 8179488295

You are being asked to participate in a clinical research study conducted at the Krishna Institute of Medical Sciences, Secunderabad. Please go through the following information carefully before you decide whether you want to take part in this study or not. Feel free to ask if you need more information about what is stated in this form and the study as a whole.

- 1) The main objective of this study is To determine the prognostic accuracy of the HEART Score as a predictor for Major Adverse Cardiac Events in patients presenting with chest pain to the Emergency department in a tertiary care hospital.
- 2) This is an prospective study which will be carried out from May23-November2024
- 3) Your participation in this research is entirely voluntary.

Whether you choose to participate or not, all the services you receive at this hospital will continue and nothing will change. You may change your mind later and stop participating even if you agreed earlier. If you agree to participate in this research study, the information will be kept confidential. Unless required by law, only the study doctor and designee, the ethics committee will have direct access to your medical records. In case of any adverse event during period of the study, there will be no compensation to cover such expenses in case it is not covered by a third party or governmental insurance.

Investigator's Statement:

I have reviewed, in detail, the informed consent document for this research study with _____
_____the purpose of the study and its risks and benefits. I have answered to
all the patient's questions clearly. I will inform the participant in case of any changes to the
research study.

Name of Investigator or designee

signature

Date:

INFORMED CONSENT

Subject identification number of this trial /study:

Title of the Project:

Name of the Principal Investigator: _____ Telephone no

I have received the information sheet on the above study and have read and/ or understood the written information.

I have been given the chance to discuss the study and ask questions. I consent to take part in the study and I am aware that my participation is voluntary. I understand that I may withdraw at any time without this affecting my future care. I understand that information collected about me from my participation in this research and sections of any of my medical notes may be looked at by responsible persons (ethics committee members / regulatory authorities).

I give my access to these individuals to have access to my records. I understand I will receive a copy of the patient information sheet and informed consent form.

Signature/Thumb Impression of subject

Date of signature

Printed name of the subject in capitals

Signature / Thumb Impression of legally
accepted representative

Date of signature

(The legally acceptable representative signature should be added if the subject is a minor or is unable to sign for themselves. The relationship between the subject and the

legally acceptable representatives should be stated. The impartial witness signature should be added if the subject / legally acceptable representative is unable to read or write and consent should be obtained in his presence)

_____Printed name of legally acceptable representative in capitals

_____Relationship of legally accepted representative to subject in capitals

Signature of the person conducting
signature the Informed consent discussion

Date of

Printed name of the person conducting the Informed consent discussion in capitals

Signature of impartial witness

Date of signature

Printed name of the impartial witness in capitals

Sno	IP NO	Age	Age Group	Sex	Giddine s	General ised Weakne ss	Syncop e	Epigastric Discomfort	Nausea	Vomitin g	Altered Mental Status	History	ECG	HTM	DM	Smokin g	Obesity	Family History	Alcohol	Risk Factors	TROPON IN	Total Source
1	7241	33	2	2	2	1	2	2	2	2	2	2	1	1	1	1	1	2	2	3	1	3
2	9465	43	2	1	1	2	2	2	1	1	1	2	2	1	1	1	2	2	2	3	3	2
3	5295	37	3	2	2	1	1	1	1	2	2	2	2	2	1	2	1	1	2	1	3	3
4	4136	40	3	1	2	1	2	1	2	1	2	2	2	1	2	2	2	2	2	1	1	2
5	1346	29	2	2	2	2	2	2	2	2	2	3	3	1	1	1	2	2	2	3	2	1
6	9381	38	3	1	2	1	2	2	2	2	2	3	2	1	1	1	1	2	2	3	2	1
7	5774	41	1	1	2	1	2	2	2	1	2	3	2	1	2	1	2	2	1	2	1	1
8	7925	30	2	2	1	1	2	2	1	1	2	3	2	1	1	2	1	2	1	2	3	2
9	5891	18	2	2	2	1	2	1	1	1	2	2	2	2	2	1	2	2	2	3	1	1
10	4126	31	3	2	1	1	2	1	2	2	2	3	2	1	1	1	1	2	1	2	1	3
11	8999	43	3	2	1	1	2	1	2	2	1	3	3	2	1	2	2	2	2	3	3	3
12	6947	22	2	2	2	1	2	1	1	1	2	3	2	1	2	2	2	2	2	1	3	1
13	8176	19	2	1	1	1	2	2	1	2	1	3	1	2	2	1	2	2	2	3	1	1
14	9788	34	2	2	1	1	2	2	2	1	2	3	2	1	1	2	2	2	2	3	3	2
15	5885	36	3	1	2	2	2	2	2	2	2	3	1	2	2	1	1	2	2	3	3	1
16	3583	28	2	2	1	1	2	1	1	2	2	3	2	1	1	2	2	2	2	2	2	3
17	4675	24	2	1	1	1	2	1	1	1	1	2	2	2	2	2	2	2	2	3	3	3
18	1983	33	3	1	1	1	2	2	1	2	2	3	2	1	1	2	2	2	2	3	1	3
19	9442	26	1	1	2	1	2	2	1	1	2	2	3	1	1	2	1	2	2	1	1	2
20	9466	34	2	1	1	2	2	2	2	1	1	3	2	2	2	2	2	2	2	2	3	1
21	1508	27	2	2	1	1	2	2	1	2	2	2	3	1	2	2	2	2	1	2	3	2
22	9218	31	2	1	1	1	2	1	1	1	2	3	2	2	1	2	1	2	2	2	2	1
23	7100	18	3	2	1	1	2	2	1	2	2	2	2	2	1	1	2	2	2	2	2	3
24	8862	21	2	2	2	1	2	1	1	2	2	1	2	1	1	2	2	2	2	3	3	3
25	9225	29	2	1	1	2	2	1	1	2	1	3	2	2	2	1	1	2	2	3	1	2
26	8964	28	2	1	2	2	2	1	1	1	2	2	2	1	2	2	2	2	2	2	3	1
27	8562	44	2	1	2	1	2	2	2	2	2	3	1	2	2	2	1	2	1	2	1	1
28	8808	31	2	2	1	2	2	1	2	1	2	3	2	1	1	2	1	2	2	1	2	2
29	9744	30	3	2	1	1	2	2	1	1	2	3	2	2	2	1	2	2	2	2	3	2
30	7961	43	3	2	2	1	1	1	2	1	2	3	2	1	2	2	1	2	2	2	1	3
31	1940	34	2	2	1	2	2	2	2	1	1	2	2	1	1	2	2	2	2	3	1	2
32	7798	26	1	1	1	2	2	2	1	1	1	2	1	1	1	2	2	2	2	2	2	1
33	1050	33	2	1	2	1	1	2	1	1	2	3	3	1	1	2	1	2	2	2	3	2
34	8293	35	2	2	1	1	2	1	2	1	1	3	2	2	2	2	2	2	2	3	1	2
35	8063	32	2	2	2	1	2	2	1	2	2	3	2	1	1	2	2	2	2	3	3	1
36	3921	20	1	1	2	1	2	1	1	2	2	3	2	1	2	2	1	2	2	2	2	2
37	1560	25	3	2	1	2	2	2	1	1	2	3	2	1	1	2	2	2	2	2	3	3
38	7597	27	2	2	2	1	2	2	1	1	2	3	2	1	1	2	2	2	2	2	2	2
39	9067	31	2	1	2	2	2	1	2	2	2	2	3	1	2	2	1	2	2	3	3	3
40	4547	42	2	1	2	1	2	2	2	2	2	3	2	2	2	2	1	2	2	2	1	2
41	1329	26	2	1	1	1	2	2	1	1	2	3	3	1	1	2	2	2	2	2	2	2
42	6832	37	2	1	2	1	1	1	1	2	2	3	2	2	2	2	2	2	2	2	2	3
43	4304	41	1	1	1	1	2	2	2	1	2	2	2	1	2	2	2	2	2	3	1	2

44	5632	36	2	1	2	2	2	1	1	1	1	3	2	2	1	2	1	2	2	3	2	3
----	------	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

45	9072	39	1	1	2	1	2	1	1	1	1	3	2	2	2	2	2	2	2	3	2
46	2748	42	2	2	2	2	2	1	1	1	2	3	2	2	2	2	2	2	1	3	2
47	2438	23	3	1	1	2	2	1	2	2	2	3	2	2	1	2	1	2	2	1	2
48	9662	23	1	2	2	1	2	2	1	2	2	2	1	2	2	2	2	2	1	1	1
49	2632	37	1	2	1	2	2	2	1	2	2	3	3	2	2	1	2	2	3	2	2
50	3158	42	2	2	2	1	2	2	2	2	1	3	2	2	1	1	1	2	1	3	2
51	5557	31	3	1	2	1	2	1	1	1	2	3	2	1	1	2	2	2	2	3	3
52	5998	35	3	1	1	2	1	1	2	2	2	2	2	1	2	2	2	2	2	2	3
53	5505	34	2	2	1	2	1	1	2	1	2	3	2	1	2	2	1	2	3	1	3
54	5727	33	1	1	2	1	2	1	2	1	1	3	2	1	1	2	2	2	3	3	2
55	1427	27	1	2	1	1	2	2	1	2	1	3	2	2	2	2	2	1	2	3	2
56	1863	35	3	1	2	1	2	1	1	1	2	3	2	2	2	2	1	2	3	1	2
57	8860	38	3	1	1	1	2	2	2	2	1	3	3	2	1	2	2	2	2	2	3
58	6929	31	2	1	2	1	2	1	2	1	2	3	2	1	1	2	2	2	3	1	1
59	4333	19	3	2	2	2	2	1	1	2	1	3	2	2	2	2	1	2	1	3	2
60	1952	34	3	2	1	2	2	2	2	2	1	3	2	1	2	1	2	2	2	3	3
61	2983	23	1	1	2	1	2	2	2	2	2	3	1	2	1	2	2	2	2	3	1
62	3953	30	3	1	2	1	2	1	2	2	1	2	2	1	1	2	1	2	2	2	2
63	1487	22	3	2	2	2	2	2	2	1	2	3	2	1	2	2	2	2	2	1	2
64	8564	20	1	1	1	2	2	2	2	1	2	2	2	1	1	1	2	2	3	1	1
65	7899	24	2	1	2	2	2	2	2	1	2	3	2	1	2	2	1	2	3	2	3
66	7870	30	1	2	1	2	2	2	2	2	1	3	2	2	1	2	1	2	1	2	3
67	2584	29	2	1	1	1	2	1	2	2	2	1	2	2	2	1	2	2	3	3	3
68	4076	21	2	1	1	2	2	1	2	1	1	3	2	2	2	2	2	1	2	3	3
69	8708	44	2	1	1	2	2	1	1	2	2	3	2	1	1	2	1	2	3	3	3
70	5192	25	3	1	2	2	2	2	1	2	2	3	1	2	1	2	2	1	2	2	2
71	4188	19	1	1	1	2	2	1	1	2	2	3	2	1	1	1	2	2	3	1	2
72	7582	41	1	2	1	2	2	1	2	2	1	2	3	2	2	2	1	2	3	1	2
73	9532	59	1	2	2	2	2	2	1	1	1	3	2	2	2	2	2	2	2	3	1
74	6999	58	2	1	1	1	2	1	2	2	1	3	2	1	2	2	2	1	1	2	3
75	4395	55	3	2	1	1	2	2	1	2	1	3	1	2	1	1	1	2	3	3	3
76	9219	51	2	1	1	1	1	2	1	1	2	3	2	1	2	1	2	2	2	1	3
77	3399	48	2	1	1	2	2	1	2	1	1	2	1	2	2	1	2	1	3	1	1
78	2679	56	2	1	1	1	2	2	2	1	2	3	2	1	2	2	1	2	2	3	3
79	9736	45	1	2	1	2	2	1	2	1	1	3	1	1	1	2	2	2	3	3	3
80	1867	56	2	2	1	1	2	2	2	1	1	3	3	2	1	2	1	2	3	3	1
81	7564	50	3	1	1	2	2	1	2	1	1	2	2	1	2	2	2	1	2	3	1
82	1801	56	1	1	1	1	2	1	1	2	2	3	2	1	2	2	1	2	2	3	1
83	9456	64	2	1	1	1	2	2	2	2	2	3	3	2	1	2	2	2	2	1	3
84	2695	46	2	2	2	2	2	2	2	2	2	3	2	1	2	2	2	2	2	2	1
85	8518	56	2	1	2	2	2	1	2	2	2	1	1	2	1	2	2	2	3	1	1
86	3311	62	2	2	1	1	2	1	2	1	1	3	2	2	2	2	1	2	1	3	3
87	8193	54	2	1	1	1	2	2	1	1	2	3	1	1	1	2	2	2	2	1	2
88	7685	45	3	1	1	1	2	2	1	2	2	3	2	2	2	2	2	1	2	1	1
89	4973	62	1	2	2	1	2	1	2	2	1	3	2	2	2	1	1	2	3	1	2
90	1943	62	1	2	1	2	2	1	2	2	1	3	2	2	1	1	2	2	2	3	2

91	5396	48	2	1	1	2	2	1	2	1	1	3	1	1	1	2	2	2	1	2	3	2
92	1079	48	1	2	1	2	2	1	1	1	2	3	1	1	1	1	1	2	1	3	1	1
93	5317	45	3	2	1	1	2	2	1	1	2	3	2	1	1	2	2	2	1	3	2	3
94	8222	58	2	1	1	1	2	1	2	2	1	3	2	2	2	2	2	2	2	2	1	2
95	3897	48	2	1	1	1	2	1	1	1	2	3	2	1	1	2	1	1	2	2	1	2
96	8114	48	3	1	1	2	2	1	2	2	2	3	1	1	1	2	2	2	2	2	3	1
97	1214	46	2	2	1	1	2	2	2	2	2	2	1	1	2	1	2	2	2	3	1	2
98	8401	55	2	1	2	1	2	1	2	1	2	3	2	2	2	2	2	2	2	2	2	3
99	2326	57	1	1	1	1	2	1	2	1	1	3	3	1	1	1	2	2	2	2	3	1
100	4718	57	2	1	1	1	2	1	1	2	2	3	1	1	2	2	2	2	2	3	1	2
101	2346	48	2	2	1	1	2	2	1	2	1	3	1	2	1	2	1	2	2	3	1	3
102	1871	46	2	2	2	2	2	2	2	1	2	3	1	2	2	2	2	2	2	3	1	3
103	7186	51	2	1	1	1	2	2	1	2	2	3	1	2	2	2	2	2	2	3	1	2
104	8104	64	2	2	1	1	2	2	1	1	1	2	2	2	2	1	2	1	2	1	1	3
105	7305	62	3	2	1	1	2	2	2	1	2	3	2	2	1	2	2	2	2	2	2	2
106	6153	58	2	2	1	1	2	1	1	1	2	3	2	2	1	1	2	2	2	2	3	1
107	8333	47	3	2	1	1	2	1	1	1	2	3	1	2	2	2	1	2	2	1	1	3
108	8097	51	3	2	1	2	2	2	1	1	2	3	2	1	2	1	2	2	2	2	1	1
109	2473	59	2	2	1	1	1	2	1	2	2	2	1	1	1	2	2	2	2	2	3	2
110	6359	47	3	1	1	1	2	1	2	1	2	3	2	2	1	2	1	2	2	2	1	3
111	3777	46	2	1	2	2	2	1	2	2	1	2	1	2	1	2	2	2	2	2	1	3
112	1752	59	2	2	1	1	2	1	1	1	1	3	1	1	2	2	2	2	2	3	2	3
113	3339	48	3	1	2	2	2	2	1	1	2	3	2	2	2	2	2	2	1	2	2	1
114	2923	53	2	2	2	2	2	1	1	2	1	3	1	2	1	2	1	2	2	3	1	1
115	6763	58	1	1	2	2	2	1	2	1	1	3	3	1	1	2	2	2	1	3	2	1
116	3619	48	2	1	1	1	2	2	2	2	2	2	2	2	1	2	2	2	1	2	1	2
117	2643	48	2	1	1	1	2	1	1	2	2	3	1	2	2	2	1	2	2	2	2	3
118	6358	48	2	1	2	1	2	2	2	1	2	2	3	1	1	2	2	2	2	3	1	3
119	7531	63	1	2	2	2	2	2	1	1	2	3	2	2	1	2	1	2	2	2	1	2
120	9919	55	2	1	2	2	2	1	1	2	2	3	3	2	2	2	2	2	2	3	1	1
121	3605	45	3	1	1	1	2	1	1	2	2	2	2	2	1	1	2	2	2	2	3	3
122	7666	47	3	2	2	2	2	1	2	1	2	3	3	1	1	2	1	2	2	1	1	3
123	7155	59	2	2	1	1	2	2	1	1	1	2	2	1	2	2	2	2	2	2	1	2
124	7239	56	3	1	1	2	2	1	1	1	1	3	2	1	2	2	2	2	1	2	2	3
125	2753	57	3	1	2	1	2	2	2	2	2	3	1	2	1	2	2	2	2	2	2	2
126	7282	47	3	1	2	1	2	1	2	1	2	3	1	1	1	2	1	2	2	2	3	3
127	9243	57	2	1	2	1	2	1	2	1	2	3	1	2	2	2	2	2	1	3	1	3
128	1194	56	3	2	1	2	2	2	2	2	2	3	1	1	1	1	2	2	2	2	1	3
129	8307	46	2	1	2	1	2	2	2	2	2	3	1	1	2	2	2	2	1	2	1	2
130	4539	49	3	2	2	1	2	1	2	1	2	3	2	1	1	2	2	2	2	1	2	3
131	6412	61	3	2	2	1	2	2	2	1	1	3	2	2	2	2	1	2	2	3	1	2
132	8428	58	2	1	1	1	2	1	2	2	2	3	2	2	1	2	2	2	2	3	1	3
133	3100	57	3	2	2	1	2	1	2	2	2	3	3	2	2	2	1	2	1	3	1	3
134	6664	55	1	1	2	1	2	2	2	2	2	2	2	1	2	1	1	2	2	2	1	1
135	8252	63	2	1	2	1	2	2	1	2	2	2	3	1	1	2	2	2	2	2	3	3
136	9393	54	3	2	2	1	2	1	1	1	2	3	2	2	1	1	2	1	2	2	1	2

137	1381	57	2	2	1	2	2	2	1	1	2	3	1	2	1	2	1	2	2	3	1	2
138	9096	53	2	1	1	1	2	1	1	2	2	2	2	2	2	2	1	2	2	2	1	3
139	4013	50	2	1	1	2	2	1	1	2	1	3	1	1	1	1	2	2	1	2	3	2
140	6894	58	2	1	1	2	2	2	1	1	2	3	3	2	1	2	2	2	2	2	1	2
141	8571	59	1	2	1	1	2	2	2	1	2	3	1	1	2	1	1	2	1	2	2	3
142	4711	55	3	1	2	2	2	1	2	2	2	3	2	1	1	1	2	2	2	2	1	2
143	2809	64	1	2	1	1	2	2	2	1	1	2	1	2	2	1	2	2	2	3	1	3
144	1662	51	2	1	1	1	2	1	2	2	1	2	3	1	1	2	1	2	1	1	2	3
145	2219	60	1	1	2	2	2	2	2	1	2	3	2	1	2	2	2	2	2	2	2	1
146	4118	62	1	1	1	2	2	2	2	2	2	2	2	1	1	2	2	2	2	3	1	2
147	2933	46	2	2	2	2	2	1	1	1	2	3	3	2	1	2	1	2	2	2	3	2
148	5031	58	3	1	1	2	2	2	2	2	1	3	1	2	2	2	2	2	1	3	1	3
149	9000	50	1	2	2	2	2	1	1	1	2	2	2	2	1	2	2	2	2	2	1	1
150	9303	56	2	2	2	1	2	2	2	1	1	3	1	1	1	2	2	2	2	1	1	3
151	8699	49	2	1	1	1	2	2	2	1	2	3	3	2	1	2	1	2	2	3	1	3
152	4469	51	2	2	1	1	2	1	2	1	1	3	2	2	2	2	2	2	2	3	1	3
153	9881	54	2	2	2	1	2	2	1	1	2	3	2	1	1	2	2	2	2	2	1	3
154	4279	60	2	1	2	2	2	1	2	1	2	2	1	2	1	2	1	2	2	2	1	1
155	4241	56	3	1	1	2	2	2	1	2	2	3	3	2	2	2	2	2	2	2	1	2
156	6079	55	2	1	1	2	2	1	2	2	2	3	3	2	2	2	2	2	1	2	1	2
157	7172	59	2	1	1	1	2	2	1	1	2	3	2	2	1	1	1	2	1	2	1	2
158	5296	49	1	1	2	2	2	2	1	2	1	3	2	2	1	2	2	1	2	2	3	3
159	1540	57	2	2	2	2	2	1	2	1	1	3	1	2	1	1	2	2	1	2	2	3
160	1538	58	2	1	2	1	2	1	1	2	2	3	1	1	2	2	2	2	2	2	3	2
161	6533	47	2	1	1	2	2	2	2	2	2	2	2	2	1	2	2	2	2	2	3	1
162	2843	52	2	2	1	1	2	2	2	2	2	3	3	2	1	1	1	1	2	2	1	2
163	5038	46	1	1	1	1	2	2	1	1	2	3	2	1	1	2	2	2	2	3	1	2
164	7002	59	1	1	2	1	2	2	1	2	1	3	2	1	1	1	2	2	1	2	3	1
165	7344	64	3	1	2	1	2	1	1	2	1	2	1	1	2	2	1	2	1	1	1	1
166	2189	53	3	1	1	2	2	1	1	1	1	3	2	1	1	2	2	2	2	1	1	2
167	2468	63	2	2	1	2	2	2	1	1	1	1	1	2	2	2	2	2	2	2	1	2
168	9941	48	3	2	1	1	2	1	1	2	1	3	2	2	1	1	1	2	2	3	1	3
169	6215	45	2	1	1	2	2	1	2	2	1	3	2	2	1	2	1	2	2	3	3	1
170	9802	62	2	1	1	1	2	1	2	2	2	3	3	1	2	2	2	1	2	2	3	3
171	6899	62	2	1	2	2	2	1	2	2	1	3	2	2	1	2	2	2	2	2	3	1
172	6752	61	3	2	1	2	2	1	1	1	1	3	1	1	2	2	2	2	2	1	2	3
173	5290	52	2	1	1	1	2	2	1	1	1	2	1	2	2	2	2	2	2	2	1	1
174	4080	61	3	1	1	1	2	1	1	1	1	2	2	2	1	2	2	2	2	3	3	2
175	8564	61	1	1	1	2	2	2	2	1	2	3	3	2	1	2	2	2	2	3	1	1
176	8824	60	2	2	1	1	2	2	1	1	2	3	1	2	2	2	2	2	2	2	1	2
177	8572	62	2	2	1	1	2	1	2	1	2	3	2	2	2	2	2	2	2	2	1	3
178	3892	60	3	1	1	1	2	2	2	1	2	2	1	2	1	2	2	1	2	1	3	3
179	2447	58	2	2	2	1	2	2	2	1	2	2	1	2	1	2	2	2	2	2	3	2
180	6676	51	2	1	2	2	2	1	2	2	2	3	1	1	1	2	2	2	2	1	3	2
181	2213	63	3	1	2	2	2	1	1	2	2	3	2	2	2	2	2	2	1	2	2	1
182	8983	51	2	1	2	1	2	1	1	1	2	3	1	2	1	2	2	2	2	2	1	1

183	6663	60	2	2	1	1	2	1	1	2	2	3	2	2	1	2	2	2	2	3	1	3
184	4508	47	3	1	1	2	1	2	2	2	2	3	2	2	1	1	2	2	2	2	1	3
185	5592	60	2	1	1	1	2	1	2	2	1	2	3	1	2	1	2	2	2	3	1	2
186	2942	53	2	2	1	2	1	2	1	1	2	3	2	2	2	2	2	2	2	2	1	2
187	4123	55	1	1	1	1	2	1	2	1	1	2	2	2	1	2	2	2	2	1	2	3
188	7934	47	1	1	1	1	2	2	2	1	2	3	2	1	1	2	2	2	2	2	3	3
189	9533	50	1	2	2	1	2	1	1	1	2	3	2	2	2	2	2	2	2	2	1	2
190	3208	51	2	1	2	1	2	2	2	1	2	3	2	1	1	1	2	2	2	2	2	2
191	9672	58	3	1	1	1	2	2	1	2	2	2	1	2	1	2	2	2	2	1	1	1
192	8828	51	3	1	1	1	2	2	2	1	1	3	2	2	2	1	2	2	2	1	1	3
193	7378	47	2	1	1	2	2	2	2	2	2	3	3	1	2	2	2	2	2	3	1	3
194	1237	63	2	2	2	2	1	1	2	2	2	2	3	2	1	2	2	2	1	2	1	2
195	8523	45	1	1	1	2	2	1	1	1	2	3	2	1	2	2	2	2	2	2	1	2
196	6290	48	1	1	2	1	2	2	1	1	2	3	1	2	1	2	2	2	2	2	3	3
197	9420	49	3	2	2	2	2	1	2	2	2	3	2	2	1	2	2	2	2	1	1	1
198	6301	53	2	1	2	1	1	2	2	2	2	3	3	1	1	1	2	2	1	1	1	2
199	9245	56	3	2	1	1	2	2	2	2	1	3	1	2	1	2	2	2	2	2	1	2
200	3854	61	2	2	1	1	2	1	2	2	1	3	3	1	1	2	2	2	2	3	1	2
201	3969	54	2	1	2	1	2	2	2	2	2	1	2	2	2	2	2	2	2	2	1	3
202	9886	48	2	1	1	1	2	2	2	2	1	3	1	2	1	2	2	2	2	1	1	2
203	2656	45	2	1	2	2	2	1	2	1	1	3	1	1	2	2	2	2	1	1	3	2
204	6281	51	3	1	2	1	2	1	1	1	2	3	1	1	1	2	2	2	2	2	1	3
205	2249	62	2	2	2	1	2	2	1	1	2	2	1	2	1	2	2	2	2	3	1	2
206	7622	48	2	2	1	1	2	1	2	1	2	3	2	2	2	2	2	2	2	2	3	2
207	4432	58	2	2	2	2	2	1	2	2	2	3	3	2	1	2	2	2	2	2	1	1
208	8796	52	1	1	2	2	1	2	2	2	2	3	1	1	2	2	2	2	2	3	1	3
209	1538	50	3	2	2	2	2	1	2	2	1	2	1	2	1	2	2	2	2	3	1	2
210	2398	48	2	1	1	1	2	2	1	2	2	3	3	2	2	1	2	2	2	1	1	2
211	5940	48	2	2	1	1	2	2	1	2	2	3	2	2	1	2	2	2	2	3	1	2
212	1090	55	3	1	1	1	2	2	1	1	2	3	2	2	1	1	2	1	2	3	1	2
213	7693	64	2	2	2	1	1	1	1	1	2	3	1	2	1	1	2	2	2	2	1	3
214	4582	63	2	2	1	1	2	1	2	2	2	3	2	1	2	1	2	2	2	1	1	2
215	7380	55	3	1	1	1	2	2	2	2	2	3	1	1	1	2	2	2	1	2	1	2
216	4411	64	3	1	1	1	2	2	2	2	1	3	2	1	2	1	2	2	1	3	3	2
217	5182	47	3	2	2	1	2	2	2	1	2	3	3	1	1	1	2	2	2	1	1	2
218	3709	62	2	1	2	2	2	2	1	1	2	3	3	1	2	1	2	2	2	2	1	2
219	2341	50	2	1	1	2	2	2	1	1	1	3	3	1	1	2	2	2	2	3	1	1
220	4762	45	2	2	1	2	2	1	1	1	1	3	2	2	1	2	2	2	2	3	1	2
221	5387	45	1	1	2	2	2	2	1	2	2	2	3	2	2	2	2	2	2	2	1	2
222	8878	52	1	1	1	2	2	2	1	1	2	3	2	2	1	2	2	2	2	1	1	1
223	8362	60	2	1	1	2	2	1	1	1	2	3	2	1	2	2	2	2	2	3	3	2
224	5510	64	2	1	1	1	2	2	1	2	1	3	2	2	1	2	2	2	2	2	3	1
225	9113	55	2	1	2	2	2	1	2	1	1	3	1	1	2	1	2	2	2	3	1	2
226	8788	51	3	1	1	2	2	2	1	2	1	3	3	1	2	2	2	2	2	2	3	2
227	6959	55	3	2	1	1	2	2	1	1	2	2	3	1	1	2	2	2	2	1	1	1
228	4906	64	2	1	2	2	2	2	2	2	1	3	2	1	1	2	2	2	2	3	1	3

229	7849	62	3	2	1	1	2	2	1	2	1	3	2	2	1	2	2	2	1	2	1	3
230	2808	52	3	1	1	1	2	2	2	1	1	3	3	2	2	2	2	2	2	1	2	2
231	3670	61	2	1	2	2	2	2	1	1	2	2	1	2	1	2	2	2	2	2	1	1
232	8218	54	1	1	2	1	2	2	1	2	2	3	3	2	1	1	2	2	2	2	3	1
233	4089	61	2	2	2	2	2	1	2	2	2	2	2	1	2	2	2	2	1	3	1	1
234	9580	57	2	1	2	2	2	2	2	2	2	3	1	2	2	2	2	2	2	3	1	1
235	9274	57	2	1	2	1	2	1	1	2	2	3	2	1	1	2	2	2	2	2	1	1
236	1476	54	3	1	1	1	2	2	1	1	1	3	3	1	1	2	2	2	2	2	1	2
237	5036	63	3	1	1	1	2	2	1	1	1	3	3	1	2	2	2	2	2	3	1	2
238	5338	53	2	1	1	1	2	2	1	1	2	3	2	1	1	2	2	2	2	2	3	3
239	7112	46	1	1	1	1	2	1	2	1	1	1	1	1	2	2	2	2	1	2	3	3
240	5984	48	3	1	1	1	2	1	1	1	1	3	2	1	1	2	2	2	2	2	1	3
241	3093	60	2	1	2	1	2	2	1	1	1	3	2	1	1	2	2	2	2	1	3	2
242	5507	52	1	1	1	1	2	1	1	2	1	3	1	1	1	2	2	2	2	1	1	1
243	3555	47	1	1	1	1	2	2	1	2	2	3	2	2	2	2	2	2	1	2	1	3
244	9078	56	3	1	2	2	2	1	1	2	1	3	2	2	1	2	2	2	2	2	1	2
245	2092	48	2	1	1	1	2	2	1	1	2	2	2	2	1	2	2	2	2	2	3	2
246	6786	45	1	1	2	2	2	2	2	2	1	3	1	2	2	1	2	2	2	2	1	1
247	5210	64	3	2	1	1	2	2	1	2	2	3	2	2	2	2	2	2	2	1	1	2
248	4813	51	2	1	2	2	2	1	1	1	2	3	2	1	1	2	2	2	1	2	1	1
249	2157	51	1	1	1	1	2	2	2	2	1	2	1	2	1	2	2	2	2	1	3	3
250	7104	46	1	2	1	1	2	1	1	2	2	2	3	1	2	1	2	2	2	3	1	2
251	3431	51	2	1	1	2	2	2	2	2	1	2	2	1	1	2	2	2	2	1	1	3
252	2895	54	1	1	1	1	2	1	2	2	2	3	1	2	1	1	2	1	2	2	3	2
253	8048	51	3	2	2	2	2	1	2	1	2	3	3	1	1	2	2	2	2	3	2	1
254	3214	61	3	2	2	2	2	1	2	2	2	3	1	2	2	2	2	2	2	1	1	1
255	1938	58	2	1	1	1	2	1	2	2	2	3	3	2	1	2	2	2	2	1	1	3
256	5195	57	2	2	2	2	2	2	2	2	1	3	2	1	2	2	2	2	2	3	1	1
257	8864	52	3	2	1	2	2	1	1	1	1	2	1	2	2	1	2	2	2	2	3	3
258	6257	57	2	2	2	2	2	2	2	2	2	2	2	2	1	1	2	2	1	2	1	3
259	6444	55	3	1	1	1	2	2	1	1	2	2	3	1	1	2	2	2	2	2	3	2
260	4825	60	2	2	2	2	2	2	2	1	1	3	1	2	1	2	2	2	1	2	3	3
261	4131	57	1	1	1	2	2	1	1	2	2	3	2	2	2	2	2	2	1	2	1	3
262	1576	48	1	2	2	2	2	2	1	2	2	3	2	2	2	1	2	2	2	2	1	1
263	5856	59	2	1	1	1	2	1	2	1	2	3	1	2	2	1	2	2	2	1	2	2
264	8752	53	2	2	1	1	2	1	2	2	1	3	3	2	2	2	2	2	1	1	1	2
265	7352	62	2	2	1	1	2	2	1	2	1	3	1	2	1	2	2	2	2	1	1	2
266	8726	64	1	2	1	2	2	1	1	1	2	3	2	1	1	1	2	2	2	2	3	2
267	9216	49	2	1	1	1	2	1	1	2	1	2	2	2	1	2	2	2	2	1	3	2
268	8953	54	2	2	2	2	2	2	1	2	1	2	1	2	2	2	2	2	2	2	1	3
269	1965	59	1	1	1	2	2	2	2	2	2	3	2	2	1	2	2	2	2	1	1	2
270	1282	47	2	2	1	2	1	1	1	1	2	3	2	1	1	2	2	2	2	3	2	1
271	1126	72	2	2	2	1	2	2	1	2	1	3	3	1	1	2	2	2	2	2	3	3
272	9931	86	2	2	2	2	2	1	1	1	1	3	2	1	1	2	2	2	2	1	1	1
273	7336	67	2	1	2	2	2	1	1	1	2	3	1	2	1	1	2	2	2	1	1	2
274	4793	65	3	2	1	1	2	2	1	1	1	3	2	1	1	2	2	2	2	2	3	3

275	8460	87	3	1	2	1	2	2	2	1	2	2	1	2	1	1	2	2	2	2	3	3
276	7514	81	3	1	1	1	2	2	2	2	1	2	2	1	2	2	2	2	2	1	1	3
277	7838	79	2	1	1	1	2	2	2	2	1	3	1	2	1	1	2	2	2	2	1	2
278	3713	77	3	1	1	1	2	2	1	1	2	3	1	2	2	2	2	2	2	3	1	3
279	1864	80	3	2	2	1	2	2	1	1	2	3	3	1	1	2	2	2	2	2	1	1
280	8348	68	1	2	2	1	2	1	2	1	2	3	2	1	2	2	1	2	2	1	1	1
281	9619	85	2	1	1	2	2	1	2	1	2	3	2	1	1	2	1	1	2	1	2	3
282	3851	72	2	2	2	1	1	2	1	2	1	2	3	1	1	1	1	2	2	1	1	2
283	1883	86	1	1	1	1	2	1	1	1	2	3	3	2	1	1	2	2	2	2	1	2
284	5365	67	3	1	2	2	2	2	1	2	2	3	1	2	2	2	2	2	2	2	2	1
285	4226	84	2	1	1	2	2	1	1	1	2	2	2	2	1	2	2	2	2	1	1	1
286	6981	88	2	1	2	1	2	1	1	1	2	2	1	2	2	2	2	2	2	3	1	1
287	4727	86	3	1	1	2	2	2	2	1	2	1	1	1	2	2	2	2	2	2	2	2
288	3121	67	2	1	2	2	2	1	2	1	2	3	2	2	1	2	2	2	2	1	1	1
289	7022	73	3	1	1	1	2	2	2	1	2	3	3	1	1	2	1	2	2	3	2	2
290	8643	84	2	2	2	2	2	2	1	1	1	2	2	1	1	2	2	2	2	3	1	3
291	1720	68	1	2	1	2	2	2	2	2	2	3	1	2	2	2	2	2	2	2	3	3
292	2033	89	2	1	2	1	2	2	1	1	1	2	1	1	1	2	2	2	2	1	1	1
293	6273	83	2	2	1	1	2	2	2	2	2	3	2	1	2	1	2	2	2	1	1	3
294	3118	81	1	2	1	2	2	1	1	1	2	3	1	1	2	2	2	2	2	1	1	2
295	8667	70	1	1	1	1	2	1	2	2	2	3	3	1	2	2	2	2	2	3	1	1
296	6345	67	3	1	2	2	2	1	1	2	1	3	3	2	2	2	2	2	2	2	3	2
297	7372	70	2	1	2	1	2	2	1	1	2	3	3	2	2	1	2	2	2	1	1	3
298	8851	80	3	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	3	1
299	4411	67	1	1	1	1	2	2	2	1	1	3	3	1	1	2	2	2	2	1	3	2
300	6321	79	2	1	1	2	2	2	2	1	1	3	1	1	2	2	2	2	2	2	3	1
301	2228	79	2	2	2	2	2	1	1	1	2	2	1	2	1	2	2	2	2	1	1	2
302	1567	74	2	2	2	1	2	2	2	1	2	3	1	2	1	2	2	2	2	1	2	2
303	3255	65	3	1	2	1	2	1	1	1	2	3	1	1	1	2	2	2	2	1	1	1
304	1085	68	2	2	1	1	2	2	2	2	1	3	2	1	2	2	2	2	2	1	3	2
305	5246	86	3	1	2	1	2	2	2	1	2	3	1	2	1	2	1	2	2	2	3	1
306	3237	66	2	1	1	1	2	1	1	2	2	2	1	1	2	1	2	2	1	1	1	3
307	5230	67	2	2	2	1	2	2	2	2	2	3	3	2	2	1	2	2	2	1	2	3
308	8656	72	2	1	2	1	2	1	2	1	2	3	1	1	1	2	2	2	2	2	1	2
309	7711	88	2	1	1	1	2	1	1	1	2	3	3	2	2	2	2	2	1	2	1	2
310	3792	73	3	2	1	2	2	1	2	1	1	3	3	1	2	2	2	2	2	2	3	2
311	6098	86	3	2	2	2	2	2	1	2	2	2	2	2	1	2	2	2	1	1	3	1
312	7680	77	2	2	1	2	2	2	2	1	2	2	3	2	1	2	2	2	2	1	2	1
313	4249	86	3	2	1	1	2	1	2	2	1	3	2	1	2	2	2	2	2	2	3	3
314	1337	79	3	1	1	2	2	1	1	1	2	2	1	1	1	2	2	2	2	2	1	2
315	3859	72	3	2	1	1	2	1	1	2	2	3	1	1	2	2	2	2	2	2	1	2
316	8852	65	1	1	1	2	2	2	2	1	2	3	1	2	1	2	2	2	2	3	3	2
317	3727	82	1	2	2	1	2	1	1	1	2	3	1	1	2	2	2	2	2	2	1	2
318	8547	71	1	1	2	1	2	1	1	1	2	3	3	2	1	2	2	2	2	2	1	3
319	8775	78	3	2	1	2	2	2	2	2	1	3	3	2	1	2	2	2	2	2	1	1
320	1621	65	1	1	1	2	2	2	2	2	2	3	3	1	2	2	2	2	2	2	1	2

321	9209	78	1	1	2	1	2	2	1	2	2	3	3	2	1	2	2	2	1	2	1	2
322	9182	85	3	2	1	1	2	2	1	1	2	1	2	1	2	2	2	2	2	2	1	2
323	4274	71	2	2	2	2	2	2	1	1	2	3	2	2	2	2	2	2	2	1	3	1
324	3917	74	3	1	2	1	1	1	1	1	1	3	1	2	1	2	2	2	2	2	1	3
325	8957	83	1	1	1	2	2	1	1	1	1	3	2	1	2	2	2	2	2	3	1	1
326	7732	82	3	1	1	1	2	2	2	2	2	2	3	1	2	2	2	2	2	1	2	2
327	6792	77	3	1	2	1	2	2	1	1	2	3	1	2	2	1	2	1	2	2	1	3
328	3502	76	2	2	1	1	2	2	1	1	2	3	1	1	2	2	2	2	2	2	3	3
329	9991	79	2	1	2	2	2	1	1	1	1	3	2	1	2	2	2	2	1	2	1	1
330	3073	78	3	1	1	2	2	1	1	1	1	3	1	1	1	2	2	2	2	3	2	3
331	4296	74	2	1	1	2	2	2	1	2	2	2	1	2	1	1	2	2	2	2	1	2
332	4933	67	2	2	2	1	2	1	1	2	1	3	1	1	2	2	2	2	2	2	1	1
333	3574	76	2	2	2	1	2	1	2	1	2	2	3	2	2	1	2	2	1	3	3	3
334	8964	69	2	1	1	2	1	2	1	2	2	3	1	2	1	2	2	2	2	3	1	3
335	3333	66	2	1	2	1	2	1	1	2	1	3	2	2	2	2	2	1	2	2	1	1
336	4345	72	2	1	1	2	2	1	2	1	1	3	2	2	2	2	2	2	2	2	1	3
337	6629	87	2	2	2	1	2	2	1	2	2	3	1	2	1	2	2	2	2	1	1	2
338	9940	80	2	2	1	1	2	2	1	2	2	3	3	1	2	2	2	2	2	2	1	2
339	7975	73	2	2	2	2	2	1	2	1	2	3	1	1	2	2	2	2	2	3	1	1
340	9497	78	3	1	1	1	2	2	1	2	2	1	3	2	2	2	2	2	2	3	2	2
341	5024	69	1	1	2	1	2	1	1	2	2	2	1	1	2	2	2	2	2	1	1	2
342	7502	69	3	1	2	1	2	1	1	2	2	3	2	1	2	2	2	2	2	3	1	1
343	6863	79	2	1	1	2	2	2	1	1	2	3	1	1	1	1	2	2	2	2	2	2
344	3368	78	1	1	2	1	2	2	2	1	2	3	1	1	2	1	2	2	2	3	3	1
345	2731	79	2	1	2	1	2	2	2	1	2	3	3	1	2	2	2	2	2	2	1	1
346	4259	67	2	1	1	1	2	1	1	2	2	2	1	1	2	2	2	2	2	1	3	2
347	9443	88	2	1	1	2	2	1	1	2	1	2	2	2	2	2	1	2	2	2	3	3
348	1363	82	2	1	2	1	2	1	2	1	2	3	1	2	1	2	2	2	2	1	1	1
349	4781	81	1	1	1	2	2	2	1	2	2	3	3	1	2	1	2	2	2	3	2	2
350	4607	86	3	2	1	2	2	1	1	1	1	3	3	2	2	1	2	2	2	1	1	1
351	5850	74	2	2	2	2	2	2	1	1	1	2	1	1	1	2	2	2	2	3	1	3
352	3097	81	2	1	1	2	2	2	1	2	2	2	3	2	2	2	2	2	2	1	1	3
353	4217	85	2	1	1	1	2	2	2	1	2	3	2	2	2	2	2	1	1	2	3	3
354	3905	69	1	2	1	1	2	2	2	1	2	3	2	2	1	2	2	2	1	3	1	3
355	6518	88	2	1	2	1	2	1	1	1	2	3	3	2	1	2	2	2	2	2	1	3
356	9763	72	2	2	1	1	2	1	2	2	2	2	1	1	2	2	2	2	1	2	3	2
357	4722	75	2	2	1	2	2	1	1	2	2	3	3	1	2	2	2	2	2	3	2	2
358	3260	76	2	1	2	2	2	1	1	2	2	3	2	1	2	2	2	2	2	3	2	2
359	1119	87	1	2	2	2	2	2	1	2	2	3	3	1	1	2	2	2	2	3	1	1
360	3932	86	3	2	1	2	2	1	1	2	1	2	1	2	2	2	2	2	2	2	1	2
361	8123	88	2	1	1	1	2	2	2	1	2	3	3	2	1	2	1	2	2	3	1	2
362	6802	83	2	2	2	2	2	2	1	1	1	3	2	1	1	2	2	2	1	1	1	3
363	6157	79	2	1	1	2	2	2	1	2	1	3	3	2	1	2	2	1	1	3	3	2
364	4432	65	3	1	1	1	2	1	2	1	1	3	2	2	2	2	2	2	2	2	1	2
365	3496	65	2	1	2	2	2	1	1	2	1	2	2	2	1	1	2	2	2	2	1	3
366	5104	88	2	1	2	2	2	1	2	2	2	2	3	2	1	2	2	2	2	2	1	1

367	5624	80	3	2	2	1	2	1	1	1	2	3	3	2	2	2	2	2	2	1	1	1
368	2480	86	2	1	2	1	2	2	2	2	2	3	3	2	1	2	2	2	2	2	1	3
369	3030	82	1	1	1	1	2	2	2	1	2	1	1	2	2	2	1	2	2	1	1	3
370	8465	70	2	1	2	2	2	1	2	2	2	3	1	2	2	2	2	2	2	1	1	2
371	3899	88	2	1	1	1	2	1	1	2	2	3	3	2	2	2	2	2	2	2	1	1
372	5534	79	2	2	1	2	2	2	2	2	2	3	1	2	2	2	2	2	1	2	3	3
373	8959	73	2	1	1	1	2	2	1	2	1	3	3	2	2	2	2	2	2	2	1	3
374	7700	88	2	1	2	1	2	1	2	1	1	2	3	1	2	2	2	2	2	1	1	2

Annexure - Thesis Protocol Approval (TPA)

KIMS/ME-275/2023

June 2, 2023

Executive Director,
National Board of Examinations in Medical Sciences
Medical Enclave, Ansari Nagar, Mahatma Gandhi Marg
(Ring Road) New Delhi - 110029

Subject - Thesis Protocol Approval Letter

Sir,

This is for your kind information that the research proposal/Thesis protocol of below listed DNB Candidate have been considered and reviewed by the Institutional Ethics Committee (IEC) in its meeting held on 04.05.2023.

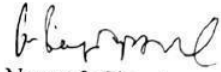
S.No	Name of DNB Candidate	Specialty	Session	Roll No	Thesis Topic Title
1	DR. SHAIK KARISHMA BANU	EMERGENCY MEDICINE	2022	2266008496	TO DETERMINE THE PROGNOSTIC ACCURACY OF THE HEART SCORE AS A PREDICTOR FOR MAJOR ADVERSE CARDIAC EVENTS IN PATIENTS PRESENTING WITH CHEST PAIN TO THE EMERGENCY DEPARTMENT IN A TERTIARY CARE HOSPITAL

The IEC which reviewed the proposal is duly registered with the department of Health Research (DHR), Ministry of Health & Family Welfare, Government of India. The authenticated copies of composition of the IEC is enclosed herewith. The IEC has approved conducting the study on above listed research proposal of DNB candidate for the purpose of writing their DNB thesis

It is further certified that the proposed research protocol have not been/shall not be submitted elsewhere for any degree, fellowship or any other titles for recognition. The minutes of aforesaid meeting of IEC are available with the hospital and can be reproduced before NBEMS, if so required, at any point of time.



Name & Signature of the
Academic Head/DNB Coordinator



Name & Signature of
Administrative Head

Prof. S. Manimala Rao

DA (Lond) M.D. FCCP, FCCM
Director, Medical Education (Academics)
Sr. Consultant Anaesthesiologist & Critical Care
Minister Road, Secunderabad-500003

Dr. P. RAJA GOPAL

Krishna Institute of Medical Sciences Limited
Minister Road, Secunderabad - 500003, Telangana, India
Phone: +91 40 4488 5000/4488 5184 | Fax: +91 40 27840986 | Email: kims@kims-hospitals.com
CIN: L55105TG2013PLC040558
Minister Road, Secunderabad-500003

Encl: Composition of Institutional Ethics Committee (IEC)

KIMS Scientific Review/ Research Committee for Thesis**COMPOSITION*****Chairperson***

Prof.S. Manimala Rao,
MD., DA., FICEM
(Head of the Institute)

Members

Mr. G. Manikanth.
M.Sc (Statistician),
(Institutional).

Dr. T. Suhasini
MD, EDIC, FICCM
Local Teaching Faculty
(Non-Institutional).

Guide

Dr. Praveen Kumar
Kulkarni

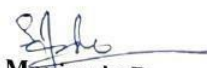
Co-Guide

Dr. Srinivas

Dr.D. Radhika Chowdary,
MD (Bio Chemistry)
Basic Sciences Faculty
Senior Consultant

THESIS PROTOCOL APPROVAL

This is to certify that DR. SHAIK KARISHMA BANU, DNB Trainee in the Discipline of EMERGENCY MEDICINE presented the thesis entitled "TO DETERMINE THE PROGNOSTIC ACCURACY OF THE HEART SCORE AS A PREDICTOR FOR MAJOR ADVERSE CARDIAC EVENTS IN PATIENTS PRESENTING WITH CHEST PAIN TO THE EMERGENCY DEPARTMENT IN A TERTIARY CARE HOSPITAL " in the KIMS Scientific Review/ Research Committee for Thesis on 15.03.2023. The same has been approved by the above committee.


Prof. S. Manimala Rao
Chairperson

Prof. S. Manimala Rao
DA (Lond) M.D. FCCP, FICCM
Director, Medical Education (Academics)
Sr. Consultant Anaesthesiologist & Critical Care
Krishna Institute of Medical Sciences Ltd.
Minister Road, Secunderabad-500003

KIMS Ethics Committee for Thesis**COMPOSITION*****Chairperson***

Dr. R. Gopinath,
MD., DM. (Anaesthesia)
(Non-Institutional)

Members

Dr. Rafia Sultana
MD(Pharmacology),
DM (Clinical Pharmacology),
Basic Medical Scientist
(Non-Institutional)

Dr. (Smt) Kanakabhushanam
MD., Clinician.
(Institutional)

Mr. G.S.L.N. Rao
Lay Person.
(Non-Institutional)

Mr.Kodali Venkaiah
Social Scientist & Statistician
(Non-Institutional)

Dr.Sairam Reddy Palicherla
Social Scientist & Philosopher
(Non-Institutional)

S. Naveen Chakravarthy.
Legal Expert
(Non-Institutional)

Zubair Fazal.
Legal Expert
(Non-Institutional)

Member Secretary

Dr. Anil Kumar Bilollikar,
MBBS., MD (Microbiology),
Member Secretary
(Institutional)

Dr. M.Sailaja,
MBBS., MD (Pathology),
Joint Member Secretary
(Institutional)

THESIS PROTOCOL APPROVAL

This is to certify that Dr. SHAIK KARISHMA BANU, DNB Trainee in the Discipline of, EMERGENCY MEDICINE presented the thesis entitled "TO DETERMINE THE PROGNOSTIC ACCURACY OF THE HEART SCORE AS A PREDICTOR FOR MAJOR ADVERSE CARDIAC EVENTS IN PATIENTS PRESENTING WITH CHEST PAIN TO THE EMERGENCY DEPARTMENT IN A TERTIARY CARE HOSPITAL " in the KIMS Ethics Committee for Thesis on 04/05/2023. The same has been approved by the above committee.



Dr. Anil Kumar Bilollikar
Member Secretary

Secretary
KIMS Ethics Committee,
Krishna Institute of Medical Sciences,
1-8-31/1, Minister Road, Secunderabad - 500003.

Krishna Institute of Medical Sciences Limited

Minister Road, Secunderabad - 500 003, Telangana, India

Phone: +91 40 4488 5000/4488 5184 | Fax: +91-40-27840980 | kims-hospitals.com

CIN: L55101TG1973PLC040558